All Polynomial Generators Preserve Distance with Mutual Correlated Agreement

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Abstract

A generator is a function that maps a random seed to a list of coefficients. We study generators that preserve distance to a linear code: the linear combination of any list of vectors using coefficients sampled by the generator has distance to the code no smaller than that of the original vectors, except for a small error. Distance preservation plays a central role in modern probabilistic proofs, and has been formalized in several ways. We study *mutual correlated agreement*, the strongest known form of distance preservation.

We initiate a systematic study of mutual correlated agreement, aiming to characterize the class of generators with this property. Towards this, we study polynomial generators, a rich class that includes all examples of generators considered in the distance preservation literature. Our main result is that *all polynomial generators guarantee mutual correlated agreement for every linear code*. This improves on prior work both in generality (the class of generators covered) and in parameters (the error bounds).

We additionally provide new results for the case where the linear code is a Reed–Solomon code, which is of particular interest in applications. We prove that all polynomial generators satisfy mutual correlated agreement for Reed–Solomon codes up to the Johnson bound. In particular, we improve upon the state-of-the-art by Ben-Sasson, Carmon, Ishai, Kopparty, and Saraf (FOCS 2020) and answer a question posed by Arnon, Chiesa, Fenzi, and Yogev (Eurocrypt 2025).

Along the way we develop a flexible and general toolbox for mutual correlated agreement, and are the first to establish distance preservation for generators that lie beyond polynomial generators.

Keywords: proximity testing, distance preservation, mutual correlated agreement

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1 Introduction

We study probabilistic reductions that preserve the distance of a set of vectors to a linear code. The reductions are captured via generators that, given a random seed, output coefficients used to linearly combine the vectors. We recall notions of distance preservation and then introduce the coding theory problem that we study.

Distance preservation. A *generator* G over a field $\mathbb{F}$ with output size ℓ is a function that maps a seed $\mathbf{x}$ in a set S to a coefficient vector in $\mathbb{F}^\ell$. In its simplest form, distance preservation for G states the following: for every $\mathbf{u}_1, \ldots, \mathbf{u}_\ell \in \mathbb{F}^n$ where at least one $\mathbf{u}_i$ is far (in Hamming distance) from a linear code $\mathcal{C} \subseteq \mathbb{F}^n$, the linear combination $\mathbf{u}_\mathbf{x} \coloneqq \sum_{j\in[\ell]} G(\mathbf{x})_j \cdot \mathbf{u}_j$ is far from $\mathcal{C}$ with high probability over a random seed $\mathbf{x} \in S$. For example, [RVW13] prove for the identity generator $G\colon \mathbb{F}^\ell \to \mathbb{F}^\ell$ where $G(\mathbf{x}) \coloneqq \mathbf{x}$ (the ℓ coefficients are random elements in $\mathbb{F}$) that the probability that the maximum distance drops by more than half is at most $\frac{1}{|\mathbb{F}|}$; they use this to design interactive proofs of proximity. Statements about distance preservation such as these are fundamental coding-theoretic properties that play an essential role in various *batch proximity testing problems*, which involve protocols for testing whether $\mathbf{u}_1, \ldots, \mathbf{u}_\ell$ are all close to $\mathcal{C}$. Applications, however, often demand forms of distance preservation that are *more powerful and flexible* than the result in [RVW13].

Improved distance preservation. Statements about distance preservation can be superior in several ways.

- *No distance drop:* [AHIV17; BKS18; BGKS20] show that for the identity generator $G(\mathbf{x}) \coloneqq \mathbf{x}$ studied in [RVW13] the maximum distance *does not drop at all* (except for a small error over the choice of seed $\mathbf{x}$).
- *More generators:* [BCIKS20] proves distance preservation for the univariate powers generator $G\colon \mathbb{F} \to \mathbb{F}^{d+1}$ where $G(x) \coloneqq (1, x, \ldots, x^d)$. Several works [BCL22; BCGL22; DP24a; MZ25; BMMS25b] establish distance preservation properties for specific generators based on low-degree polynomials. These other specific generators are chosen to best fit the application and setting studied in those papers.
- *Quality of distance preservation:* [BCIKS20] introduces distance preservation with *correlated agreement* (CA), which states that if $\mathbf{u}_\mathbf{x}$ is close to $\mathcal{C} \subseteq \mathbb{F}^n$ for many $\mathbf{x} \in S$ then there is a large subset of indices $T \subseteq [n]$ over which $\mathbf{u}_1, \ldots, \mathbf{u}_\ell$ agree with corresponding codewords in $\mathcal{C}$. [BCIKS20] shows that the identity generator and the univariate powers generator have correlated agreement if $\mathcal{C}$ is a Reed–Solomon code. [ACFY25] introduces distance preservation with *mutual correlated agreement* (MCA), a strengthening that we study in this paper and discuss below. These stronger forms of distance preservation are useful in applications because they provide information on *how* the vectors agree with the code $\mathcal{C}$.

Mutual correlated agreement. We study distance preservation with *mutual correlated agreement* (MCA) [ACFY25]; informally, for many seeds $\mathbf{x} \in S$, if $\mathbf{u}_\mathbf{x} \coloneqq \sum_{j\in[\ell]} G(\mathbf{x})_j \cdot \mathbf{u}_j$ agrees with an element in $\mathcal{C}$ over a large subset $T \subseteq [n]$ of indices, then each $\mathbf{u}_j$ agrees with a corresponding codeword in $\mathcal{C}$ *over the same set of indices* T. The fraction of $\mathbf{x} \in S$ for which this does not hold is the *MCA error of* G *for the code* $\mathcal{C}$.

Definition 1. $G\colon S \to \mathbb{F}^\ell$ *has* **mutual correlated agreement** *for a linear code* $\mathcal{C} \subseteq \mathbb{F}^n$ *with error* ϵ_{MCA} *if*

$$\forall\, \mathbf{U} = \begin{pmatrix} \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_\ell \end{pmatrix} \in \mathbb{F}^{\ell\times n},\ \forall\, \gamma \in [0,1], \quad \Pr_{\mathbf{x}\in S}\left[\exists\, T \subseteq [n] :\ \begin{array}{l} |T| \geq n\cdot(1-\gamma) \\ \wedge\ (G(\mathbf{x})\cdot\mathbf{U})|_T \in \mathcal{C}|_T \\ \wedge\ \vee_{j\in[\ell]}\, \mathbf{u}_j|_T \notin \mathcal{C}|_T \end{array}\right] \leq \epsilon_{\mathsf{MCA}}(\gamma)\ .$$

Of course, any generator has MCA for any code with the trivial error bound $\epsilon_{\mathsf{MCA}}(\gamma) \coloneqq 1$; hence when we say that a generator "has MCA" we mean that the error bound $\epsilon_{\mathsf{MCA}}(\gamma)$ is not trivial.

MCA is related to mathematically elegant properties of linear codes. For example, MCA implies a natural *list-decoding commutativity*: if G has MCA for the code $\mathcal{C}$ then, except for the MCA error over a random

seed $\mathbf{x}$, the list of codewords close to the combined word $\mathbf{u}_{\mathbf{x}} \coloneqq \sum_{j\in[\ell]} G(\mathbf{x})_j \cdot \mathbf{u}_j$ equals the list obtained by taking the linear combination according to $G(\mathbf{x})$ of the codewords in the lists of each word $\mathbf{u}_j$; that is, "linearly combine and then list decode" equals "list decode and then linearly combine". (See Section A.)

Properties such as list-decoding commutativity are at the heart of the power of MCA for the design and analysis of testing protocols like interactive oracle proofs (IOPs) and interactive oracle reductions (IORs). MCA has thus rapidly become the "gold standard" for distance preservation, by enabling highly efficient protocols and facilitating simpler security analyses (see, e.g., [ACFY25; BCFW25; BMMS25b; GMW25]).

Unfortunately, MCA remains *poorly understood*. The MCA error of the affine space generator $G\colon \mathbb{F}^s \to \mathbb{F}^{s+1}$ where $G(x_1,\ldots,x_s) \coloneqq (1,x_1,\ldots,x_s)$ for every linear code is studied in [GKL24; GCXK25]. However, which generators G have small MCA error and how this error depends on properties of the code $\mathcal{C}$ remains wide open. In fact, the study of distance preservation for several specific generators has led to a medley of different and suboptimal analyses (see [BCL22; BCGL22; DP24a; MZ25; BMMS25b]).

The prominent practical applications of MCA motivate a deeper mathematical understanding of MCA. For example, all known analyses that establish distance preservation for generators (including those in this paper) are limited by the list-decoding bound of the given code $\mathcal{C}$. Yet, establishing MCA beyond this bound (if at all possible) for codes such as Reed–Solomon codes is closely related to desirable efficiency improvements in practical hash-based succinct arguments (e.g., see [ACFY25]), so much so that Ethereum Foundation has allocated over $1M towards resolving certain conjectures about MCA [E25].

Which generators have MCA? We initiate a systematic study of generators with MCA: what are the necessary and sufficient properties for a generator $G\colon S \to \mathbb{F}^\ell$ to have MCA? An initial observation is that G must be *zero-evading*, namely, the zero-evading error $\epsilon_{\mathsf{ZE}} \coloneqq \max_{\mathbf{v}\in\mathbb{F}^\ell\setminus\{\mathbf{0}\}} \Pr_{\mathbf{x}\in S}\left[G(\mathbf{x})\cdot\mathbf{v}^\intercal = 0\right]$ (maximum probability of obtaining a zero for a non-zero vector) must be small because $\epsilon_{\mathsf{ZE}} \le \min_{\gamma\in[0,1]} \epsilon_{\mathsf{MCA}}(\gamma)$ (see Lemmas 3.16 and 3.18). We wish to understand the reverse direction:

Do all zero-evading generators have mutual correlated agreement?

In other words, can one upper bound the MCA error $\epsilon_{\mathsf{MCA}}(\gamma)$ of a generator G in terms of its zero-evading error ϵ_{ZE} (perhaps up to some loss beyond ϵ_{ZE})? We seek a fundamental understanding of why and how a zero-evading generator G has a non-trivial MCA error $\epsilon_{\mathsf{MCA}}(\gamma)$.

In this paper we focus on the distance preservation properties of *polynomial generators*, a large and flexible class of zero-evading generators that includes all examples of generators previously studied in the distance preservation literature.

Definition 2. *For $S_1,\ldots,S_s \subseteq \mathbb{F}$, $G\colon S_1\times\cdots\times S_s \to \mathbb{F}^\ell$ is a* **polynomial generator** *if $G(\mathbf{x}) = (P_j(\mathbf{x}))_{j\in[\ell]}$ for every $\mathbf{x}\in S_1\times\cdots\times S_s$, where $P_1,\ldots,P_\ell$ are linearly independent polynomials in $\mathbb{F}[\mathtt{X}_1,\ldots,\mathtt{X}_s]$.*

1.1 Our results

Our main result is that **all polynomial generators have MCA for every linear code**. Moreover, for every polynomial generator we establish **better MCA errors for Reed–Solomon codes**, linear codes whose MCA properties have significant practical implications. Tables 1 to 3 compare our results and prior art in the distance preservation literature; our error bounds **match or improve all bounds for every generator** that was previously studied. We obtain our results thanks to a flexible and general framework of tools for studying MCA that we deem of independent interest. Next we discuss our results in more detail.

MCA for arbitrary linear codes. The polynomial generator $G\colon S_1\times\cdots\times S_s \to \mathbb{F}^\ell$ where $G(\mathbf{x}) = (P_j(\mathbf{x}))_{j\in[\ell]}$ has zero-evading error $\epsilon_{\mathsf{ZE}} \le \sum_{i\in[s]} \frac{d_i}{|S_i|}$ and $d_i \coloneqq \max_{j\in[\ell]}\{\deg_{\mathtt{X}_i}(P_j)\}$ is the maximum degree

of $\mathtt{X}_i$ in $(P_j)_{j\in[\ell]}$. Hence for G we expect that $\min_{\gamma\in[0,1]} \epsilon_{\mathsf{MCA}}(\gamma) \geq \sum_{i\in[s]} \frac{d_i}{|S_i|}$. The theorem below bounds the MCA error ϵ_{MCA} of this polynomial generator *for every linear code* $\mathcal{C}$, with a bound corresponding to a unique-decoding analysis and another corresponding to a list-decoding analysis. (While in introductory sections we use big-O notation, all technical sections include explicit constants in the upper bounds.)

Theorem 1 (informal)**.** *Let* $G\colon S_1 \times S_2 \times \cdots \times S_s \to \mathbb{F}^\ell$ *be a polynomial generator where* $G(\mathbf{x}) = (P_j(\mathbf{x}))_{j\in[\ell]}$*; let* $d_i \coloneqq \max_{j\in[\ell]}\{\deg_{\mathtt{X}_i}(P_j)\}$ *for every* $i \in [s]$*. Then for every linear code* $\mathcal{C} \subseteq \mathbb{F}^n$ *with relative distance* $\delta_{\mathcal{C}}$ *and tradeoff parameter* $\eta \in (0,1)$*,* G *has MCA for* $\mathcal{C}$ *with error*

$$\epsilon_{\mathsf{MCA}}(\gamma) \coloneqq \begin{cases} \max\{n\cdot\gamma, 1\} \cdot \sum_{i\in[s]} \frac{d_i}{|S_i|} & \text{if } \gamma \leq \delta_{\mathcal{C}}/(\max_{i\in[s]} d_i + 2) \\ O\left(\frac{n}{\eta}\right) \cdot \sum_{i\in[s]} \frac{d_i}{|S_i|} & \text{if } \gamma \leq 1 - (1-\delta_{\mathcal{C}}+\eta)^{\frac{1}{\max_{i\in[s]} d_i + 2}} \end{cases}.$$

Both upper bounds are the zero-evading error of G with a multiplicative overhead; the two bounds are incomparable, both in applicability (whether the bound holds based on γ) and the achieved error bound. Theorem 1 *matches or improves all bounds previously established for specific generators*; see Tables 1 and 2 for a comparison. We prove that the error bound in the unique-decoding setting (the upper one) is *tight* for the univariate powers generator $G(x) = (1, x, \ldots, x^s)$ (see the discussion in Section 2.3).

The affine space generator $G\colon \mathbb{F}^s \to \mathbb{F}^{s+1}$ where $G(x_1, \ldots, x_s) \coloneqq (1, x_1, \ldots, x_s)$ is a notable special case; its zero-evading error is $\epsilon_{\mathsf{ZE}} = \frac{1}{|\mathbb{F}|}$. Numerous works obtain error bounds on its distance preservation properties [RVW13; AHIV17; BGKS20; BCIKS20; GKL24; GCXK25]. For any given linear code $\mathcal{C}$, directly applying Theorem 1 to this generator recovers the best bounds in prior work [GKL24]:

$$\epsilon_{\mathsf{MCA}}(\gamma) \coloneqq \begin{cases} \max\{n\cdot\gamma, 1\} \cdot \frac{s}{|\mathbb{F}|} & \text{if } \gamma \leq \delta_{\mathcal{C}}/3 \\ O\left(\frac{n}{\eta}\right) \cdot \frac{s}{|\mathbb{F}|} & \text{if } \gamma \leq 1 - (1-\delta_{\mathcal{C}}+\eta)^{\frac{1}{3}} \end{cases}.$$

In fact, for any linear code $\mathcal{C} \subseteq \mathbb{F}^n$, we show that the MCA error of the affine space generator is equal to the MCA error of the affine line generator $\mathbb{F} \to \mathbb{F}^2 ; x \mapsto (1, x)$ up to a small, constant multiplicative overhead.

Lemma 1 (informal)**.** *Let* $\mathcal{C} \subseteq \mathbb{F}^n$ *be a linear code. Then the generator* $G\colon \mathbb{F}^s \to \mathbb{F}^{s+1}$ *such that* $G(x_1, \ldots, x_s) \coloneqq (1, x_1, \ldots, x_s)$ *has MCA for* $\mathcal{C}$ *with error*

$$\epsilon_{\mathsf{MCA}}(\gamma) \coloneqq \begin{cases} \max\{n\cdot\gamma, 1\} \cdot \frac{1}{|\mathbb{F}|} & \text{if } \gamma \leq \delta_{\mathcal{C}}/3 \\ O\left(\frac{n}{\eta}\right) \cdot \frac{1}{|\mathbb{F}|} & \text{if } \gamma \leq 1 - (1-\delta_{\mathcal{C}}+\eta)^{\frac{1}{3}} \end{cases}.$$

This gives an MCA error for the affine space generator that is *independent of the number of variables* s, and leads to the tightest bounds on the MCA error for this generator. This agrees with the fact that $\epsilon_{\mathsf{ZE}} = \frac{1}{|\mathbb{F}|}$ is independent of s. Moreover, our reduction does not impose a lower bound on the field size, contrary to prior analyses for CA [GG25] (see also Section C).

[BGKS20] constructs a linear code $\mathcal{C}$ and words $\mathbf{u}_1, \mathbf{u}_2$ that are far from $\mathcal{C}$ such that their linear combination via the affine line generator $x \mapsto (1, x)$ drops with probability almost one. This gives intuition for why a result that holds for every linear code, such as the above lemma, holds only up to some distance γ and incurs errors that get close to 1 in some parameter settings (e.g., when the blocklength n of $\mathcal{C}$ approaches $|\mathbb{F}|$).

MCA for Reed–Solomon codes. Numerous highly efficient probabilistic proofs rely on distance preservation properties of Reed–Solomon codes (e.g., [AHIV17; BBHR18; BGKS20; BFHVXZ20; DP24a; ACFY24; ZCF24; ACFY25; BCFRRZ25; WHV24; DP24b; MZ25; BMNW25a; BMNW25b; BCFW25; BMMS25a;

BMMS25b; NA25; GMW25]). For Reed–Solomon codes, [BCIKS20] proves *correlated agreement* for the univariate powers generator and the affine space generator, achieving better error bounds than what is known for arbitrary linear codes. We strengthen their result to achieve MCA and use our framework to extend it to hold for all polynomial generators. Below, $\mathsf{RS}[\mathbb{F}, D, k]$ denotes the Reed–Solomon code consisting of evaluations of polynomials over $\mathbb{F}$ of degree less than k over $D \subseteq \mathbb{F}$; we let $n \coloneqq |D|$ and $\rho \coloneqq k/n$.

Theorem 2 (informal)**.** *Let $G\colon \mathbb{F}^s \to \mathbb{F}^\ell$ be a polynomial generator where $G(\mathbf{x}) = (P_j(\mathbf{x}))_{j\in[\ell]}$; let $d_i \coloneqq \max_{j\in[\ell]}\{\deg_{\mathbf{X}_i}(P_j)\}$ for every $i \in [s]$. Then for every tradeoff parameter $m \geq 3$, G has MCA for $\mathsf{RS}[\mathbb{F}, D, k]$ with error*

$$\epsilon_{\mathsf{MCA}}(\gamma) \coloneqq O\left(\frac{m^7 \cdot n^2}{\rho^{3/2}}\right) \cdot \frac{\sum_{i\in[s]} d_i}{|\mathbb{F}|} \quad \text{if } \gamma \leq 1 - \left(1 + \frac{1}{2m}\right) \cdot \sqrt{\rho}\ .$$

The error bound in Theorem 2 matches the one for correlated agreement in [BCIKS20] (when fixing the specific generators studied therein). Table 3 gives a comparison of our bound on Reed–Solomon codes with prior art (and similarly for Reed–Muller codes). As a special case, Theorem 2 resolves a question asked in [ACFY25] on the MCA error for the univariate powers generator $x \mapsto (1, x, \ldots, x^d)$ for Reed–Solomon codes; prior to this work [Hab25] resolves this question for the affine line generator $x \mapsto (1, x)$ ($d = 1$).

List decoding and MCA. If a linear code $\mathcal{C}$ is list-decodable with list size L up to distance δ, then the affine space generator has a small MCA error $\epsilon_{\mathsf{MCA}}(\gamma)$ for $\mathcal{C}$ provided $\gamma < 1 - \sqrt{1-\delta}$ [GCXK25]. This connection is particularly interesting for code families that are list-decodable up to capacity, e.g., random linear codes [GHSZ02; GHK11; CGV14; LW18], random Reed–Solomon codes [RW14; ST20; GLSTW21; FKS22; GST23; BGM23; GZ23; AGL24; LMS25], folded Reed–Solomon codes, and univariate multiplicity codes [CZ25]. However, for codes in this paper, namely arbitrary Reed–Solomon codes and arbitrary linear codes, the best known list-decoding bound is the Johnson bound, which is $\approx 1 - (1 - \delta_{\mathcal{C}})^{1/2}$ if the code $\mathcal{C}$ has distance $\delta_{\mathcal{C}}$. Applying [GCXK25]'s result would yield a non-trivial MCA error for the affine space generator only for γ within distance $\approx 1 - (1 - \delta_{\mathcal{C}})^{1/4}$ (the "double Johnson bound"). Our results for the affine space generator (implied by Lemma 1 and Theorem 2) subsume the application of [GCXK25] to these codes.

Operations on codes. Most of this work investigates which generators have MCA for general linear codes (and sometimes specific ones like Reed–Solomon codes). An orthogonal question is what happens to MCA when we fix an arbitrary generator and apply various operations on codes. We prove for example that if any generator G has MCA for some linear code $\mathcal{C}$ with error ϵ_{MCA} then G has MCA for the k-interleaving $\mathcal{C}^k$ with error at most $k \cdot \epsilon_{\mathsf{MCA}}$. Similarly we obtain upper bounds on the MCA error of arbitrary generators when we take the direct product of codes, or when we concatenate them. See Section 10 for details.

Open questions. Theorem 1 makes progress towards a characterization of generators with MCA by providing a large class of generators with MCA. Yet we already know that more can be said: we prove that a zero-evading generator in [AGHP92], which is *not* a polynomial generator, has MCA (see Section 2.6). This tells us that MCA holds beyond polynomial generators and a characterization of MCA remains open.

The behavior of MCA under natural operations on codes remains to be understood. Of particular interest are tensor products of codes, and distance amplification procedures on codes.

For the special case of Reed–Solomon codes, Theorem 2 states that any polynomial generator has MCA for γ close to the code's list-decoding bound $1 - \sqrt{\rho}$. In probabilistic proofs, distance preservation at a larger distance typically implies more efficient verification. Can one bound the MCA error for distances γ beyond $1 - \sqrt{\rho}$? [ACFY25] conjectures MCA for the univariate powers generator for γ close to the code's capacity $1 - \rho$, and similar conjectures on Reed–Solomon codes are sometimes used in practice for efficiency. Yet [DG25; CS25] shows that this is too much to hope for: there is a family of Reed–Solomon codes for which

distance preservation fails for the affine line generator $x \mapsto (1, x)$ when γ is close to the code's capacity $1 - \rho$. Understanding whether MCA holds between the list-decoding and capacity bounds remains open.

1.2 Comparison with prior work

We give a comparison with prior work that studies distance preservation with CA and with MCA.

Table 1: arbitrary linear codes in the unique decoding setting. [AHIV17] proves that the identity generator has CA error $n \cdot \delta_{\mathcal{C}} \cdot \frac{1}{|\mathbb{F}|}$ within distance $\frac{\delta_{\mathcal{C}}}{3}$. Our results imply that the affine space generator $(x_1, \ldots, x_s) \mapsto (1, x_1, \ldots, x_s)$ has MCA (hence, CA) with the improved error $\frac{n \cdot \delta_{\mathcal{C}}}{3} \cdot \frac{1}{|\mathbb{F}|}$ within distance $\frac{\delta_{\mathcal{C}}}{3}$.

Table 2: arbitrary linear codes in the list-decoding setting. [BGKS20] shows that the affine line generator $x \mapsto (1, x)$ has CA for every linear code almost up to distance $1 - (1 - \delta_{\mathcal{C}})^{1/3}$, known as the "1.5 Johnson bound". [GKL24] obtains the stronger MCA for the affine space generator $(x_1, \ldots, x_s) \mapsto (1, x_1, \ldots, x_s)$ for every linear code almost up to the 1.5 Johnson bound. [Zei25] improves upon the constant in the error of [GKL24] in the case of the affine line generator $x \mapsto (1, x)$. [BMMS25b] uses [Zei25] to prove MCA for the multilinear equality generator $\mathbf{x} \mapsto (\widehat{\mathsf{eq}}(\mathbf{v}, \mathbf{x}))_{\mathbf{v} \in \{0,1\}^s}$ almost up to the 1.5 Johnson bound. We recover all of these specific results via our Theorem 1 (MCA for all polynomial generators). Moreover, for the affine space generator $(x_1, \ldots, x_s) \mapsto (1, x_1, \ldots, x_s)$, we improve over prior bounds by removing the dependency on s.

Table 3: Reed–Solomon and Reed–Muller. [BCIKS20] shows that both the affine space generator $(x_1, \ldots, x_s) \mapsto (1, x_1, \ldots, x_s)$ and the univariate powers generator $x \mapsto (1, x, \ldots, x^d)$ have CA for Reed–Solomon codes up to the Johnson bound (i.e., beyond the 1.5 Johnson bound known for arbitrary linear codes). We show that the same bound holds for MCA and for every polynomial generator.

[MZ25] uses [BCIKS20] to show that the univariate powers generator has CA for Reed–Muller codes up to some distance parameter well below the list-decoding bound. Our results imply a tighter bound on the MCA error for the univariate powers generator for Reed–Muller codes.

	Generator	(M)CA	Distance	Error
[AHIV17]	$(\mathtt{X}_1, \ldots, \mathtt{X}_s)$	CA	$\delta_{\mathcal{C}}/3$	$n \cdot \delta_{\mathcal{C}} \cdot \frac{1}{\lvert\mathbb{F}\rvert}$
This work (Lemma C.5)	$(1, \mathtt{X}_1, \ldots, \mathtt{X}_s)$	CA	$\delta_{\mathcal{C}}/3$	$\frac{n \cdot \delta_{\mathcal{C}}}{3} \cdot \frac{1}{\lvert\mathbb{F}\rvert}$
This work (Corollary 8.4)	$(1, \mathtt{X}_1, \ldots, \mathtt{X}_s)$	MCA	$\delta_{\mathcal{C}}/3$	$\frac{n \cdot \delta_{\mathcal{C}}}{3} \cdot \frac{1}{\lvert\mathbb{F}\rvert}$

Table 1: Results on correlated agreement (CA) and mutual correlated agreement (MCA) based on unique-decoding analyses, for linear/affine space generators and arbitrary linear codes $\mathcal{C} \subseteq \mathbb{F}^n$ of relative distance $\delta_{\mathcal{C}}$.

	Generator	(M)CA	Distance	Error
[BGKS20]	$(1, \mathtt{X})$	CA	$1 - (1 - \delta_{\mathcal{C}} + \eta)^{1/3}$	$O\left(\frac{1}{\eta^2}\right) \cdot \frac{1}{\lvert\mathbb{F}\rvert}$
[ABN22]	$(\vec{\mathtt{X}}^{\mathbf{v}})_{\mathbf{v} \in \{0,1\}^s}$	CA	$1 - (1 - \delta_{\mathcal{C}} + \eta)^{1/3}$	$O\left(\frac{2s}{\eta^2}\right) \cdot \frac{1}{\lvert\mathbb{F}\rvert}$
[GKL24]	$(1, \mathtt{X}_1, \ldots, \mathtt{X}_s)$	MCA	$1 - (1 - \delta_{\mathcal{C}} + \eta)^{1/3}$	$O\left(\frac{n \cdot s}{\eta}\right) \cdot \frac{1}{\lvert\mathbb{F}\rvert}$
[Zei25]	$(1, \mathtt{X})$	MCA	$1 - (1 - \delta_{\mathcal{C}} + \eta)^{1/3}$	$O\left(\frac{n}{\eta}\right) \cdot \frac{1}{\lvert\mathbb{F}\rvert}$
[BMMS25b]	$(\widehat{\mathsf{eq}}(\mathbf{v}, \vec{\mathtt{X}}))_{\mathbf{v} \in \{0,1\}^s}$	MCA	$1 - (1 - \delta_{\mathcal{C}} + \eta)^{1/3} - \eta$	$O\left(\frac{3^s}{s \cdot \eta^2}\right) \cdot \frac{s}{\lvert\mathbb{F}\rvert}$
This work (Theorem 8.2)	$(P_j(\vec{\mathtt{X}}))_{j \in [\ell]}$, $\max_{j \in [\ell]} \deg_{\mathtt{X}_i}(P_j) \leq d_i$	MCA	$1 - (1 - \delta_{\mathcal{C}} + \eta)^{\frac{1}{\max_{i \in [s]} d_i + 2}}$	$O\left(\frac{n}{\eta}\right) \cdot \frac{\sum_i d_i}{\lvert\mathbb{F}\rvert}$
This work (Corollary 8.4)	$(1, \mathtt{X}_1, \ldots, \mathtt{X}_s)$	MCA	$1 - (1 - \delta_{\mathcal{C}} + \eta)^{1/3}$	$O\left(\frac{n}{\eta}\right) \cdot \frac{1}{\lvert\mathbb{F}\rvert}$
This work (Corollary 8.6)	$(\widehat{\mathsf{eq}}(\mathbf{v}, \vec{\mathtt{X}}))_{\mathbf{v} \in \{0,1\}^s}$	MCA	$1 - (1 - \delta_{\mathcal{C}} + \eta)^{1/3}$	$O\left(\frac{n \cdot s}{\eta}\right) \cdot \frac{s}{\lvert\mathbb{F}\rvert}$

Table 2: Comparison of results on correlated agreement (CA) and mutual correlated agreement (MCA) for several generators and arbitrary linear codes $\mathcal{C} \subseteq \mathbb{F}^n$ of relative distance $\delta_{\mathcal{C}}$. The parameter $\eta \in (0, 1)$ is a tradeoff between distance and error. In the table we assume that $d_i \leq n$ for every i. The polynomial $\widehat{\mathsf{eq}}$ is the multilinear equality polynomial (see Corollary 8.6 for a definition).

	Code	Generator	(M)CA	Distance	Error
[BCIKS20]	$\mathsf{RS}[\mathbb{F}, n, \rho]$	$(\mathtt{X}_1, \ldots, \mathtt{X}_s)$	CA	$1 - \left(1 + \frac{1}{2m}\right)\sqrt{\rho}$ for $m \geq 3$	$O\left(\frac{m^7 \cdot n^2}{\rho^{3/2}}\right) \cdot \frac{1}{\lvert\mathbb{F}\rvert}$
[BCIKS20]		$(1, \mathtt{X}, \ldots, \mathtt{X}^d)$	CA		$O\left(\frac{m^7 \cdot n^2}{\rho^{3/2}}\right) \cdot \frac{d}{\lvert\mathbb{F}\rvert}$
[Hab25]		$(1, \mathtt{X})$	MCA		$O\left(\frac{m^7 \cdot n^2}{\rho^{3/2}}\right) \cdot \frac{1}{\lvert\mathbb{F}\rvert}$
This work (Theorem 9.2)		$(P_j(\vec{\mathtt{X}}))_{j \in [\ell]}$, $\max_j \deg_{\mathtt{X}_i}(P_j) \leq d_i$	MCA		$O\left(\frac{m^7 \cdot n^2}{\rho^{3/2}}\right) \cdot \frac{\sum_i d_i}{\lvert\mathbb{F}\rvert}$
[MZ25]	$\mathsf{RM}[\mathbb{F}, \mathbb{B}^2, k]$	$(1, \mathtt{X}, \ldots, \mathtt{X}^d)$	CA	$1 - O\left(\frac{d}{\lvert\mathbb{B}\rvert^{\frac{1}{14d+2}}}\right)$	$\frac{\lvert\mathbb{B}\rvert^4}{k^2 \cdot d} \cdot \frac{d}{\lvert\mathbb{F}\rvert}$
This work (Corollary 8.7)		$(1, \mathtt{X}, \ldots, \mathtt{X}^d)$	MCA	$1 - \left(\frac{k}{\lvert\mathbb{B}\rvert} + \eta\right)^{\frac{1}{d+2}}$	$O\left(\frac{\lvert\mathbb{B}\rvert^2}{\eta}\right) \cdot \frac{d}{\lvert\mathbb{F}\rvert}$

Table 3: Comparison of results on correlated agreement (CA) and mutual correlated agreement (MCA) for some generators and specific linear codes. $\mathsf{RS}[\mathbb{F}, n, \rho]$ is the Reed–Solomon code of evaluations of univariate polynomials over $\mathbb{F}$ of degree at most $\rho \cdot n - 1$ over a domain of size n. For an extension $\mathbb{B} \subseteq \mathbb{F}$, $\mathsf{RM}[\mathbb{F}, \mathbb{B}^2, k]$ consists of bivariate polynomials $f : \mathbb{B}^2 \to \mathbb{F}$ of total degree at most k. The integer $m \geq 3$ is a tradeoff between distance and error. In the table we assume that $d \leq \lvert\mathbb{B}\rvert^2$.

2 Technical overview

Prior work with bounds for MCA does not offer suitable starting points towards Theorem 1 and Theorem 2, as existing techniques typically target either specific generators or specific codes. To overcome these limitations, we develop a toolbox for MCA for our setting and explain how it helps us prove Theorem 1 and Theorem 2.

In Section 2.1 we compare the MCA and zero-evading errors. In Section 2.2 we reduce proving Theorem 1 to establishing MCA for the univariate powers generator. In Section 2.3 we sketch how to obtain MCA for the univariate powers generator for all linear codes. In Section 2.4 we sketch the proof of Lemma 1. In Section 2.5 we explain the main idea underlying Theorem 2. Lastly, in Section 2.6 we obtain a bound on the MCA error of a non-polynomial generator for all linear codes and summarize our toolbox for MCA.

2.1 On mutual correlated agreement vs. zero-evading error

We outline why the MCA error is lower bounded by the zero-evading error. This justifies our focus on the class of zero-evading generators, and serves as an opportunity to build intuition on the definition of MCA. A simple initial observation is that the MCA error of $G\colon S \to \mathbb{F}^\ell$ is monotonically non-decreasing, almost by definition. Therefore, it suffices to show that the MCA error at 0 is lower bounded by the zero-evading error:

$$\epsilon_{\mathsf{MCA}}(0) \geq \Pr_{\mathbf{x}\in S}\left[\begin{array}{l} G(\mathbf{x})\cdot\mathbf{U}\in\mathcal{C} \\ \wedge\, \exists\, j\in[\ell],\ \mathbf{u}_j\notin\mathcal{C} \end{array}\right] \geq \max_{\mathbf{v}\in\mathbb{F}^\ell\setminus\{\mathbf{0}\}} \Pr_{\mathbf{x}\in S}\left[G(\mathbf{x})\cdot\mathbf{v}^\intercal = 0\right] \quad .$$

The right-hand side is mainly about cancellations: it measures the maximum probability that a non-trivial linear combination of $G(\mathbf{x})$ equals zero. Proving the lower bound corresponds to the intuition that if the combined word $G(\mathbf{x})\cdot\mathbf{U}$ falls into the code while some of the individual words $\mathbf{u}_j$ do not, this must be due to similar cancellations. This can be concretized as follows. Without loss of generality the code is not the entire space, so there exists a word $\mathbf{u}\in\mathbb{F}^n\setminus\mathcal{C}$. Fix a non-zero vector $\mathbf{v}\in\mathbb{F}^\ell\setminus\{\mathbf{0}\}$ and define words $\mathbf{u}_1,\ldots,\mathbf{u}_\ell$ such that, for every $j\in[\ell]$, $\mathbf{u}_j \coloneqq \mathbf{v}_j\cdot\mathbf{u}$. Since $\mathbf{v}$ is not the zero vector, there is j such that $\mathbf{u}_j\notin\mathcal{C}$. Then, for every seed $\mathbf{x}\in S$, the linear combination $G(\mathbf{x})\cdot\mathbf{U}$ is equal to $(G(\mathbf{x})\cdot\mathbf{v}^\intercal)\cdot\mathbf{u}$ and thus belongs to the code if and only if $G(\mathbf{x})\cdot\mathbf{v}^\intercal = 0$. Since $\mathbf{v}$ was arbitrary, this establishes the desired lower bound.

With a bit more effort, we can show that the MCA error at 0 in fact *equals* the zero-evading error. This amounts to showing that there exists, for every $\mathbf{u}_1,\ldots,\mathbf{u}_\ell\in\mathbb{F}^n$, a non-zero vector $\mathbf{v}\in\mathbb{F}^\ell\setminus\{\mathbf{0}\}$ such that

$$\Pr_{\mathbf{x}\in S}\left[\begin{array}{l} G(\mathbf{x})\cdot\mathbf{U}\in\mathcal{C} \\ \wedge\, \exists\, j\in[\ell],\ \mathbf{u}_j\notin\mathcal{C} \end{array}\right] \leq \Pr_{\mathbf{x}\in S}\left[G(\mathbf{x})\cdot\mathbf{v}^\intercal = 0\right] \quad .$$

This is neatly achieved by introducing parity checks for the code $\mathcal{C}$ and considering the linear combinations of these parity checks that are violated by the words $\mathbf{u}_1,\ldots,\mathbf{u}_\ell$. We refer to Lemma 3.18 for details.

Towards upper bounds on ϵ_{MCA}. The previous result gives hope that one might, in general, be able to upper bound the MCA error ϵ_{MCA} of a generator $G\colon S\to\mathbb{F}^\ell$ via (some function of) its zero-evading error ϵ_{ZE}. We do not know this to be true. In this paper we establish results for the MCA error ϵ_{MCA} when G is a polynomial generator (or some other special cases). For these generators, we show that the zero-evading error *does* play a useful role in upper bounding ϵ_{MCA}. We explain this connection in Section 2.3.

2.2 All polynomials generators have MCA for all linear codes

Our approach to prove Theorem 1 consists of two main steps: (i) we develop tools to reduce MCA of polynomial generators to MCA of the univariate powers generator, with minimum parameter loss; and (ii) we prove that the univariate powers generator has MCA for all linear codes.

Let $G(\mathbf{x}) \coloneqq (P_j(\mathbf{x}))_{j\in[s]}$ be a polynomial generator. Each polynomial P_j is a linear combination of monomials in the variables $\mathtt{X}_1, \ldots, \mathtt{X}_s$, where the individual degree of $\mathtt{X}_i$ is at most $\deg_{\mathtt{X}_i}(P_j)$. Define $d_i \coloneqq \max_{j\in[\ell]}\{\deg_{\mathtt{X}_i}(P_j)\}$ and $k \coloneqq \prod_{i\in[s]}(1+d_i)$. There exists a matrix $\mathbf{A} \in \mathbb{F}^{k\times\ell}$ such that

$$(P_j(\mathtt{X}_1, \ldots, \mathtt{X}_s))_{j\in[\ell]} = \left(\bigotimes_{i\in[s]}(1, \mathtt{X}_i, \ldots, \mathtt{X}_i^{d_i})\right) \cdot \mathbf{A} \ .$$

The expression $\bigotimes_{i\in[s]}(1, \mathtt{X}_i, \ldots, \mathtt{X}_i^{d_i})$ represents the tensor product of the univariate powers generators. The polynomials $(P_j)_{j\in[\ell]}$ are linearly independent, so $\mathbf{A}$ has a left inverse (any non-trivial linear dependence on the rows of $\mathbf{A}$ yields a non-trivial linear dependence on $(P_j)_{j\in[\ell]}$). In sum, every polynomial generator can be viewed as a linear transformation of the tensor products of univariate powers generators. In light of this, we prove lemmas to track MCA across these two operations.

Lemma 2 (tensor products preserve MCA)**.** *Let $G\colon S \to \mathbb{F}^\ell$, $G'\colon S' \to \mathbb{F}^{\ell'}$ be generators that have MCA for a linear code $\mathcal{C} \subseteq \mathbb{F}^n$ with respective errors $\epsilon_{\mathsf{MCA}}, \epsilon'_{\mathsf{MCA}}$. Then the tensor product generator $G \otimes G'$ has MCA for $\mathcal{C}$ with error $\epsilon_{\mathsf{MCA}} + \epsilon'_{\mathsf{MCA}}$.*

Lemma 3 (linear transformations preserve MCA)**.** *Let $G\colon S \to \mathbb{F}^\ell$ be a generator that has MCA for a linear code $\mathcal{C} \subseteq \mathbb{F}^n$ with error ϵ_{MCA}. For every $\tilde{\ell} \in \mathbb{N}$ and $\mathbf{A} \in \mathbb{F}^{\ell\times\tilde{\ell}}$ with a left inverse, let $\tilde{G}\colon S \to \mathbb{F}^{\tilde{\ell}}$ be the generator defined as $\tilde{G}(\mathbf{x}) \coloneqq G(\mathbf{x}) \cdot \mathbf{A}$. Then $\tilde{G}$ has MCA for $\mathcal{C}$ with error ϵ_{MCA}.*

From Lemmas 2 and 3 we deduce that the MCA error $\epsilon_{\mathsf{MCA}}(\gamma)$ for $\mathcal{C}$ of the polynomial generator G is

$$\epsilon_{\mathsf{MCA}}(\gamma) = \sum_{i\in[s]} \xi_{\mathcal{C},d_i,|S_i|}(\gamma)$$

where $\xi_{\mathcal{C},d,|S|}$ is the MCA error for $\mathcal{C}$ of the univariate powers generator $G\colon S \to \mathbb{F}^{d+1}$ where $G(x) \coloneqq (1, x, \ldots, x^d)$. We refer the reader to Section 4 for proofs of the above lemmas. Next, we turn our attention to establishing the MCA error of the univariate powers generator.

Remark 1 (MDS generators)**.** The univariate powers generator is a special case of an *MDS generator*. One can associate to each generator G a corresponding linear code $\mathcal{C}_G$; we say G is MDS if $\mathcal{C}_G$ meets the Singleton bound (see Definition 3.12). In Section 6 we in fact prove a generalization of Theorem 3: every *MDS generator* has MCA for arbitrary linear codes. We believe this generalization is of independent interest as MDS generators are not necessarily polynomial generators.

2.3 The univariate powers generator

Our theorem below establishes MCA of the univariate powers generator for every linear code.

Theorem 3. *Let $G\colon S \subseteq \mathbb{F} \to \mathbb{F}^{d+1}$ be the univariate powers generator where $G(x) \coloneqq (1, x, \ldots, x^d)$. Let $\mathcal{C} \subseteq \mathbb{F}^n$ be a linear code with relative distance $\delta_{\mathcal{C}}$. Then, for every tradeoff parameter $\eta \in (0, 1)$, G has MCA for $\mathcal{C}$ with error $\epsilon_{\mathsf{MCA}}(\gamma) \coloneqq \xi_{\mathcal{C},d,|S|}(\gamma)$ where*

$$\xi_{\mathcal{C},d,|S|}(\gamma) \coloneqq \begin{cases} n \cdot \gamma \cdot \frac{d}{|S|} & \text{if } \gamma \le \delta_{\mathcal{C}}/(d+2) \\ O\left(\frac{n}{\eta}\right) \cdot \frac{d}{|S|} & \text{if } \gamma \le 1 - (1 - \delta_{\mathcal{C}} + \eta)^{\frac{1}{d+2}} \end{cases} \ .$$

Moreover, we prove that Theorem 3 is *tight* for the regime $\gamma \leq \delta_{\mathcal{C}}/(d+2)$: we show that for every linear code $\mathcal{C} \subseteq \mathbb{F}^n$ with relative minimum distance $\delta_{\mathcal{C}}$ and $\gamma < \delta_{\mathcal{C}}/2$, the MCA error of the univariate powers generator (when $S = \mathbb{F}$) is lower bounded by $n \cdot \gamma \cdot \frac{d}{|\mathbb{F}|}$. In fact, we prove the lower bound for the weakest form of distance-preservation instead of MCA; see Lemma B.1.

Next, we outline the main ideas behind the result: in Section 2.3.1 we discuss the upper bound arising in the unique-decoding setting, and in Section 2.3.2 we discuss the upper bound arising in the list-decoding setting. Our analysis in the list-decoding setting is directly inspired by the combinatorial techniques in [GKL24; Zei25]. Technical details for this theorem are in Section 6.

2.3.1 Unique-decoding setting

Fix $\mathbf{u}_1, \ldots, \mathbf{u}_{d+1} \in \mathbb{F}^n$. We define the set of "bad seeds":

$$\mathsf{B} \coloneqq \left\{ x \in S \;\middle|\; \exists\, T_x \subseteq [n] : \begin{array}{l} |T_x| \geq n \cdot (1-\gamma) \\ \wedge\ (\sum_{j \in [d+1]} G(x)_j \cdot \mathbf{u}_j)|_{T_x} \in \mathcal{C}|_{T_x} \\ \wedge\ \vee_{j \in [d+1]}\, \mathbf{u}_j|_{T_x} \notin \mathcal{C}|_{T_x} \end{array} \right\} \ .$$

The goal is to show $|\mathsf{B}| \leq n \cdot \gamma \cdot d$. Assume without loss of generality that $|\mathsf{B}| \geq d+1$ (else the claim holds trivially). For every $x \in \mathsf{B}$, define $\mathbf{u}_x \coloneqq \sum_{j \in [d+1]} G(x)_j \cdot \mathbf{u}_j$ and let T_x be a set specified in the definition of B (break ties arbitrarily); as we are within the unique decoding bound, there is a unique codeword $\mathbf{c}_x \in \mathcal{C}$ such that $\mathbf{u}_x|_{T_x} = \mathbf{c}_x|_{T_x}$. We first argue that it suffices to find codewords $\mathbf{c}^*_1, \ldots, \mathbf{c}^*_{d+1} \in \mathcal{C}$ such that

$$\forall\, x \in \mathsf{B},\ \mathbf{c}_x = \sum_{j \in [d+1]} G(x)_j \cdot \mathbf{c}^*_j \ . \tag{1}$$

Consider the agreement set T defined as

$$T \coloneqq \{ i \in [n] : \forall\, j \in [d+1], \mathbf{u}_j[i] = \mathbf{c}^*_j[i] \} \ .$$

By definition of B, for every $x \in \mathsf{B}$ there exists $i \in T_x$ with $i \notin T$. Moreover, note that $T \subseteq T_x$ because

$$\mathbf{u}_x|_T = \sum_{j \in [d+1]} G(x)_j \cdot \mathbf{u}_j|_T = \sum_{j \in [d+1]} G(x)_j \cdot \mathbf{c}^*_j|_T = \mathbf{c}_x|_T \ .$$

We deduce that, $T \subsetneq T_x$.

Set $\mathbf{v}_i \coloneqq (\mathbf{u}_1[i] - \mathbf{c}^*_1[i], \ldots, \mathbf{u}_{d+1}[i] - \mathbf{c}^*_{d+1}[i])$ for every $i \in [n]$. The vector $\mathbf{v}_i$ is non-zero precisely when $i \notin T$. For every $i \in T_x$, $\mathbf{u}_x[i] = \mathbf{c}_x[i]$ by definition, so $G(x) \cdot \mathbf{v}_i^\intercal = \mathbf{u}_x[i] - \mathbf{c}_x[i] = 0$. Hence, for every $i \in T_x \setminus T$, it holds that (i) $\mathbf{v}_i$ is non-zero and (ii) $G(x) \cdot \mathbf{v}_i^\intercal = 0$. This yields the following equivalent characterization of the set of bad seeds:

$$\mathsf{B} = \{ x \in \mathsf{B} : \exists\, i \in T_x \setminus T,\ G(x) \cdot \mathbf{v}_i^\intercal = 0 \} \ .$$

We use this characterization to bound $|\mathsf{B}|$ as follows:

$$\begin{aligned} |\mathsf{B}| &= |\{ x \in \mathsf{B} : \exists\, i \in T_x \setminus T,\ G(x) \cdot \mathbf{v}_i^\intercal = 0 \}| \\ &\leq \sum_{i \in [n] \setminus T} |\{ x \in S : G(x) \cdot \mathbf{v}_i^\intercal = 0 \}| \\ &\leq n \cdot d \ , \end{aligned} \tag{2}$$

where the last inequality follows from the zero-evading error of G. In fact, with more effort in the technical sections we can lower bound the size of T by $n \cdot (1-\gamma)$, which leads to the claimed upper bound $|\mathsf{B}| \leq n \cdot \gamma \cdot d$.

In sum, we reduce the problem of bounding the size of B to bounding the size of $T_x \setminus T$ and the zero-evading error of G, once there is an agreement set T such that $T \subsetneq T_x$ for all $x \in \mathsf{B}$. We use this idea also to bound the MCA error in the list-decoding regime in Section 2.3.2 and to show Theorem 2 in Section 2.5.

We are left to show the existence of $\mathbf{c}_1^*, \ldots, \mathbf{c}_{d+1}^*$ satisfying Equation 1. Fix distinct $x_1, \ldots, x_{d+1} \in B$. Let $\mathbf{M} \in \mathbb{F}^{(d+1)\times(d+1)}$ be the matrix whose i-th row is $(1, x_i, \ldots, x_i^d)$; $\mathbf{M}$ is invertible because it is a Vandermonde matrix. For every $i \in [d+1]$, let $\mathbf{c}_i$ be the (unique) closest codeword to $\mathbf{u}_{x_i}$. We construct $\mathbf{c}_1^*, \ldots, \mathbf{c}_{d+1}^*$ as follows:

$$\begin{pmatrix} \mathbf{c}_1^* \\ \vdots \\ \mathbf{c}_{d+1}^* \end{pmatrix} \coloneqq \mathbf{M}^{-1} \cdot \begin{pmatrix} \mathbf{c}_1 \\ \vdots \\ \mathbf{c}_{d+1} \end{pmatrix} \ .$$

For every $x \in \mathsf{B}$, a repeated application of the triangle inequality shows that

$$\delta\left(\mathbf{c}_x, G(x) \cdot \begin{pmatrix} \mathbf{c}_1^* \\ \vdots \\ \mathbf{c}_{d+1}^* \end{pmatrix}\right) < (d+2) \cdot \gamma \leq \delta_{\mathcal{C}} \ .$$

The last inequality uses the fact that $\gamma \leq \delta_{\mathcal{C}}/(d+2)$. Since $\delta_{\mathcal{C}}$ is $\mathcal{C}$'s minimum distance, we deduce the desired property:

$$\mathbf{c}_x = G(x) \cdot \begin{pmatrix} \mathbf{c}_1^* \\ \vdots \\ \mathbf{c}_{d+1}^* \end{pmatrix} \ .$$

2.3.2 List-decoding setting

We sketch our proof that, when $\gamma \leq 1 - (1 - \delta_{\mathcal{C}} + \eta)^{\frac{1}{d+2}}$, the set of bad seeds can be upper bounded as

$$|\mathsf{B}| \leq O\left(\frac{n}{\eta}\right) \cdot d \ . \tag{3}$$

The main challenge is that, as we are not guaranteed to be in the unique decoding setting, there is no notion of a unique closest codeword. We cannot find fixed $\mathbf{c}_1^*, \ldots, \mathbf{c}_{d+1}^* \in \mathcal{C}$ such that $\mathbf{c}_x = \sum_{j\in[d+1]} G(x) \cdot \mathbf{c}_j^*$ for every $x \in \mathsf{B}$. Therefore, we cannot define an agreement set T that is included in T_x for every bad seed x in B.

Our solution relies on a "cutoff" parameter $\gamma' \coloneqq 1 - (1 - \delta_{\mathcal{C}} + \eta)^{\frac{1}{d+1}}$ ($\delta_{\mathcal{C}} > \gamma' > \gamma$), adapted from [GKL24]. We partition B into two disjoint sets B_1 and B_2: (1) B_1 contains all seeds $x \in \mathsf{B}$ for which there exist $\tilde{\mathbf{c}}_1, \ldots, \tilde{\mathbf{c}}_{d+1} \in \mathcal{C}$ such that their agreement set $\tilde{T} \coloneqq \{i \in [n] : \wedge_{j\in[d+1]} \mathbf{u}_j[i] = \tilde{\mathbf{c}}_j[i]\}$ satisfies $\tilde{T} \subsetneq T_x$ and $|\tilde{T}| \geq n \cdot (1-\gamma')$; and (2) $\mathsf{B}_2 \coloneqq \mathsf{B} \setminus \mathsf{B}_1$. We then separately upper bound the sizes of B_1 and B_2.

Bounding the size of B_1. The definition of B_1 allows us to apply our counting argument similarly as in Section 2.3.1. For every $\tilde{T}$, define $\mathsf{B}_{1,\tilde{T}} \coloneqq \{x \in \mathsf{B}_1 : \tilde{T} \subsetneq T_x\}$. Note that $\mathsf{B}_1 \subseteq \bigcup_{\tilde{T}} \mathsf{B}_{1,\tilde{T}}$, which implies that

$$|\mathsf{B}_1| \leq |\{\tilde{T} \subseteq [n] : \exists\, \tilde{\mathbf{c}}_1, \ldots, \tilde{\mathbf{c}}_{d+1} \in \mathcal{C},\ \forall\, j \in [d+1],\ \mathbf{u}_j|_{\tilde{T}} = \tilde{\mathbf{c}}_j|_{\tilde{T}}\}| \cdot \max_{\tilde{T}}\{|\mathsf{B}_{1,\tilde{T}}|\} \ .$$

The number of $\tilde{T}$ is upper bounded by $1/\eta$ by a combinatorial argument based on the Corrádi lemma (see Lemma 3.23). For every fixed $\tilde{T}$, we bound the size of $\mathsf{B}_{1,\tilde{T}}$ via a similar argument as in Equation 2, namely,

to bound the size of $T_x \setminus \tilde{T}$ and the zero-evading error of G. Together these yield

$$|\mathsf{B}_1| \leq \frac{1}{\eta} \cdot n \cdot d \ .$$

In fact, because we have a lower bound $|\tilde{T}| \geq n \cdot (1 - \gamma')$, the upper bound on $|\mathsf{B}_1|$ can be improved to $\frac{1}{\eta} \cdot n \cdot \gamma' \cdot d$ as in Section 2.3.1 (see details in Section 6).

Bounding the size of B_2. Since $\mathsf{B}_2 = \mathsf{B} \setminus \mathsf{B}_1$, we do not have large agreement sets $\tilde{T}$. Nevertheless, we can bound the size of B_2 using a similar counting argument as before (Equation 2).

Assume for contradiction that B_2 is large. Using a probabilistic argument, one can show that there exists a set $\hat{T}$ where (i) $n \cdot (1 - \delta_{\mathcal{C}}) < |\hat{T}| < n \cdot (1 - \gamma')$; (ii) there exist $\hat{\mathbf{c}}_1, \ldots, \hat{\mathbf{c}}_{d+1} \in \mathcal{C}$ with $\mathbf{u}_j|_{\hat{T}} = \hat{\mathbf{c}}_j|_{\hat{T}}$; and (iii) $|\{x \in \mathsf{B}_2 : \hat{T} \subsetneq T_x\}| \geq C(d, \eta, \delta_{\mathcal{C}})$ for some quantity $C(d, \eta, \delta_{\mathcal{C}})$.

Note that properties of $\hat{T}$ described in Items i and ii allow us to adapt our usual counting argument (Equation 2) to obtain an upper bound on the possible number of T_x such that $\hat{T} \subsetneq T_x$. Specifically,

$$|\{x \in \mathsf{B}_2 : \hat{T} \subsetneq T_x\}| < C(d, \eta, \delta_{\mathcal{C}}) \ ,$$

which contradicts Item iii above. Hence, B_2 cannot be large.

We omit technical details for the above exposition; see Section 6.2 for more.

2.4 The affine space generator

The affine space generator is $G\colon \mathbb{F}^s \to \mathbb{F}^{s+1}$ where $G(x_1, \ldots, x_s) \coloneqq (1, x_1, \ldots, x_s)$. We show that, for any linear code $\mathcal{C} \subseteq \mathbb{F}^n$, the affine space generator has the same MCA error as the affine line generator $x \mapsto (1, x)$, up to a small constant multiplicative overhead. This implies our Lemma 1.

Let $\mathbf{u}_0, \ldots, \mathbf{u}_s \in \mathbb{F}^n$ be words, and define $U \coloneqq \mathsf{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_s\}$ and $V \coloneqq \mathbf{u}_0 + U$. The intuition for the proof is fairly simple: if the "bad seed" set for the affine space generator is large, then we can find a line of the form $\mathbf{u}' + t \cdot \tilde{\mathbf{u}}$, with $\mathbf{u}' \in V$ and $\tilde{\mathbf{u}} \in U$ that contradicts the MCA error of the affine line generator. More precisely, there are many $t \in \mathbb{F}$ such that there exists $T \subseteq [n]$ with $(\mathbf{u}' + t \cdot \tilde{\mathbf{u}})|_T \in \mathcal{C}|_T$ and $\tilde{\mathbf{u}}|_T \notin \mathcal{C}|_T$.

Here we outline this reduction; for details see Lemma 7.1. The set of bad seeds for the affine space generator is

$$\mathsf{B} \coloneqq \left\{ \mathbf{x} = (x_1, \ldots, x_s) \in \mathbb{F}^s \ : \begin{array}{l} \exists\, T_{\mathbf{x}} \subseteq [n] : \\ \quad |T_{\mathbf{x}}| \geq n \cdot (1 - \gamma) \\ \quad \wedge\ (\mathbf{u}_0 + \sum_{j \in [s]} x_j \cdot \mathbf{u}_j)|_{T_{\mathbf{x}}} \in \mathcal{C}|_{T_{\mathbf{x}}} \\ \quad \wedge\ \exists j \in \{0, \ldots, s\},\ \mathbf{u}_j|_{T_{\mathbf{x}}} \notin \mathcal{C}|_{T_{\mathbf{x}}} \end{array} \right\} \ .$$

For any $\mathbf{x} \in \mathsf{B}$, fix $T_{\mathbf{x}}$ as in the definition of B. The first step is to find $(\lambda_1^\star, \ldots, \lambda_s^\star) \in \mathbb{F}^s$ such that if $\tilde{\mathbf{u}} \coloneqq \sum_{j \in [s]} \lambda_j^\star \cdot \mathbf{u}_j \in U$ then $\mathsf{B}' \coloneqq \{\mathbf{x} \in \mathsf{B} : \tilde{\mathbf{u}}|_{T_{\mathbf{x}}} \notin \mathcal{C}|_{T_{\mathbf{x}}}\} \subseteq \mathsf{B}$ is large, which we do via a linear algebraic observation. Consider the quotient linear space $\mathbb{F}^{|T_{\mathbf{x}}|}/\mathcal{C}|_{T_{\mathbf{x}}}$ and the linear map

$$\mathcal{L} : \mathbb{F}^s \to \mathbb{F}^{|T_{\mathbf{x}}|}/\mathcal{C}|_{T_{\mathbf{x}}}, \quad (\lambda_1, \ldots, \lambda_s) \mapsto \sum_{j=1}^{s} \lambda_j \cdot [\mathbf{u}_j] \ ,$$

where $[\mathbf{u}]$ denotes the class of $\mathbf{u}|_{T_{\mathbf{x}}}$ in $\mathbb{F}^{|T_{\mathbf{x}}|}/\mathcal{C}|_{T_{\mathbf{x}}}$. For every $\mathbf{x} \in \mathsf{B}$, the kernel of $\mathcal{L}$ is a proper subspace of $\mathbb{F}^s$. By an averaging argument over the choice of $(\lambda_1, \ldots, \lambda_s)$, we deduce that there exists $(\lambda_1^\star, \ldots, \lambda_s^\star) \in \mathbb{F}^s$ such that B' has size at least $|\mathsf{B}|\,(1 - 1/|\mathbb{F}|)$.

We are left to construct $\mathbf{u}' \in V$ such that for many $t \in \mathbb{F}$, $\mathbf{u}' + t \cdot \tilde{\mathbf{u}}$ breaks the MCA condition. The key idea is to consider the following line *in seed space* for $\mathbf{v} \in \mathbb{F}^s$:

$$\mathbf{x}_{\mathbf{v}}(t) \coloneqq \mathbf{v} + t \cdot (\lambda_1^\star, \ldots, \lambda_s^\star) \ .$$

While there is no guarantee that $\mathbf{x}_{\mathbf{v}}(t) \in \mathsf{B}'$, by using another averaging argument, there exists $\mathbf{v}^\star \in \mathbb{F}^s$ such that:

$$\Pr_{t \in \mathbb{F}}[\mathbf{x}_{\mathbf{v}^\star}(t) \in \mathsf{B}'] \geq \frac{|\mathsf{B}'|}{|\mathbb{F}^s|} \ .$$

In other words, the density of $t \in \mathbb{F}$ such that $\mathbf{x}_{\mathbf{v}^\star}(t) \in \mathsf{B}'$ is at least the density of B' in $\mathbb{F}^s$. Set $\mathbf{u}' \coloneqq \mathbf{u}_0 + \sum_{j \in [s]} \mathbf{v}_j^\star \cdot \mathbf{u}_j$. By the definition of B', for every $t \in \mathbb{F}$ such that $\mathbf{x}_{\mathbf{v}^\star}(t) = \mathbf{x} \in \mathsf{B}'$, we have:

$$(\mathbf{u}_0 + \sum_{j \in [s]} x_j \cdot \mathbf{u}_j)|_{T_{\mathbf{x}}} \in \mathcal{C}|_{T_{\mathbf{x}}} \quad \text{and} \quad \tilde{\mathbf{u}}|_{T_{\mathbf{x}}} \notin \mathcal{C}|_{T_{\mathbf{x}}} \ .$$

But here $\mathbf{u}_0 + \sum_{j \in [s]} x_j \cdot \mathbf{u}_j = \mathbf{u}' + t \cdot \tilde{\mathbf{u}}$. Let ϵ_{MCA} be the error of the affine line generator for the code $\mathcal{C}$. By our discussion,

$$\epsilon_{\mathsf{MCA}}(\gamma) \geq \Pr_{t \in \mathbb{F}} \left[\begin{array}{l} \exists\, T_t \subseteq [n] : \\ \quad |T_t| \geq n \cdot (1 - \gamma) \\ \quad \wedge\ (\mathbf{u}' + t \cdot \tilde{\mathbf{u}})|_{T_t} \in \mathcal{C}|_{T_t} \\ \quad \wedge\ \tilde{\mathbf{u}}|_{T_t} \notin \mathcal{C}|_{T_t} \end{array} \right] \geq \Pr_{t \in \mathbb{F}}[\mathbf{x}_{\mathbf{v}^\star}(t) \in \mathsf{B}'] \geq \frac{|\mathsf{B}'|}{|\mathbb{F}^s|} \geq \frac{|\mathsf{B}|\,(1 - 1/|\mathbb{F}|)}{|\mathbb{F}^s|} \ .$$

In turn, the size of B is at most $\epsilon_{\mathsf{MCA}}(\gamma) \cdot |\mathbb{F}|^s \cdot (1 - 1/|\mathbb{F}|)^{-1}$, which concludes the proof.

2.5 Polynomial generators and Reed–Solomon codes

We outline our proof of Theorem 2. Similarly to Theorem 1, first we establish that the univariate powers generator has MCA for Reed–Solomon codes up to the list-decoding bound and thereafter rely on our MCA-preserving transformations to establish MCA for all polynomial generators. We discussed the second (generic) step in Section 2.2 (see Lemmas 2 and 3); below we focus on the first step.

We prove that the univariate powers generator $G(x) \coloneqq (1, x, \ldots, x^d)$ has MCA for $\mathcal{C} \coloneqq \mathsf{RS}[\mathbb{F}, D, k]$ with error

$$\epsilon_{\mathsf{MCA}}(\gamma) \coloneqq O\left(\frac{m^7 \cdot n^2}{\rho^{3/2}}\right) \cdot \frac{d}{|\mathbb{F}|} \quad \text{if } \gamma \leq 1 - \left(1 + \frac{1}{2m}\right) \cdot \sqrt{\rho} \ .$$

We bound the size of the set of bad seeds B defined as follows:

$$\mathsf{B} \coloneqq \left\{ x \in \mathbb{F} \;\middle|\; \exists\, T_x \subseteq D : \begin{array}{l} |T_x| \geq n \cdot (1 + 1/2m) \cdot \sqrt{\rho} \\ \wedge\ \left(\sum_{i=0}^{d} z^i \cdot \mathbf{u}_i\right)\Big|_{T_x} \in \mathcal{C}|_{T_x} \\ \wedge\, \exists\, \ell \in \{0, \ldots, d\}, \mathbf{u}_\ell|_{T_x} \notin \mathcal{C}|_{T_x} \end{array} \right\} \ .$$

Adapting techniques from [BCIKS20; Hab25], B can be partitioned into sets $(\mathsf{B}_i)_{i \in [\frac{m+1/2}{\sqrt{\rho}}]}$ where, for each i, if $|\mathsf{B}_i| > \frac{(m+1/2)^6 \cdot d \cdot n^2}{3\rho}$ then there exist codewords $\mathbf{c}_1, \ldots, \mathbf{c}_{d+1}$ such that

$$\mathsf{B}_i = \left\{ x \in \mathbb{F} \;\middle|\; \exists\, T_x \subseteq D : \begin{array}{l} |T_x| \geq n \cdot (1 + 1/2m) \cdot \sqrt{\rho} \\ \wedge\ \left(\sum_{j=0}^{d} x^j \cdot \mathbf{u}_j\right)\Big|_{T_x} = \left(\sum_{j=0}^{d} x^j \cdot \mathbf{c}_j\right)\Big|_{T_x} \\ \wedge\, \exists\, \ell \in \{0, \ldots, d\}, \mathbf{u}_\ell|_{T_x} \notin \mathcal{C}|_{T_x} \end{array} \right\} \ .$$

Letting $\tilde{T} \coloneqq \{i \in [n] : \forall j \in [d+1],\ \mathbf{u}_j[i] = \mathbf{c}_j[i]\}$, we deduce that $\tilde{T} \subsetneq T_x$ for every $x \in \mathsf{B}_i$. Hence, using the idea explained in Section 2.3 (specifically the counting argument captured in Equation 2) we show that the number of $x \in \mathbb{F}$ with $\tilde{T} \subsetneq T_x$ is at most $n \cdot d$, which implies that $|\mathsf{B}_i| \leq n \cdot d$. Since $|\mathsf{B}_i| > \frac{(m+1/2)^6 \cdot d \cdot n^2}{3\rho} > n \cdot d$, we reach a contradiction. Therefore,

$$|\mathsf{B}| = \sum_{i \in [\frac{m+1/2}{\sqrt{\rho}}]} |\mathsf{B}_i| \leq \frac{m+1/2}{\sqrt{\rho}} \cdot \frac{(m+1/2)^6 \cdot d \cdot n^2}{3\rho} = \frac{(m+1/2)^7 \cdot d \cdot n^2}{3\rho^{3/2}} \ .$$

See Section 9 for technical details.

2.6 An MCA toolbox, and going beyond polynomial generators

Our results rely, in particular, on several transformations that we prove preserve MCA. In fact, via such transformations, we prove that a zero-evading generator in [AGHP92], which is *not* a polynomial generator, has MCA for arbitrary linear codes; no prior work shows any distance preservation property for generators that are not polynomial generators. This shows that the "MCA toolbox" that we develop enables going beyond polynomial generators. Below we briefly discuss the generator in [AGHP92] and our MCA toolbox.

MCA for a non-polynomial generator. Fix a finite field $\mathbb{F}_q$ with q elements and let $\mathbb{F}_{q^k}$ be its extension of degree k. Fix an isomorphism ϕ of $\mathbb{F}$-vector spaces that maps $\mathbb{F}_{q^k}$ to $\mathbb{F}_q^k$. The generator $G\colon \mathbb{F}_{q^k} \times \mathbb{F}_q^k \to \mathbb{F}_q^\ell$ in [AGHP92] is defined as

$$G(x, \mathbf{v}) \coloneqq (\phi(x^j) \cdot \mathbf{v}^\intercal)_{j \in [\ell]} \ .$$

The zero-evading error of this generator is $\epsilon_{\mathsf{ZE}} = \frac{\ell-1}{q^k} + \frac{1}{q}$.

While G is not a polynomial generator, it can be viewed as the *matrix-vector composition* of two polynomial generators in the following sense. Consider $G_{\mathsf{in}}\colon \mathbb{F}_q^k \to \mathbb{F}_q^k$ and $G_{\mathsf{out}}\colon \mathbb{F}_{q^k} \to (\mathbb{F}_{q^k})^\ell$ defined as

$$\begin{aligned} &\forall\, \mathbf{v} \in \mathbb{F}_q^k, G_{\mathsf{in}}(\mathbf{v}) \coloneqq \mathbf{v} \ ; \\ &\forall\, x \in \mathbb{F}_{q^k}, G_{\mathsf{out}}(x) \coloneqq (x, x^2, \ldots, x^\ell) \ . \end{aligned}$$

Then

$$\forall\, j \in [\ell], G(x, \mathbf{v})[j] = \langle \phi(G_{\mathsf{out}}(x)[j]), G_{\mathsf{in}}(\mathbf{v}) \rangle \ .$$

In other words, $G(x, \mathbf{v}) = \mathbf{A}_x \cdot G_{\mathsf{in}}(\mathbf{v})^\intercal$ where $\mathbf{A}_x \in \mathbb{F}^{\ell \times k}$ is the matrix whose j-th row equals $\phi(G_{\mathsf{out}}(x)[j])$.

In Section 4.3, we show that the matrix-vector composition of two generators as above results in a generator whose MCA error is the sum of their two MCA errors. Moreover, in Corollary 4.2, we show that removing some coordinates in the output of a generator (to construct a sub-generator) does not change its MCA error. Since G_{in} is a sub-generator of the affine space generator and G_{out} is a sub-generator of the univariate powers generator, we can bound the MCA error of G using Theorem 1.

Toolbox for MCA. We summarize the transformations that we prove preserve MCA. Formal statements and proofs for these are provided in Section 4.

- A sub-generator of a generator has the same MCA error for any code as the generator.
- The linear transformation of a generator has the same MCA error for any code as the generator.
- The tensor product of two generators has MCA error that is the sum of the MCA errors of the original generators.

- If the affine line generator $x \mapsto (1, x)$ has MCA error for a code, then the affine space generator $(x_1, \ldots, x_s) \mapsto (1, x_1, \ldots, x_s)$ has the same MCA error for the code.
- The matrix-vector composition of two generators has MCA error that is the sum of the MCA errors of the original generators.

These transformations enable us to capture all polynomial generators, and even beyond that as [AGHP92]'s generator illustrates. The MCA properties of these transformations are likely to be useful beyond this paper.

3 Preliminaries

We denote by $\mathbb{N}$ the set of integers greater than 0. For $n \in \mathbb{N}$, we use the shorthand $[n] \coloneqq \{1, 2, \ldots, n\}$.

Finite fields. Every field considered in this paper is finite. We use $\mathbb{F}$ to refer to any finite field, and denote by $\mathbb{F}_q$ the finite field with q elements.

Vectors and matrices. Vectors and matrices are denoted in bold. For every $n \in \mathbb{N}$, a vector $\mathbf{v} \in \mathbb{F}^n$ is a row vector and we denote its i-th entry by $\mathbf{v}[i]$, for every $i \in [n]$. Similarly, consider $\ell \in \mathbb{N}$, for every $(i, j) \in [\ell] \times [n]$, the (i, j)-th entry of a matrix $\mathbf{M} \in \mathbb{F}^{\ell \times n}$ is denoted by $\mathbf{M}[i, j]$. For every set of vectors $\mathbf{v}_1, \ldots, \mathbf{v}_\ell \in \mathbb{F}^n$, we denote by $\mathsf{Mat}(\mathbf{v}_1, \ldots, \mathbf{v}_\ell) \in \mathbb{F}^{\ell \times n}$ the matrix whose i-th row is $\mathbf{v}_i$ for every $i \in [\ell]$. For every nonempty subset $T \subseteq [n]$, we denote by $\mathbf{v}|_T$ the vector in $\mathbb{F}^{|T|}$ defined as $(\mathbf{v}[i])_{i \in T}$ (with the natural order on T inherited from $[n]$).

Polynomials. For $s \in \mathbb{N}$, the ring of s-variate polynomials is denoted by $\mathbb{F}[\mathtt{X}_1, \ldots, \mathtt{X}_s]$. When we evaluate an s-variate polynomial at a vector of field elements $\mathbf{x} \in \mathbb{F}^s$, we denote by x_i the i-th entry of $\mathbf{x}$, for $i \in [s]$. When s is clear from context, we use $\vec{\mathtt{X}}$ for the vector of variables $(\mathtt{X}_1, \ldots, \mathtt{X}_s)$.

Distance. Let $n \in \mathbb{N}$ be a length and Σ be an alphabet. The (Hamming) distance between $\mathbf{u}_1, \mathbf{u}_2 \in \Sigma^n$ is

$$\Delta_\Sigma(\mathbf{u}_1, \mathbf{u}_2) \coloneqq |\{i \in [n] : \mathbf{u}_1[i] \neq \mathbf{u}_2[i]\}| \ .$$

The relative (Hamming) distance between them is

$$\delta_\Sigma(\mathbf{u}_1, \mathbf{u}_2) \coloneqq \frac{\Delta_\Sigma(\mathbf{u}_1, \mathbf{u}_2)}{n} \ .$$

We omit the alphabet from the distance notation when it is clear from context. For $\delta \in [0, 1]$, we say that $\mathbf{u}_1$ is δ-*far* from $\mathbf{u}_2$ if $\delta_\Sigma(\mathbf{u}_1, \mathbf{u}_2) > \delta$ (equivalently, $\Delta_\Sigma(\mathbf{u}_1, \mathbf{u}_2) > \delta \cdot n$); otherwise, if $\delta_\Sigma(\mathbf{u}_1, \mathbf{u}_2) \leq \delta$ (equivalently, $\Delta_\Sigma(\mathbf{u}_1, \mathbf{u}_2) \leq \delta \cdot n$), we say that $\mathbf{u}_1$ is δ-*close* to $\mathbf{u}_2$. For every set $S \subseteq \Sigma^n$, we define:

$$\Delta_\Sigma(\mathbf{u}_1, S) \coloneqq \min_{\mathbf{u}_2 \in S}\{\Delta_\Sigma(\mathbf{u}_1, \mathbf{u}_2)\} \quad \text{and} \quad \delta_\Sigma(\mathbf{u}_1, S) \coloneqq \min_{\mathbf{u}_2 \in S}\{\delta_\Sigma(\mathbf{u}_1, \mathbf{u}_2)\} \ .$$

3.1 Linear codes

Definition 3.1. *A* **code** *of blocklength $n \in \mathbb{N}$ over a finite alphabet Σ is a subset $\mathcal{C} \subseteq \Sigma^n$. Arbitrary elements of Σ^n are called* **words** *and elements of $\mathcal{C}$ are called* **codewords***. The* **(minimum) distance** *$\Delta_\mathcal{C}$ of $\mathcal{C}$ is the minimum distance between two distinct codewords of $\mathcal{C}$. The* **relative (minimum) distance** *of $\mathcal{C}$ is $\delta_\mathcal{C} \coloneqq \frac{\Delta_\mathcal{C}}{n}$.*

Definition 3.2. *A code $\mathcal{C}$ of blocklength $n \in \mathbb{N}$ is* **$\mathbb{F}$-linear** *if Σ is a vector space over a field $\mathbb{F}$ and $\mathcal{C}$ is a linear subspace of Σ^n.*

If $\Sigma = \mathbb{F}$ then $\mathcal{C}$ is **linear***. The* **dimension** *k of a linear code $\mathcal{C} \subseteq \mathbb{F}^n$ is its dimension as an $\mathbb{F}$-vector space. A linear code $\mathcal{C}$ with blocklength n, dimension k, and distance $\Delta_\mathcal{C}$ is an $[n, k, \Delta_\mathcal{C}]_\mathbb{F}$-code (or simply a $[n, k]_\mathbb{F}$-code when the distance is not specified); a* **generator matrix** *for $\mathcal{C}$ is a matrix $\mathbf{G}_\mathcal{C} \in \mathbb{F}^{n \times k}$ whose columns form a basis of $\mathcal{C}$.*

Definition 3.3. *For $k \in \mathbb{N}$, the* **k-interleaving of a code** *$\mathcal{C} \subseteq \mathbb{F}^n$, denoted by $\mathcal{C}^k$, is the $\mathbb{F}$-linear code over alphabet $\Sigma = \mathbb{F}^k$ that consists of matrices in $\mathbb{F}^{k \times n}$ whose rows are codewords in $\mathcal{C}$. The* **interleaved distance** *is simply the Hamming distance Δ_Σ.*

Remark 3.4. One may consider $\mathcal{C}^k$ as a linear code, by identifying $(\mathbb{F}^k)^n$ and $\mathbb{F}^{kn}$, but in that case the Hamming distance over $\mathbb{F}$ does not capture the correct notion of (column-wise) distance.

Definition 3.5. *A linear code $\mathcal{C} \subseteq \mathbb{F}^n$ is* **maximum distance separable** *(MDS) if $\Delta_{\mathcal{C}} = n - k + 1$ (i.e., its parameters meet the Singleton bound).*

Lemma 3.6 (See [GRS23])**.** *A $[n, k]_{\mathbb{F}}$-code $\mathcal{C}$ is MDS if and only if any k rows of a generator matrix for $\mathcal{C}$ are linearly independent.*

Definition 3.7. *Let $\mathcal{C} \subseteq \Sigma^n$ be a code. For every nonempty subset $T \subseteq [n]$, we define*

$$\mathcal{C}|_T \coloneqq \{\mathbf{c}|_T : \mathbf{c} \in \mathcal{C}\} \ .$$

Note that $\mathcal{C}|_T$ is a code with blocklength $|T|$.

Definition 3.8. *Fix a field $\mathbb{F}$, a domain $D \subseteq \mathbb{F}$, and an integer $k \in \mathbb{N}$. The* **Reed–Solomon code** $\mathsf{RS}[\mathbb{F}, D, k] \subseteq \mathbb{F}^{|D|}$ *is the $[|D|, k, |D|-k+1]_{\mathbb{F}}$-code whose codewords are evaluations over D of polynomials in $\mathbb{F}[\mathtt{X}]$ of degree at most $k-1$. Its rate is $\rho \coloneqq k/|D|$.*

Definition 3.9. *Fix a base field $\mathbb{B}$, a field extension $\mathbb{F}$, and an integer $k \in \mathbb{N}$. The* **Reed–Muller code** $\mathsf{RM}[\mathbb{F}, \mathbb{B}^2, k] \subseteq \mathbb{F}^{|\mathbb{B}|^2}$ *is the linear code whose codewords are evaluations of over $\mathbb{B}^2$ of bivariate polynomials in $\mathbb{F}[\mathtt{X}, \mathtt{Y}]$ with total degree at most k.*

3.2 Generators

Definition 3.10. *A* **generator** *with output size $\ell \in \mathbb{N}$ over a field $\mathbb{F}$ is a function $G \colon S \to \mathbb{F}^\ell$ for a set S.*

Definition 3.11. *A generator $G \colon S \to \mathbb{F}^\ell$ is* **zero-evading with error** $\epsilon_{\mathsf{ZE}} \in [0, 1]$ *if*

$$\max_{\mathbf{v} \in \mathbb{F}^\ell \setminus \{\mathbf{0}\}} \Pr_{\mathbf{x} \in S} [G(\mathbf{x}) \cdot \mathbf{v}^\intercal = 0] \leq \epsilon_{\mathsf{ZE}} \ .$$

Definition 3.12. *Let $G \colon S \to \mathbb{F}^\ell$ be a generator. Let $\mathbf{M}_G$ be the $(|S| \times \ell)$-matrix with rows $\{G(\mathbf{x}) : \mathbf{x} \in S\}$ (with some ordering on S). Let $k \coloneqq \mathsf{rank}(\mathbf{M}_G)$ and let $\mathcal{C}_G$ be the $[|S|, k]_{\mathbb{F}}$-code generated by $\mathbf{M}_G$:*

$$\mathcal{C}_G \coloneqq \left\{\mathbf{M}_G \cdot \mathbf{v}^\intercal : \mathbf{v} \in \mathbb{F}^\ell\right\} \ .$$

We say that G is an **MDS generator** *if $\mathcal{C}_G$ is MDS.*

Lemma 3.13. *Let $G \colon S \to \mathbb{F}^\ell$ be a generator. If $\mathcal{C}_G$ has distance $\Delta_{\mathcal{C}_G}$ then G is zero-evading with error $1 - \frac{\Delta_{\mathcal{C}_G}}{|S|}$. If $\mathcal{C}_G$ is MDS with dimension k then $\Delta_{\mathcal{C}_G} = |S| - k + 1$, so G is zero-evading with error $\frac{k-1}{|S|}$.*

Definition 3.14. *A generator $G \colon S \to \mathbb{F}^\ell$ has* **mutual correlated agreement (MCA)** *for an $\mathbb{F}$-linear code $\mathcal{C} \subseteq \Sigma^n$ with error $\epsilon_{\mathsf{MCA}} \colon [0, 1] \to [0, 1]$ if for every $\mathbf{u}_1, \ldots, \mathbf{u}_\ell \in \Sigma^n$ and $\gamma \in [0, 1]$, defining the matrix $\mathbf{U} \coloneqq \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell) \in \Sigma^{\ell \times n}$, the following holds:*

$$\Pr_{\mathbf{x} \in S} \left[\begin{array}{l} \exists\, T \subseteq [n] \text{ s.t.} \\ \quad |T| \geq n \cdot (1 - \gamma) \\ \quad \wedge\, (G(\mathbf{x}) \cdot \mathbf{U})|_T \in \mathcal{C}|_T \\ \quad \wedge\, \exists\, j \in [\ell], \mathbf{u}_j|_T \notin \mathcal{C}|_T \end{array} \right] \leq \epsilon_{\mathsf{MCA}}(\gamma) \ .$$

Sometimes we omit the code $\mathcal{C}$ and error ϵ_{MCA}, and we call G an MCA generator.

The MCA error of the trivial "all-in code" $\mathcal{C} \subseteq \Sigma^n$ such that $\mathcal{C} = \Sigma^n$ is $\epsilon_{\mathsf{MCA}}(\gamma) = 0$. Hence, when convenient in stating results, we may assume without loss of generality that $\mathcal{C} \neq \Sigma^n$.

Remark 3.15. MCA analyses typically break down above a certain distance $\gamma_0 < 1$. This is captured by the above definition by defining the error ϵ_{MCA} as a piecewise function such that $\epsilon_{\mathsf{MCA}}(\gamma) = 1$ for $\gamma > \gamma_0$.

Lemma 3.16. *Let $G\colon S \to \mathbb{F}^\ell$ be any generator that has MCA for an $\mathbb{F}$-linear code $\mathcal{C} \subseteq \Sigma^n$ with error $\epsilon_{\mathsf{MCA}}\colon [0,1] \to [0,1]$. Then ϵ_{MCA} is monotonically increasing.*

Proof. For every $0 \leq \gamma \leq \gamma' \leq 1$, $\mathbf{U} \coloneqq \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell) \in \Sigma^{\ell \times n}$ and $\mathbf{x} \in S$:

$$\left[\begin{array}{l} \exists T \subseteq [n] \text{ s.t.} \\ \quad |T| \geq n \cdot (1-\gamma) \\ \quad \wedge\, (G(\mathbf{x}) \cdot \mathbf{U})|_T \in \mathcal{C}|_T \\ \quad \wedge\, \exists\, j \in [\ell], \mathbf{u}_j|_T \notin \mathcal{C}|_T \end{array}\right] \implies \left[\begin{array}{l} \exists T \subseteq [n] \text{ s.t.} \\ \quad |T| \geq n \cdot (1-\gamma') \\ \quad \wedge\, (G(\mathbf{x}) \cdot \mathbf{U})|_T \in \mathcal{C}|_T \\ \quad \wedge\, \exists\, j \in [\ell], \mathbf{u}_j|_T \notin \mathcal{C}|_T \end{array}\right] .$$

which follows because $n \cdot (1-\gamma) \geq n \cdot (1-\gamma')$. □

Lemma 3.17. *Let $G\colon S \to \mathbb{F}^\ell$ be any generator that has MCA for an $\mathbb{F}$-linear code $\mathcal{C} \subseteq \Sigma^n$ with error $\epsilon_{\mathsf{MCA}}\colon [0,1] \to [0,1]$. For any linear subcode $\mathcal{C}' \subseteq \mathcal{C}$, if G has MCA for $\mathcal{C}'$ with error ϵ'_{MCA}, then:*

$$\epsilon_{\mathsf{MCA}}(0) \geq \epsilon'_{\mathsf{MCA}}(0) \ .$$

Proof. Let $\mathbf{U} \coloneqq \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell) \in \Sigma^{\ell \times n}$ be any set of words. Note the equality:

$$\begin{aligned} &\left\{\mathbf{x} \in S : \begin{array}{l} \exists\, \mathbf{c} \in \mathcal{C},\ G(\mathbf{x}) \cdot \mathbf{U} = \mathbf{c} \\ \exists\, j \in [\ell],\ \mathbf{u}_j \notin \mathcal{C} \end{array}\right\} \\ &= \left\{\mathbf{x} \in S : \begin{array}{l} \exists\, \mathbf{c}' \in \mathcal{C}',\ G(\mathbf{x}) \cdot \mathbf{U} = \mathbf{c}' \\ \exists\, j \in [\ell],\ \mathbf{u}_j \notin \mathcal{C} \end{array}\right\} \bigsqcup \left\{\mathbf{x} \in S : \begin{array}{l} \exists\, \mathbf{c} \in \mathcal{C} \setminus \mathcal{C}',\ G(\mathbf{x}) \cdot \mathbf{U} = \mathbf{c} \\ \exists\, j \in [\ell],\ \mathbf{u}_j \notin \mathcal{C} \end{array}\right\} \\ &= \left\{\mathbf{x} \in S : \begin{array}{l} \exists\, \mathbf{c}' \in \mathcal{C}',\ G(\mathbf{x}) \cdot \mathbf{U} = \mathbf{c}' \\ \exists\, j \in [\ell],\ \mathbf{u}_j \notin \mathcal{C}' \end{array}\right\} \bigsqcup \left\{\mathbf{x} \in S : \begin{array}{l} \exists\, \mathbf{c} \in \mathcal{C} \setminus \mathcal{C}',\ G(\mathbf{x}) \cdot \mathbf{U} = \mathbf{c} \\ \exists\, j \in [\ell],\ \mathbf{u}_j \notin \mathcal{C} \end{array}\right\} \ , \end{aligned}$$

where the last equality holds as $\mathbf{u}_j \notin \mathcal{C}$ implies that $\mathbf{u}_j \notin \mathcal{C}'$. □

Lemma 3.18. *Let $G\colon S \to \mathbb{F}^\ell$ be any generator that has MCA for an $\mathbb{F}$-linear code $\mathcal{C} \subsetneq \Sigma^n$ with error $\epsilon_{\mathsf{MCA}}\colon [0,1] \to [0,1]$. Let $\epsilon_{\mathsf{ZE}} \in [0,1]$ be the zero-evading error of G. Then, $\epsilon_{\mathsf{MCA}}(0) = \epsilon_{\mathsf{ZE}}$.*

Proof. Let $\Sigma = \mathbb{F}^k$ for some $k \in \mathbb{N}$.

Proof that $\epsilon_{\mathsf{MCA}}(0) \leq \epsilon_{\mathsf{ZE}}$. A parity check matrix $\mathbf{M}$ for $\mathcal{C}$ can be described as a matrix in $(\Sigma^\vee)^{r \times n}$, where $\Sigma^\vee$ is the dual space of Σ and r is the co-dimension of $\mathcal{C}$ as a vector space over $\mathbb{F}$. Let $\mathbf{U} \coloneqq \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell) \in \Sigma^{\ell \times n}$, and let $\mathbf{w}_j \coloneqq \mathbf{M} \cdot \mathbf{u}_j^\intercal \in \mathbb{F}^r$ for every j. Then, for every $\mathbf{x} \in S$,

$$\begin{aligned} G(\mathbf{x}) \cdot \mathbf{U} \in \mathcal{C} &\iff \mathbf{M} \cdot (G(\mathbf{x}) \cdot \mathbf{U})^\intercal = \mathbf{0}^r \\ &\iff \sum_{j \in [\ell]} G(\mathbf{x})[j] \cdot \mathbf{w}_j = \mathbf{0}^r \ . \end{aligned}$$

Assuming that there exists j such that $\mathbf{u}_j \notin \mathcal{C}$, we have that $\mathbf{w}_j \neq \mathbf{0}^r$. Choose an index $i \in [r]$ such that $\mathbf{w}_j[i] \neq 0$, and set $\mathbf{v} \coloneqq (\mathbf{w}_1[i], \ldots, \mathbf{w}_\ell[i])$. It then follows that:

$$\Pr_{\mathbf{x} \in S}\left[\begin{array}{l} G(\mathbf{x}) \cdot \mathbf{U} \in \mathcal{C} \\ \wedge\, \exists\, j \in [\ell],\ \mathbf{u}_j \notin \mathcal{C} \end{array}\right] = \Pr_{\mathbf{x} \in S}\left[\sum_{j \in [\ell]} G(\mathbf{x})[j] \cdot \mathbf{w}_j = \mathbf{0}^r\right]$$

$$\leq \Pr_{\mathbf{x}\in S}[G(\mathbf{x})\cdot\mathbf{v}^{\mathsf{T}}=0]$$
$$\leq \epsilon_{\mathsf{ZE}} \ .$$

Since $\mathbf{U}$ was arbitrary, we conclude that $\epsilon_{\mathsf{MCA}}(0)\leq\epsilon_{\mathsf{ZE}}$.

Proof that $\epsilon_{\mathsf{MCA}}(0)\geq\epsilon_{\mathsf{ZE}}$. Fix any non-zero vector $\mathbf{v}\in\mathbb{F}^{\ell}$. Since $\mathcal{C}\subsetneq\Sigma^n$, there exists $\mathbf{u}\in\Sigma^n\setminus\mathcal{C}$. Define $\mathbf{U}\coloneqq\mathsf{Mat}(\mathbf{u}_1,\ldots,\mathbf{u}_\ell)$ as follows:

$$\forall\, j\in[\ell],\ \mathbf{u}_j\coloneqq\mathbf{v}_j\cdot\mathbf{u} \ .$$

Since at least one $\mathbf{v}_j$ is non-zero, there exists j such that $\mathbf{u}_j\notin\mathcal{C}$. Also, for every $\mathbf{x}\in S$,

$$G(\mathbf{x})\cdot\mathbf{U}\in\mathcal{C} \iff G(\mathbf{x})\cdot\mathbf{U}=\left(\sum_{j\in[\ell]}G(\mathbf{x})[j]\cdot\mathbf{v}_j\right)\cdot\mathbf{u}\in\mathcal{C}$$
$$\iff \sum_{j\in[\ell]}G(\mathbf{x})[j]\cdot\mathbf{v}_j=0 \ .$$

The last equivalence follows from linearity of $\mathcal{C}$. It follows that:

$$\Pr_{\mathbf{x}\in S}\left[\begin{array}{l} G(\mathbf{x})\cdot\mathbf{U}\in\mathcal{C} \\ \wedge\ \exists\, j\in[\ell],\ \mathbf{u}_j\notin\mathcal{C}\end{array}\right]=\Pr_{\mathbf{x}\in S}\left[\sum_{j\in[\ell]}G(\mathbf{x})[j]\cdot\mathbf{v}_j=0\right]$$
$$=\Pr_{\mathbf{x}\in S}[G(\mathbf{x})\cdot\mathbf{v}^{\mathsf{T}}=0] \ .$$

Since $\mathbf{v}$ was arbitrary, we conclude that $\epsilon_{\mathsf{MCA}}(0)\geq\epsilon_{\mathsf{ZE}}$. □

Definition 3.19. *Fix a finite field $\mathbb{F}$. Consider $s,\ell\in\mathbb{N}$ and $S_1,\ldots,S_s\subseteq\mathbb{F}$. Let $S\coloneqq S_1\times\cdots\times S_s$. A mapping $G\colon S\to\mathbb{F}^{\ell}$ is a* **polynomial generator** *if there exist ℓ linearly independent multivariate polynomials $P_1,\ldots,P_\ell$ in $\mathbb{F}[\mathtt{X}_1,\ldots,\mathtt{X}_s]$ such that for every $\mathbf{x}\in S$,*

$$G(\mathbf{x})=(P_j(\mathbf{x}))_{j\in[\ell]} \ .$$

We call G a **univariate generator** *if $s=1$, and a* **multivariate generator** *if $s>1$.*

Remark 3.20. If a polynomial generator G has individual degrees $(d_i)_{i\in[s]}$ then G has zero-evading error $\sum_{i\in[s]}d_i/|S_i|$. This is because for every non-zero $\mathbf{v}\in\mathbb{F}^{\ell}$ the dot product of $\mathbf{v}$ and the output of G is a non-zero polynomial (due to linear independence) of individual degrees $(d_i)_{i\in[s]}$, which vanishes at a random point in S with probability at most $\sum_{i\in[s]}d_i/|S_i|$ by the polynomial identity lemma. (One can also apply the polynomial identity lemma with the total degree d of G, which gives $d/\min_{i\in[s]}|S_i|$ as the zero-evading error.)

3.3 CA vs. MCA

We prove our main results for the notion of mutual correlated agreement. Most prior work [AHIV17; BGKS20; BCIKS20; MZ25] focuses on the weaker notion of correlated agreement (CA). Below we define a more granular notion that we still call correlated agreement, in which we are able to control how much degradation the generator introduces to the interleaved distance of a given set of words. In this more fine-tuned version, it is clearer to work with absolute distance instead of relative distance.

Definition 3.21 (Correlated agreement)**.** *Given a function* $\epsilon_{\mathsf{CA}}\colon \{(e,t) : 1 \le t < e \le n\} \to [0,1]$*, a generator* $G : S \to \mathbb{F}^\ell$ *has* **correlated agreement (CA)** *for an* $\mathbb{F}$*-linear code* $\mathcal{C} \subseteq \Sigma^n$ *with error* ϵ_{CA} *if it satisfies the following property. For every* $1 \le t < e \le n$*, and words* $\mathbf{U} \in \Sigma^{\ell \times n}$ *such that* $\Delta_{\Sigma^\ell}(\mathbf{U}, \mathcal{C}^\ell) \ge e$*,*

$$\Pr_{\mathbf{x}\in S}\left[\Delta\left(G(\mathbf{x})\cdot\mathbf{U}, \mathcal{C}\right) \le e - t\right] \le \epsilon_{\mathsf{CA}}(e,t) \ .$$

Sometimes we omit $\mathcal{C}$ *and* ϵ_{CA}*, and we call* G *a CA generator.*

Lemma 3.22 (MCA implies CA)**.** *Suppose that a generator* $G\colon S \to \mathbb{F}^\ell$ *has MCA for an* $\mathbb{F}$*-linear code* $\mathcal{C} \subseteq \Sigma^n$ *with error* ϵ_{MCA}*. Then* G *has CA for* $\mathcal{C}$ *with error* $\epsilon_{\mathsf{CA}}(e,t) \coloneqq \epsilon_{\mathsf{MCA}}(\frac{e-1}{n})$*.*

Proof. Let $\mathbf{U} \coloneqq \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell) \in \Sigma^{\ell\times n}$ be any words such that $\Delta_{\Sigma^\ell}(\mathbf{U}, \mathcal{C}^\ell) \ge e$. This last condition means in particular that for every $T \subseteq [n]$ with $|T| \ge n - e + 1$, there exists $j \in [\ell]$ such that $\mathbf{u}_j|_T \notin \mathcal{C}|_T$. Then,

$$\begin{aligned}
\Pr_{\mathbf{x}\in S}\left[\Delta\left(G(\mathbf{x})\cdot\mathbf{U}, \mathcal{C}\right) \le e - t\right] &\le \Pr_{\mathbf{x}\in S}\left[\Delta\left(G(\mathbf{x})\cdot\mathbf{U}, \mathcal{C}\right) < e\right] \\
&= \Pr_{\mathbf{x}\in S}\left[\begin{array}{l} \exists\, T \subseteq [n] \text{ s.t.} \\ \quad |T| \ge n - e + 1, \\ \quad \wedge\, (G(\mathbf{x})\cdot\mathbf{U})|_T \in \mathcal{C}|_T, \\ \quad \wedge\, \exists\, j \in [\ell], \mathbf{u}_j|_T \notin \mathcal{C}|_T \end{array}\right] \\
&\le \epsilon_{\mathsf{MCA}}((e-1)/n) \ .
\end{aligned}$$

□

It is also known that CA implies maximum distance preservation, defined in [AHIV17], which states that with high probability over the choice of seed, the distance between $G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell)$ and $\mathcal{C}$ is at least $\max_{j\in[\ell]}\{\Delta(\mathbf{u}_j, \mathcal{C})\}$. In Lemma B.1, we obtain lower bounds for the maximum distance preservation of the univariate powers generator of degree $d \ge 1$ up to the unique decoding radius (i.e., $\delta_{\mathcal{C}}/2$) that match our upper bounds for its MCA error within $\delta_{\mathcal{C}}/(d+1)$ (see Theorem 6.1). This shows that our MCA error bounds are tight in general in this regime, and gives hope that one might be able to extend them to the unique decoding radius.

3.4 Corrádi lemma

The Corrádi lemma [Juk11] bounds the number of subsets of a set that have density α and pairwise intersection density at most ρ.

Lemma 3.23. *Let* $A_1, \ldots, A_m$ *be subsets of a set* A*. Let* $\alpha, \rho \in [0,1]$*. Suppose the following holds:*
• *for every* $i \in [m]$*,* $|A_i| = \alpha \cdot |A|$*;*
• *for every* $i \ne j$*,* $|A_i \cap A_j| \le \rho \cdot |A|$*.*
If $\alpha^2 > \rho$ *then* $m \le \frac{\alpha-\rho}{\alpha^2-\rho}$*.*

We prove a generalization of the Corrádi lemma for intersections of $k \ge 2$ sets in Section D, which may be of independent interest.

4 A toolbox for MCA generators

In Section 4.1, we show that linear transformations of MCA generators are also MCA. In Section 4.2, we show that the tensor product of two MCA generators is also MCA. In Section 4.3, we show that a certain composition of two MCA generators is MCA.

4.1 Linear transformations

Lemma 4.1. *Let $G\colon S \to \mathbb{F}^\ell$ be a generator that has MCA for an $\mathbb{F}$-linear code $\mathcal{C} \subseteq \Sigma^n$ with error ϵ_{MCA}. For every $\ell' \in \mathbb{N}$ and $\mathbf{A} \in \mathbb{F}^{\ell \times \ell'}$ that has a left inverse, the generator $G'\colon S \to \mathbb{F}^{\ell'}$ defined for all $\mathbf{x} \in S$ as $G'(\mathbf{x}) \coloneqq G(\mathbf{x}) \cdot \mathbf{A}$ has MCA for $\mathcal{C}$ with error ϵ_{MCA}.*

Proof. Fix words $\mathbf{u}'_1, \ldots, \mathbf{u}'_{\ell'} \in \Sigma^n$, and let $\mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell) \coloneqq \mathbf{A} \cdot \mathsf{Mat}(\mathbf{u}'_1, \ldots, \mathbf{u}'_{\ell'})$. For every $\gamma \in [0,1]$,

$$\Pr_{\mathbf{x}\in S}\left[\begin{array}{l}\exists\, T \subseteq [n] \text{ s.t.}\\ \quad |T| \geq n\cdot(1-\gamma),\\ \quad \wedge\, (G'(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{u}'_1, \ldots, \mathbf{u}'_\ell))|_T \in \mathcal{C}|_T\\ \quad \wedge\, \exists\, j \in [\ell'], \mathbf{u}'_j|_T \notin \mathcal{C}|_T\end{array}\right] = \Pr_{\mathbf{x}\in S}\left[\begin{array}{l}\exists\, T \subseteq [n] \text{ s.t.}\\ \quad |T| \geq n\cdot(1-\gamma),\\ \quad \wedge\, (G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell))|_T \in \mathcal{C}|_T\\ \quad \wedge\, \exists\, j \in [\ell'], \mathbf{u}'_j|_T \notin \mathcal{C}|_T\end{array}\right] .$$

For every $T \subseteq [n]$, assume that for every $i \in [\ell]$, there exists $\mathbf{c}_i \in \mathcal{C}$ such that $\mathbf{c}_i|_T = \mathbf{u}_i|_T$. Let A_L be the left inverse of A, and set $\mathsf{Mat}(\mathbf{c}'_1, \ldots, \mathbf{c}'_\ell) \coloneqq A_L \cdot \mathsf{Mat}(\mathbf{c}_1, \ldots, \mathbf{c}_\ell)$. We can deduce that $\mathbf{c}'_i|_T = \mathbf{u}'_i|_T$ for every $i \in [\ell']$. Hence, by the contrapositive, we know for every $T \subseteq [n]$,

$$\left[\exists\, i \in [\ell'], \mathbf{u}'_i|_T \notin \mathcal{C}|_T\right] \implies \left[\exists\, i \in [\ell], \mathbf{u}_i|_T \notin \mathcal{C}|_T\right] .$$

Therefore, the probability above can be upper-bounded by the following expression:

$$\Pr_{\mathbf{x}\in S}\left[\begin{array}{l}\exists\, T \subseteq [n] \text{ s.t.}\\ \quad |T| \geq n\cdot(1-\gamma),\\ \quad \wedge\, (G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell))|_T \in \mathcal{C}|_T\\ \quad \wedge\, \exists\, j \in [\ell], \mathbf{u}_j|_T \notin \mathcal{C}|_T\end{array}\right] \leq \epsilon_{\mathsf{MCA}}(\gamma) ,$$

where the inequality holds because of the MCA error of G. □

The previous lemma immediately implies the following corollary.

Corollary 4.2. *Let $G\colon S \to \mathbb{F}^\ell$ be a generator that has MCA for an $\mathbb{F}$-linear code $\mathcal{C} \subseteq \Sigma^n$ with error ϵ_{MCA}. Then, for every non-empty $A \subseteq [\ell]$, the generator defined as $G'(\mathbf{x}) \coloneqq (G(\mathbf{x})_j)_{j\in A}$ for all $\mathbf{x} \in S$ has MCA for $\mathcal{C}$ with error ϵ_{MCA}.*

Proof. For every $\mathbf{x} \in S$, $G'(\mathbf{x}) = G(\mathbf{x}) \cdot \mathbf{A}$, where $\mathbf{A} \in \mathbb{F}^{\ell \times |A|}$ is the matrix that selects the corresponding entry in G. Note that $\mathbf{A}$ has a left inverse. The result then follows from Lemma 4.1. □

4.2 Tensor products

Definition 4.3. *Let $G\colon S \to \mathbb{F}^\ell$, $G'\colon S' \to \mathbb{F}^{\ell'}$ be two generators. The* **tensor product** *of G and G' is the generator $G \otimes G'\colon S \times S' \longrightarrow \mathbb{F}^\ell \otimes \mathbb{F}^{\ell'} \cong \mathbb{F}^{\ell\times\ell'}$ where*

$$G \otimes G'(\mathbf{x}, \mathbf{x}') \coloneqq G(\mathbf{x}) \otimes G'(\mathbf{x}') .$$

Lemma 4.4. *Let $G\colon S \to \mathbb{F}^{\ell}$ and $G'\colon S' \to \mathbb{F}^{\ell'}$ be generators with MCA for an $\mathbb{F}$-linear code $\mathcal{C} \subseteq \Sigma^n$ with errors ϵ_{MCA} and ϵ'_{MCA}, respectively. Then $G \otimes G'$ has MCA for $\mathcal{C}$ with error $\epsilon_{\mathsf{MCA}} + \epsilon'_{\mathsf{MCA}}$.*

Proof. Fix $\mathbf{u}_{(1,1)}, \ldots, \mathbf{u}_{(\ell,\ell')} \in \Sigma^n$ and $\gamma \in [0,1]$. Our goal is to bound the following probability:

$$\Pr_{(\mathbf{x},\mathbf{x}')\in S\times S'}\left[\begin{array}{l}\exists\, T \subseteq [n] \text{ s.t.} \\ \quad |T| \geq n\cdot(1-\gamma), \\ \quad \wedge\, ((G\otimes G')(\mathbf{x},\mathbf{x}')\cdot \mathsf{Mat}(\mathbf{u}_{(1,1)},\ldots,\mathbf{u}_{(\ell,\ell')}))|_T \in \mathcal{C}|_T \\ \quad \wedge\, \exists\, k \in [\ell]\times[\ell'], \mathbf{u}_k|_T \notin \mathcal{C}|_T\end{array}\right],$$

which equals to the following by definition of the tensor product of vectors:

$$\Pr_{(\mathbf{x},\mathbf{x}')\in S\times S'}\left[\begin{array}{l}\exists\, T \subseteq [n] \text{ s.t.} \\ \quad |T| \geq n\cdot(1-\gamma), \\ \quad \wedge\, \left(\sum_{i\in[\ell]}\sum_{j\in[\ell']} G(\mathbf{x})_i G'(\mathbf{x}')_j u_{(i,j)}\right)\Big|_T \in \mathcal{C}|_T \\ \quad \wedge\, \exists\, k \in [\ell]\times[\ell'], \mathbf{u}_k|_T \notin \mathcal{C}|_T\end{array}\right].$$

By the law of total probability, it suffices to bound the following two probabilities:

$$\Pr_{(\mathbf{x},\mathbf{x}')\in S\times S'}\left[\begin{array}{l}\exists\, T \subseteq [n] \text{ s.t.} \\ \quad |T| \geq n\cdot(1-\gamma), \\ \quad \wedge\, \left(\sum_{i\in[\ell]}\sum_{j\in[\ell']} G(\mathbf{x})_i G'(\mathbf{x}')_j u_{(i,j)}\right)\Big|_T \in \mathcal{C}|_T \\ \quad \wedge\, \exists\, k \in [\ell]\times[\ell'], u_k|_T \notin \mathcal{C}|_T \\ \quad \wedge\, \exists\, i \in [\ell], \left(\sum_{j\in[\ell']} G'(\mathbf{x}')_j \mathbf{u}_{(i,j)}\right)\Big|_T \notin \mathcal{C}|_T\end{array}\right], \text{ and} \tag{4}$$

$$\Pr_{(\mathbf{x},\mathbf{x}')\in S\times S'}\left[\begin{array}{l}\exists\, T \subseteq [n] \text{ s.t.} \\ \quad |T| \geq n\cdot(1-\gamma), \\ \quad \wedge\, \left(\sum_{i\in[\ell]}\sum_{j\in[\ell']} G(\mathbf{x})_i G'(\mathbf{x}')_j u_{(i,j)}\right)\Big|_T \in \mathcal{C}|_T \\ \quad \wedge\, \exists\, k \in [\ell]\times[\ell'], u_k|_T \notin \mathcal{C}|_T \\ \quad \wedge\, \forall\, i \in [\ell], \left(\sum_{j\in[\ell']} G'(\mathbf{x}')_j \mathbf{u}_{(i,j)}\right)\Big|_T \in \mathcal{C}|_T\end{array}\right]. \tag{5}$$

Equation 4 can be rewritten as follows:

$$\mathop{\mathbb{E}}_{\mathbf{x}'\in S'}\left[\Pr_{\mathbf{x}\in S}\left[\begin{array}{l}\exists\, T \subseteq [n] \text{ s.t.} \\ \quad |T| \geq n\cdot(1-\gamma), \\ \quad \wedge\, \left(\sum_{i\in[\ell]} G(\mathbf{x})_i \sum_{j\in[\ell']} G'(\mathbf{x}')_j u_{(i,j)}\right)\Big|_T \in \mathcal{C}|_T \\ \quad \wedge\, \exists\, i \in [\ell], \left(\sum_{j\in[\ell']} G'(\mathbf{x}')_j \mathbf{u}_{(i,j)}\right)\Big|_T \notin \mathcal{C}|_T\end{array}\,\middle|\, \mathbf{x}'\right]\right].$$

The probability inside the expectation can be bounded above by the MCA error of G applied to the words $\sum_{j\in[\ell']} G'(\mathbf{x}')_j \mathbf{u}_{(i,j)}$ for $i \in [\ell]$. Therefore, Equation 4 is bounded above by $\epsilon_{\mathsf{MCA}}(\gamma)$.

Equation 5 can be upper-bounded by the following:

$$\Pr_{\mathbf{x}'\in S'}\left[\begin{array}{l}\exists\, T \subseteq [n] \text{ s.t.} \\ \quad |T| \geq n\cdot(1-\gamma), \\ \quad \wedge\, \exists\, k \in [\ell]\times[\ell'], u_k|_T \notin \mathcal{C}|_T \\ \quad \wedge\, \forall\, i \in [\ell], \left(\sum_{j\in[\ell']} G'(\mathbf{x}')_j \mathbf{u}_{(i,j)}\right)\Big|_T \in \mathcal{C}|_T\end{array}\right],$$

which is bounded above by $\epsilon'_{\mathsf{MCA}}(\gamma)$, the MCA error of G'. □

4.3 Matrix-vector composition

Lemma 4.5. *Let $\mathbb{F} \subseteq \mathbb{K}$ be a finite field extension of degree $[\mathbb{K} : \mathbb{F}] = k$, and let $\phi\colon \mathbb{K} \to \mathbb{F}^k$ be an isomorphism of $\mathbb{F}$-vector spaces. Fix an $\mathbb{F}$-linear code $\mathcal{C} \subseteq \Sigma^n$ and two generators $G_{\mathsf{in}}\colon S_{\mathsf{in}} \to \mathbb{F}^k$, $G_{\mathsf{out}}\colon S_{\mathsf{out}} \to \mathbb{K}^\ell$. Assume that G_{in} has MCA for $\mathcal{C}$ with error ϵ_{MCA} and G_{out} has MCA for $\mathcal{C} \otimes \mathbb{K} \subseteq (\Sigma \otimes \mathbb{K})^n$ with error ϵ'_{MCA}. Then the generator $G\colon S_{\mathsf{in}} \times S_{\mathsf{out}} \to \mathbb{F}^\ell$ defined below has MCA for $\mathcal{C}$ with error $\epsilon_{\mathsf{MCA}} + \epsilon'_{\mathsf{MCA}}$:*

$$G(\mathbf{x}, \mathbf{x}') \coloneqq (\langle \phi(G_{\mathsf{out}}(\mathbf{x}')_j), G_{\mathsf{in}}(\mathbf{x}) \rangle)_{j \in [\ell]} \ .$$

Proof. Fix $\mathbf{U} \coloneqq \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell) \in \Sigma^{\ell \times n}$ and $\gamma \in [0, 1]$. Our goal is to bound the following probability:

$$\Pr_{(\mathbf{x},\mathbf{x}') \in S_{\mathsf{in}} \times S_{\mathsf{out}}} \left[\begin{array}{l} \exists T \subseteq [n] \\ \quad |T| \geq n \cdot (1 - \gamma) \\ \quad \wedge\ (G(\mathbf{x}, \mathbf{x}') \cdot \mathbf{U})|_T \in \mathcal{C}|_T \\ \quad \wedge\ \exists\, w \in [\ell], \mathbf{u}_w \notin \mathcal{C}|_T \end{array} \right]$$

$$= \Pr_{(\mathbf{x},\mathbf{x}') \in S_{\mathsf{in}} \times S_{\mathsf{out}}} \left[\begin{array}{l} \exists T \subseteq [n] \\ \quad |T| \geq n \cdot (1 - \gamma) \\ \quad \wedge\ (\sum_{j \in [k]} G_{\mathsf{in}}(\mathbf{x})_j \sum_{i \in [\ell]} \phi(G_{\mathsf{out}}(\mathbf{x}')_i)_j \mathbf{u}_i)|_T \in \mathcal{C}|_T \\ \quad \wedge\ \exists\, w \in [\ell], \mathbf{u}_w \notin \mathcal{C}|_T \end{array} \right] .$$

By the law of total probability, the previous probability is equal to the following:

$$\Pr_{(\mathbf{x},\mathbf{x}') \in S_{\mathsf{in}} \times S_{\mathsf{out}}} \left[\begin{array}{l} \exists T \subseteq [n] \\ \quad |T| \geq n \cdot (1 - \gamma) \\ \quad \wedge\ (\sum_{j \in [k]} G_{\mathsf{in}}(\mathbf{x})_j \sum_{i \in [\ell]} \phi(G_{\mathsf{out}}(\mathbf{x}')_i)_j \mathbf{u}_i)|_T \in \mathcal{C}|_T \\ \quad \wedge\ \exists\, j \in [k], (\sum_{i \in [\ell]} \phi(G_{\mathsf{out}}(\mathbf{x}')_i)_j \mathbf{u}_i)|_T \notin \mathcal{C}|_T \\ \quad \wedge\ \exists\, w \in [\ell], \mathbf{u}_w \notin \mathcal{C}|_T \end{array} \right] \tag{6}$$

$$+ \Pr_{(\mathbf{x},\mathbf{x}') \in S_{\mathsf{in}} \times S_{\mathsf{out}}} \left[\begin{array}{l} \exists T \subseteq [n] \\ \quad |T| \geq n \cdot (1 - \gamma) \\ \quad \wedge\ (\sum_{j \in [k]} G_{\mathsf{in}}(\mathbf{x})_j \sum_{i \in [\ell]} \phi(G_{\mathsf{out}}(\mathbf{x}')_i)_j \mathbf{u}_i)|_T \in \mathcal{C}|_T \\ \quad \wedge\ \forall\, j \in [k], (\sum_{i \in [\ell]} \phi(G_{\mathsf{out}}(\mathbf{x}')_i)_j \mathbf{u}_i)|_T \in \mathcal{C}|_T \\ \quad \wedge\ \exists\, w \in [\ell], \mathbf{u}_w \notin \mathcal{C}|_T \end{array} \right] . \tag{7}$$

The term in Equation 6 can be rewritten as follows:

$$\mathop{\mathbb{E}}_{\mathbf{x}' \in S_{\mathsf{out}}} \left[\Pr_{\mathbf{x} \in S_{\mathsf{in}}} \left[\begin{array}{l} \exists T \subseteq [n] \\ \quad |T| \geq n \cdot (1 - \gamma) \\ \quad \wedge\ (\sum_{j \in [k]} G_{\mathsf{in}}(\mathbf{x})_j \sum_{i \in [\ell]} \phi(G_{\mathsf{out}}(\mathbf{x}')_i)_j \mathbf{u}_i)|_T \in \mathcal{C}|_T \\ \quad \wedge\ \exists\, j \in [k], (\sum_{i \in [\ell]} \phi(G_{\mathsf{out}}(\mathbf{x}')_i)_j \mathbf{u}_i)|_T \notin \mathcal{C}|_T \\ \quad \wedge\ \exists\, w \in [\ell], \mathbf{u}_w \notin \mathcal{C}|_T \end{array} \right] \middle| \mathbf{x}' \right] .$$

The term inside of the expectation can be bounded above by the MCA error $\epsilon_{\mathsf{MCA}}(\gamma)$ of G_{in} applied to the words $(\sum_{i \in [\ell]} \phi(G_{\mathsf{out}}(\mathbf{x}')_i)_j \mathbf{u}_i)_{j \in [k]}$.

Now we bound the term in Equation 7. Let $(\alpha_j)_{j \in [k]}$ be a basis for $\mathbb{K}$ as a vector space over $\mathbb{F}$ such that ϕ maps any element in $\mathbb{K}$ to its decomposition in this basis. Let ϕ^n we mean the natural extension of ϕ to $\mathbb{K}^n$ that applies ϕ coordinate-wise. We observe that if for every $j \in [k]$, we have $(\sum_{i \in [\ell]} \phi(G_{\mathsf{out}}(\mathbf{x}')_i)_j \mathbf{u}_i)|_T \in \mathcal{C}|_T$,

then $(\sum_{j\in[k]}\sum_{i\in[\ell]}\phi(G_{\mathsf{out}}(\mathbf{x}')_i)_j\mathbf{u}_i\alpha_j)|_T \in \phi^n((\mathcal{C}\otimes\mathbb{K})|_T)$. Moreover, notice that if there exists $w\in[\ell]$ such that $\mathbf{u}_w \notin \mathcal{C}|_T$, then $\mathbf{u}_w \notin (\mathcal{C}\otimes\mathbb{K})|_T$. Finally, note that

$$\begin{aligned}\sum_{j\in[k]}\sum_{i\in[\ell]}\phi(G_{\mathsf{out}}(\mathbf{x}')_i)_j\mathbf{u}_i\alpha_j &= \sum_i\left(\sum_j\phi(G_{\mathsf{out}}(\mathbf{x}')_i)_j\alpha_j\right)\mathbf{u}_i\\ &= \sum_i\phi(G_{\mathsf{out}}(\mathbf{x}')_i)\mathbf{u}_i\\ &= \phi^n(G_{\mathsf{out}}(\mathbf{x}')\cdot\mathbf{U})\ .\end{aligned}$$

Hence, it follows that the term in Equation 7 is bounded above by

$$\Pr_{(\mathbf{x},\mathbf{x}')\in S_{\mathsf{in}}\times S_{\mathsf{out}}}\left[\begin{array}{l}\exists T\subseteq[n]:\\ \quad |T|\geq n\cdot(1-\gamma)\\ \quad \wedge\ (G_{\mathsf{out}}(\mathbf{x}')\cdot\mathbf{U})|_T\in(\mathcal{C}\otimes\mathbb{K})|_T\\ \quad \wedge\ \exists\, w\in[\ell],\mathbf{u}_w\notin(\mathcal{C}\otimes\mathbb{K})|_T\end{array}\right] \leq \epsilon'_{\mathsf{MCA}}(\gamma)\ .$$

□

5 Two counting lemmas

We prove two counting lemmas which are useful in the proofs of Theorem 6.1 and Theorem 9.2.

Definition 5.1. *Let $G\colon S \to \mathbb{F}^\ell$ be a generator, and let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code. The* **codeword span** *of codewords $\mathbf{c}_1, \ldots, \mathbf{c}_\ell \in \mathcal{C}$ associated with G, denoted as $\mathsf{Span}_G(\mathbf{c}_1, \ldots, \mathbf{c}_\ell)$, is defined as*

$$\mathsf{Span}_G(\mathbf{c}_1, \ldots, \mathbf{c}_\ell) \coloneqq \{G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{c}_1, \ldots, \mathbf{c}_\ell) : \mathbf{x} \in S\} \subseteq \mathcal{C} \ .$$

For every $\mathbf{x} \in S$, we also define $\mathsf{Span}_G(\mathbf{c}_1, \ldots, \mathbf{c}_\ell)[\mathbf{x}] \coloneqq G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{c}_1, \ldots, \mathbf{c}_\ell) \in \mathcal{C}$.

- In Lemma 5.2, we show that if for $\ell+1$ distinct seeds $\mathbf{x}_1, \ldots, \mathbf{x}_{\ell+1}$, there exist codewords $\mathbf{c}_1, \ldots, \mathbf{c}_{\ell+1} \in \mathcal{C}$ close to $G(\mathbf{x}_i) \cdot \mathbf{U}$ whose agreement sets have large intersection; then there must exist $\mathbf{c}_1^\star, \ldots, \mathbf{c}_\ell^\star \in \mathcal{C}$ such that for all $i \in [\ell+1]$, $\mathbf{c}_i = \mathsf{Span}_G(\mathbf{c}_1^\star, \ldots, \mathbf{c}_\ell^\star)[\mathbf{x}_i]$.
- In Lemma 5.3, we show that if $G(\mathbf{x}) \cdot \mathbf{U}$ is close to $\mathsf{Span}_G(\mathbf{c}_1^\star, \ldots, \mathbf{c}_\ell^\star)[\mathbf{x}]$ for a large enough number of seeds $\mathbf{x}$, then $\mathbf{U}$ has correlated agreement with $\mathbf{c}_1^\star, \ldots, \mathbf{c}_\ell^\star$.

Lemma 5.2. *Let $G\colon S \to \mathbb{F}^\ell$ be an MDS generator such that $\mathcal{C}_G$ has dimension ℓ. Let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code with distance $\Delta_\mathcal{C}$. Let $\mathbf{U} \coloneqq \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell) \in \Sigma^{\ell \times n}$ be a set of words. Fix distinct $\mathbf{x}_1, \ldots, \mathbf{x}_\ell \in S$, and any $\mathbf{x}_{\ell+1} \in S$. Consider $\mathbf{c}_1, \ldots, \mathbf{c}_{\ell+1} \in \mathcal{C}$. For every $j \in [\ell+1]$, define*

$$A_j \coloneqq \{i \in [n] : (G(\mathbf{x}_j) \cdot \mathbf{U})[i] = \mathbf{c}_j[i]\} \ .$$

If $|A_1 \cap \cdots \cap A_{\ell+1}| > n - \Delta_\mathcal{C}$, then there exists codewords $\mathbf{c}_1^\star, \ldots, \mathbf{c}_\ell^\star \in \mathcal{C}$, uniquely determined by $\mathbf{c}_1, \ldots, \mathbf{c}_\ell$ and $\mathbf{x}_1, \ldots, \mathbf{x}_\ell$, such that

$$\forall\, j \in [\ell+1], \mathbf{c}_j = \mathsf{Span}_G(\mathbf{c}_1^\star, \ldots, \mathbf{c}_\ell^\star)[\mathbf{x}_j] \ .$$

Proof. We first construct $\mathbf{c}_1^\star, \ldots, \mathbf{c}_\ell^\star$. Let $\mathbf{M} \coloneqq \mathsf{Mat}(G(\mathbf{x}_1), \ldots, G(\mathbf{x}_\ell))$. Recall that $\mathcal{C}_G$ is MDS with dimension ℓ, and $\mathbf{x}_1, \cdots, \mathbf{x}_\ell$ are distinct ; hence, by Lemma 3.6, $\mathbf{M}$ is invertible. Let $\mathbf{c}_1^\star, \ldots, \mathbf{c}_\ell^\star$ be defined as follows:

$$\mathsf{Mat}(\mathbf{c}_1^\star, \ldots, \mathbf{c}_\ell^\star) \coloneqq \mathbf{M}^{-1} \cdot \mathsf{Mat}(\mathbf{c}_1, \ldots, \mathbf{c}_\ell) \ .$$

The $\mathbf{c}_1^\star, \ldots, \mathbf{c}_\ell^\star$ are codewords since linear combinations of the c_j remain in $\mathcal{C}$. This also shows that the $\mathbf{c}_1^\star, \ldots, \mathbf{c}_\ell^\star$ are uniquely determined by $\mathbf{c}_1, \ldots, \mathbf{c}_\ell$ and $\mathbf{x}_1, \ldots, \mathbf{x}_\ell$.

Let $A \coloneqq A_1 \cap \cdots \cap A_\ell$. By construction, for every $i \in [\ell]$, $\mathbf{c}_i = \mathsf{Span}_G(\mathbf{c}_1^\star, \ldots, \mathbf{c}_\ell^\star)[\mathbf{x}_i]$ as desired. It remains to show that $\mathbf{c}_{\ell+1} = \mathsf{Span}_G(\mathbf{c}_1^\star, \ldots, \mathbf{c}_\ell^\star)[\mathbf{x}_{\ell+1}]$. We have that for every $j \in [\ell]$:

$$\mathbf{c}_j|_A = (G(\mathbf{x}_j) \cdot \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell))|_A \ ,$$

which implies that for every $j \in [\ell]$, $\mathbf{u}_j|_A = \mathbf{c}_j^*|_A$ (as $\mathbf{x}_1 \neq \cdots \neq \mathbf{x}_\ell$ and $\mathcal{C}_G$ is MDS). Over the $(\ell+1)$-wise intersection $A \cap A_{\ell+1}$,

$$\begin{aligned} \mathbf{c}_{\ell+1}|_{A \cap A_{\mathbf{x}_{\ell+1}}} &= (G(\mathbf{x}_{\ell+1}) \cdot \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell))|_{A \cap A_{\ell+1}} \\ &= (G(\mathbf{x}_{\ell+1}) \cdot \mathsf{Mat}(\mathbf{c}_1^\star, \ldots, \mathbf{c}_\ell^\star))|_{A \cap A_{\ell+1}} \ . \end{aligned}$$

Since $|A \cap A_{\ell+1}| > n - \Delta_\mathcal{C}$, we conclude that $\mathbf{c}_{\ell+1} = G(\mathbf{x}_{\ell+1}) \cdot \mathsf{Mat}(\mathbf{c}_1^\star, \ldots, \mathbf{c}_\ell^\star)$. □

Lemma 5.3. *Fix a zero-evading generator $G\colon S \to \mathbb{F}^\ell$ with error $\epsilon_{\mathsf{ZE}} \in [0,1]$, an $\mathbb{F}$-linear code $\mathcal{C} \subseteq \Sigma^n$, $e \in \{1,\ldots,n\}$, $1 \le t \le e$, a set of words $\mathbf{U} = \mathsf{Mat}(\mathbf{u}_1,\ldots,\mathbf{u}_\ell) \in \Sigma^{\ell\times n}$, and codewords $\mathbf{c}_1,\ldots,\mathbf{c}_\ell \in \mathcal{C}$. Define the set A as follows:*

$$A \coloneqq \{\mathbf{x} \in S : \Delta\left(G(\mathbf{x})\cdot\mathbf{U}, \mathsf{Span}_G(\mathbf{c}_1,\ldots,\mathbf{c}_\ell)[\mathbf{x}]\right) \le e - t\} \quad .$$

If $|A| > \frac{e}{t}\cdot\epsilon_{\mathsf{ZE}}\cdot|S|$ then $\Delta_{\Sigma^\ell}(\mathbf{U}, \mathsf{Mat}(\mathbf{c}_1,\ldots\mathbf{c}_\ell)) < e$.

Proof. Define T to be the set of columns $i \in [n]$ of $\mathbf{U}$ where $\mathbf{u}_j$ and $\mathbf{c}_j$ agree at the i-th location for every $j \in [\ell]$, i.e.,

$$T \coloneqq \left\{ i \in [n] : \bigwedge_{j=1}^{\ell} (\mathbf{u}_j[i] = \mathbf{c}_j[i]) \right\} \quad .$$

For every $i \in [n]$, define $\mathbf{v}(i) \in \mathbb{F}^\ell$ as follows:

$$\mathbf{v}(i) \coloneqq (\mathbf{c}_1[i] - \mathbf{u}_1[i], \ldots, \mathbf{c}_\ell[i] - \mathbf{u}_\ell[i]) \quad .$$

Note that for every $i \in [n]\setminus T$, $\mathbf{v}(i)$ is a nonzero vector. Since G is zero-evading with error ϵ_{ZE}, it follows that for every $i \in [n]\setminus T$:

$$|\{\mathbf{x} \in S : G(\mathbf{x})\cdot\mathbf{v}(i)^\intercal = 0\}| \le \epsilon_{\mathsf{ZE}}\cdot|S| \quad .$$

Hence, we deduce that:

$$\begin{aligned}
e - t &\ge \mathop{\mathbb{E}}_{\mathbf{x}\in A}\left[\Delta\left(\sum_{j\in[\ell]} G(\mathbf{x})[j]\cdot(\mathbf{c}_j - \mathbf{u}_j), \mathbf{0}\right)\right] \\
&= \sum_{\mathbf{x}\in A}\frac{1}{|A|}\sum_{i\in[n]\setminus T}\mathbf{1}\left\{G(\mathbf{x})\cdot\mathbf{v}(i)^\intercal \neq 0\right\} \\
&= \frac{1}{|A|}\sum_{i\in[n]\setminus T}\sum_{\mathbf{x}\in A}\mathbf{1}\left\{G(\mathbf{x})\cdot\mathbf{v}(i)^\intercal \neq 0\right\} \\
&\ge \frac{1}{|A|}\cdot(n - |T|)\cdot(|A| - \epsilon_{\mathsf{ZE}}\cdot|S|) \\
&> (n - |T|)\cdot\left(1 - \frac{t}{e}\right) \quad .
\end{aligned}$$

Rearranging, we obtain that $|T| > n - e$ which implies $\Delta_{\Sigma^\ell}(\mathbf{U}, \mathsf{Mat}(\mathbf{c}_1,\ldots\mathbf{c}_\ell)) < e$ by definition of T. □

Remark 5.4. The previous proof also has the following interpretation. For a set of words $\mathbf{U} = \mathsf{Mat}(\mathbf{u}_1,\ldots,\mathbf{u}_\ell) \in \Sigma^{\ell\times n}$, and codewords $\mathbf{c}_1,\ldots,\mathbf{c}_\ell \in \mathcal{C}$, define:

$$T \coloneqq \left\{ i \in [n] : \bigwedge_{j=1}^{\ell} (\mathbf{u}_j[i] = \mathbf{c}_j[i]) \right\} \quad .$$

Fix a zero-evading generator $G\colon S \to \mathbb{F}^\ell$ with error $\epsilon_{\mathsf{ZE}} \in [0,1]$. For any $\mathbf{x} \in S$ set

$$T_{\mathbf{x}} \coloneqq \left\{ i \in [n] : \left(\sum_{j\in[\ell]} G(\mathbf{x})[j]\cdot\mathbf{u}_j\right)[i] = \left(\sum_{j\in[\ell]} G(\mathbf{x})[j]\cdot\mathbf{c}_j\right)[i] \right\} \quad .$$

Then, with our previous notation for $\mathbf{v}(i)$:

$$\{T_{\mathbf{x}} : T \subsetneq T_{\mathbf{x}}\} = \{\mathbf{x} : \exists\, i \in [n] \setminus T,\ G(\mathbf{x}) \cdot \mathbf{v}(i)^{\mathsf{T}} = 0\} \ .$$

It follows that the number of $T_{\mathbf{x}}$ that strictly contain T is bounded above by $(n - |T|) \cdot \epsilon_{\mathsf{ZE}} \cdot |S|$.

6 MCA for MDS generators

We prove that MDS generators have MCA for every linear code.

Theorem 6.1 (MCA for MDS generators)**.** *Let $G\colon S \to \mathbb{F}^\ell$ be an MDS generator such that $\mathcal{C}_G$ has dimension $\ell \geq 2$. Let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code with relative distance $\delta_{\mathcal{C}}$. Set $\rho_{\mathcal{C}} \coloneqq 1 - \delta_{\mathcal{C}}$. For every $\eta \in (0,1)$, G has mutual correlated agreement for $\mathcal{C}$ with error $\epsilon_{\mathsf{MCA}}\colon [0,1] \to [0,1]$ where*

$$\epsilon_{\mathsf{MCA}}(\gamma) \coloneqq \begin{cases} \max\{n\cdot\gamma, 1\} \cdot \frac{(\ell-1)}{|S|} & \text{if } \gamma < \delta_{\mathcal{C}}/(\ell+1) \\ \frac{n\cdot\gamma_\ell}{\eta} \cdot \frac{(\ell-1)}{|S|} + \max\left\{ \frac{2\cdot(\ell-1)}{\eta\cdot((\rho_{\mathcal{C}}+\eta)^{1/(\ell+1)} - (\rho_{\mathcal{C}}+\eta)^{1/\ell})\cdot|S|}, \frac{\ell\cdot(\ell+1)}{\eta\cdot|S|} \right\} & \text{if } \gamma \leq 1-(\rho_{\mathcal{C}}+\eta)^{1/(\ell+1)} \\ 1 & \text{otherwise} \end{cases} .$$

Here, $\gamma_\ell \coloneqq 1 - (\rho_{\mathcal{C}} + \eta)^{1/\ell}$.

The MCA error ϵ_{MCA} in the theorem has two cases corresponding to the unique decoding setting and the list decoding setting, respectively. We prove the case of unique decoding in Section 6.1 and list decoding in Section 6.2.

6.1 Unique decoding

Lemma 6.2. *Let $G\colon S \to \mathbb{F}^\ell$ be an MDS generator such that $\mathcal{C}_G$ has dimension $\ell \geq 2$. Let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code with relative distance $\delta_{\mathcal{C}}$. Then, G has MCA for $\mathcal{C}$ with error:*

$$\epsilon_{\mathsf{MCA}}(\gamma) \coloneqq \max\{n\cdot\gamma, 1\} \cdot \frac{(\ell-1)}{|S|} \quad \text{if } \gamma < \delta_{\mathcal{C}}/(\ell+1) \ .$$

Proof. Let $\mathbf{U} \coloneqq \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell) \in \Sigma^{\ell\times n}$.

The case $\gamma < 1/n$. In this case, we are trying to bound the size of the set :

$$\mathsf{B} \coloneqq \left\{ \mathbf{x} \in S : \begin{array}{l} G(\mathbf{x}) \cdot \mathbf{U} \in \mathcal{C} \\ \wedge\, \exists\, j \in [\ell], \mathbf{u}_j \notin \mathcal{C} \end{array} \right\} .$$

Assume for contradiction that $|\mathsf{B}| > \ell - 1$. Consider distinct $\mathbf{x}_1, \ldots, \mathbf{x}_\ell \in \mathsf{B}$, and for all $j \in [\ell]$, let $\mathbf{c}_j \in \mathcal{C}$ be such that $G(\mathbf{x}_j) \cdot \mathbf{U} = \mathbf{c}_j$. By assumption, the matrix $\mathbf{M} \coloneqq \mathsf{Mat}(G(\mathbf{x}_1), \ldots, G(\mathbf{x}_\ell)) \in \mathbb{F}^{\ell\times\ell}$ is invertible, and therefore:

$$\mathbf{U} = \mathbf{M}^{-1} \cdot \mathsf{Mat}(\mathbf{c}_1, \ldots, \mathbf{c}_\ell) \ .$$

This shows that $\mathbf{U} \in \mathcal{C}^\ell$, a contradiction to the fact that $|\mathsf{B}| > 0$.

The case $\gamma \geq 1/n$. In this case, we are trying to bound the size of the set :

$$\mathsf{B} \coloneqq \left\{ \mathbf{x} \in S : \begin{array}{l} \exists\, T \subseteq [n] \\ \quad |T| \geq n\cdot(1-\gamma) \\ \quad \wedge\, (G(\mathbf{x}) \cdot \mathbf{U})|_T \in \mathcal{C}|_T \\ \quad \wedge\, \exists\, j \in [\ell], (\mathbf{u}_j)|_T \notin \mathcal{C}|_T \end{array} \right\} .$$

Assume for contradiction that $|\mathsf{B}| > n\cdot\gamma\cdot(\ell-1)$. For any $\mathbf{x} \in B$, define $\mathbf{u}_{\mathbf{x}} \coloneqq G(\mathbf{x}) \cdot \mathbf{U}$. By assumption $|\mathsf{B}| \geq \ell$, therefore, we can choose ℓ distinct elements $\mathbf{x}_1, \ldots, \mathbf{x}_\ell$ in B. Let $\mathbf{c}_1, \ldots, \mathbf{c}_\ell \in \mathcal{C}$ be the unique

closest codewords to $\mathbf{u}_{\mathbf{x}_1}, \ldots, \mathbf{u}_{\mathbf{x}_\ell}$, respectively (these codewords exist by definition of B). By assumption on G, the matrix $\mathbf{M} \coloneqq \mathsf{Mat}(G(\mathbf{x}_1), \ldots, G(\mathbf{x}_\ell))$ is invertible. Define $\mathbf{c}_1^*, \ldots, \mathbf{c}_\ell^* \in \mathcal{C}$ as follows:

$$\mathsf{Mat}(\mathbf{c}_1^*, \ldots, \mathbf{c}_\ell^*) \coloneqq \mathbf{M}^{-1} \cdot \mathsf{Mat}(\mathbf{c}_1, \ldots, \mathbf{c}_\ell)$$

Note that by applying the triangle inequality we obtain the following, for any $\mathbf{x} \in \mathsf{B}$:

$$\begin{aligned}
\Delta(\mathbf{u}_{\mathbf{x}}, G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{c}_1^*, \ldots, \mathbf{c}_\ell^*)) &= \Delta(G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell), G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{c}_1^*, \ldots, \mathbf{c}_\ell^*)) \\
&= \Delta(G(\mathbf{x}) \cdot \mathbf{M}^{-1} \cdot \mathsf{Mat}(\mathbf{u}_{\mathbf{x}_1}, \ldots, \mathbf{u}_{\mathbf{x}_\ell}), G(\mathbf{x}) \cdot \mathbf{M}^{-1} \cdot \mathsf{Mat}(\mathbf{c}_1, \ldots, \mathbf{c}_\ell)) \\
&\leq \sum_{i=1}^{\ell} \Delta(\mathbf{u}_{\mathbf{x}_i}, \mathbf{c}_i) \\
&\leq n \cdot \gamma \cdot \ell \ .
\end{aligned}$$

For any $\mathbf{x} \in \mathsf{B}$, let $\mathbf{c}_{\mathbf{x}}$ be the unique closest codeword to $\mathbf{u}_{\mathbf{x}}$. By applying the triangle inequality again:

$$\begin{aligned}
\Delta(\mathbf{c}_{\mathbf{x}}, G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{c}_1^*, \ldots, \mathbf{c}_\ell^*)) &\leq \Delta(\mathbf{c}_{\mathbf{x}}, \mathbf{u}_{\mathbf{x}}) + \Delta(\mathbf{u}_{\mathbf{x}}, G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{c}_1^*, \ldots, \mathbf{c}_\ell^*)) \\
&\leq n \cdot \gamma \cdot (\ell + 1) \\
&< n \cdot \delta_{\mathcal{C}} \ .
\end{aligned}$$

This last inequality holds by assumption on γ. By definition of the minimum distance, we find that for any $\mathbf{x} \in \mathsf{B}$, $\mathbf{c}_{\mathbf{x}} = \mathsf{Span}_G(\mathbf{c}_1^*, \ldots, \mathbf{c}_\ell^*)[\mathbf{x}]$. By Lemma 5.3 and its proof, this implies that the correlated agreement set $\tilde{T} \coloneqq \{i \in [n] : \forall j \in [\ell],\ \mathbf{u}_j[i] = \mathbf{c}_j^*[i]\}$ has size $|\tilde{T}| > n \cdot (1 - \gamma)$. At the same time, for any $\mathbf{x} \in \mathsf{B}$, the agreement set $T_{\mathbf{x}}$ between $\mathbf{u}_{\mathbf{x}}$ and $\mathbf{c}_{\mathbf{x}}$ must strictly contain $\tilde{T}$. We saw in Remark 5.4 that the number of such $T_{\mathbf{x}}$ is less than $\left|[n] \setminus \tilde{T}\right| \cdot (\ell - 1) < n \cdot \gamma \cdot (\ell - 1)$, a contradiction to our assumption on the size of B. □

6.2 List decoding

We rely on several lemmas adapted from [GKL24].

Definition 6.3. *Let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code. For every set of words $\mathbf{u}, \mathbf{u}_1, \ldots, \mathbf{u}_\ell \in \Sigma^n$, a subset $T \subseteq [n]$ is said to be:*

- *A* **maximal agreement domain** *between $\mathbf{u}$ and $\mathcal{C}$ if it is maximal for inclusion among subsets A of $[n]$ such that $\mathbf{u}|_A \in \mathcal{C}|_A$.*
- *A* **maximal CA domain** *between $\mathbf{u}_1, \ldots, \mathbf{u}_\ell$ and $\mathcal{C}$ if it is maximal for inclusion among subsets A of $[n]$ such that $\mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell)|_A \in \mathcal{C}^\ell|_A$.*

Lemma 6.4. *Let $G\colon S \to \mathbb{F}^\ell$ be an MDS generator such that $\mathcal{C}_G$ has dimension ℓ. Let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code with minimum distance $\Delta_{\mathcal{C}}$. Let $\mathbf{U} \coloneqq \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell) \in \Sigma^{\ell \times n}$ be a set of words. For distinct $\mathbf{x}_1, \ldots, \mathbf{x}_\ell \in S$, let $\mathbf{c}_{\mathbf{x}_1}, \ldots, \mathbf{c}_{\mathbf{x}_\ell} \in \mathcal{C}$, be such that for every $j \in [\ell]$:*

$$A_{\mathbf{x}_j} \coloneqq \{i \in [n] : (G(\mathbf{x}_j) \cdot \mathbf{U})[i] = \mathbf{c}_{\mathbf{x}_j}[i]\}$$

is a maximal agreement domain between $G(\mathbf{x}_j) \cdot \mathbf{U}$ and $\mathcal{C}$. Suppose further that $|A_{\mathbf{x}_1} \cap \cdots \cap A_{\mathbf{x}_\ell}| > n - \Delta_{\mathcal{C}}$. Then, $A_{\mathbf{x}_1} \cap \cdots \cap A_{\mathbf{x}_\ell}$ is a maximal CA domain between $\mathbf{u}_1, \ldots, \mathbf{u}_\ell$ and $\mathcal{C}$.

Proof. This follows from the proof of Lemma 5.2. Let $A \subseteq [n]$ be such that $\mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell)|_A \in \mathcal{C}^\ell|_A$ and $A_{\mathbf{x}_1} \cap \cdots \cap A_{\mathbf{x}_\ell} \subseteq A$. If $\mathbf{c}_1, \ldots, \mathbf{c}_\ell$ are the codewords uniquely determined by $\mathbf{c}_{\mathbf{x}_1}, \ldots, \mathbf{c}_{\mathbf{x}_\ell}$ and $\mathbf{x}_1, \ldots, \mathbf{x}_\ell$, then it follows from the fact that $|A_{\mathbf{x}_1} \cap \cdots \cap A_{\mathbf{x}_\ell} \cap A| > n - \Delta_{\mathcal{C}}$ that for every $j \in [\ell]$, $\mathbf{u}_j|_A = \mathbf{c}_j|_A$. Therefore, $A \subseteq A_{\mathbf{x}_j}$ for every $j \in [\ell]$. □

Lemma 6.5. *Let $G\colon S \to \mathbb{F}^\ell$ be an MDS generator such that $\mathcal{C}_G$ has dimension ℓ. Let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code with minimal distance $\Delta_{\mathcal{C}}$. Let $\mathbf{u}_1, \ldots, \mathbf{u}_\ell \in \Sigma^n$ be a set of words. Let $A \subseteq [n]$ be a maximal CA domain between $\mathbf{u}_1, \ldots, \mathbf{u}_\ell$ and $\mathcal{C}$ with $|A| > n - \Delta_{\mathcal{C}}$. For distinct $\mathbf{x}_1, \ldots, \mathbf{x}_t \in S$, let $B_1, \ldots, B_t \subseteq [n]$ be maximal agreement domains between $G(\mathbf{x}_i) \cdot \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_t)$ and $\mathcal{C}$ (respectively). Suppose that for every $i \in [t]$, $A \subseteq B_i$. Then, for every distinct $i_1, \ldots, i_\ell \in [t]$, we have:*

$$B_{i_1} \cap \cdots \cap B_{i_\ell} = A \ .$$

Proof. Since $A \subseteq B_{i_1} \cap \cdots \cap B_{i_\ell}$, we have $|B_{i_1} \cap \cdots \cap B_{i_\ell}| > n - \Delta_{\mathcal{C}}$. By Lemma 6.4, $B_{i_1} \cap \cdots \cap B_{i_\ell}$ is a maximal CA domain between $\mathbf{u}_1, \ldots, \mathbf{u}_\ell$ and $\mathcal{C}$. By maximality of A, the reverse inclusion also holds, and this concludes the proof. □

Lemma 6.6. *In the situation of Lemma 6.5, assume that $A \subsetneq B_i$. Then, it holds that $t \leq (\ell - 1) \cdot (n - |A|)$.*

Proof. The conclusion of Lemma 6.5 implies that for every distinct $i_1, \ldots, i_\ell \in [t]$,

$$(B_{i_1} \setminus A) \cap \cdots \cap (B_{i_\ell} \setminus A) = \emptyset \ .$$

In other words, no element $j \in [n]$ can be in more than ℓ sets of the form $B_i \setminus A$ (for $i \in [t]$), meaning that if $d(j) \coloneqq |\{i \in [t] : j \in B_i \setminus A\}|$ we have $d(j) \leq \ell - 1$. Therefore,

$$\sum_{i=1}^{t} |B_i \setminus A| = \sum_{j \in \cup_i B_i \setminus A} d(j) \leq (\ell - 1) \cdot \left|\cup_{i \in [t]} B_i \setminus A\right| \ .$$

We compute:

$$\begin{aligned} n - |A| &= |[n] \setminus A| \\ &\geq \left|\cup_{i \in [t]} B_i \setminus A\right| \\ &\geq \frac{1}{\ell - 1} \cdot \sum_{i=1}^{t} |B_i \setminus A| \\ &\geq \frac{t}{\ell - 1} \ . \end{aligned}$$

□

Proof of Theorem 6.1. Recall that $\Delta_{\mathcal{C}} = n \cdot \delta_{\mathcal{C}}$. Let $\mathbf{U} \coloneqq \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell)$. In order to prove the theorem, it remains to show that if $\gamma = 1 - (\rho_{\mathcal{C}} + \eta)^{1/(\ell+1)}$ for some $\eta \in (0, 1)$ and we set:

$$B_\gamma \coloneqq \left\{ \mathbf{x} \in S : \begin{array}{l} \exists\, T \subseteq [n] \text{ s.t.} \\ \quad |T| \geq n \cdot (1 - \gamma), \\ \quad \wedge\, (G(\mathbf{x}) \cdot \mathbf{U})|_T \in \mathcal{C}|_T, \\ \quad \wedge\, \exists\, j \in [\ell], \mathbf{u}_j|_T \notin \mathcal{C}|_T \end{array} \right\} \ ,$$

then:

$$|B_\gamma| \leq \frac{(\ell-1)\cdot n}{\eta} + \max\left\{\frac{2\cdot(\ell-1)}{\eta\cdot((\rho_{\mathcal{C}}+\eta)^{1/(\ell+1)}-(\rho_{\mathcal{C}}+\eta)^{1/\ell})}, \frac{\ell\cdot(\ell+1)}{\eta}\right\} \ .$$

To do so, we further partition the set B_e into the following two sets:

$$B^{(1)}_{\gamma,\gamma_\ell} \coloneqq \left\{ \mathbf{x}\in S : \begin{array}{c} \exists\, A \subsetneq T \subseteq [n] \text{ s.t.} \\ |T| \geq n\cdot(1-\gamma), \\ \wedge\, (G(\mathbf{x})\cdot\mathbf{U})|_T \in \mathcal{C}|_T, \\ \wedge\, \exists\, j\in[\ell], \mathbf{u}_j|_T \notin \mathcal{C}|_T, \\ \wedge\ |A| \geq n\cdot(1-\gamma_\ell), \\ \wedge\, \mathbf{U}|_A \in \mathcal{C}^\ell|_A, \\ \wedge\ A \text{ is a maximal CA domain} \end{array} \right\}$$

$$B^{2}_{\gamma,\gamma_\ell} \coloneqq B_\gamma \setminus B^{(1)}_{\gamma,\gamma_\ell} \ .$$

Here, $\gamma_\ell \coloneqq 1-(\rho_{\mathcal{C}}+\eta)^{1/\ell}$. We bound the size of each set individually.

Bounding the size of $B^{(1)}_{\gamma,\gamma_\ell}$. We further partition $B^{(1)}_{\gamma,\gamma_\ell}$ depending on the maximal CA domain A that is included in T. That is, for every maximal CA domain A between $\mathbf{U}$ and $\mathcal{C}$ with $|A| \geq n\cdot(1-\gamma_\ell)$, we define:

$$B_A \coloneqq \left\{ \mathbf{x}\in S : \begin{array}{c} \exists\, T \subseteq [n] \text{ s.t.} \\ |T| \geq n\cdot(1-\gamma), \\ \wedge\, (G(\mathbf{x})\cdot\mathbf{U})|_T \in \mathcal{C}|_T, \\ \wedge\, \exists\, j\in[\ell], \mathbf{u}_j|_T \notin \mathcal{C}|_T, \\ \wedge\, A \subsetneq T \end{array} \right\}$$

It follows that $B^{(1)}_{\gamma,\gamma_\ell} = \cup_A B_A$ where the union is taken over maximal CA domain A between $\mathbf{U}$ and $\mathcal{C}$ with $|A| \geq n\cdot(1-\gamma_\ell)$. We prove a series of claims that will first bound the number of possible such A, and then the size of each B_A.

Claim 6.7. *The number of maximal CA domains A between $\mathbf{U}$ and $\mathcal{C}$ with $|A| \geq n\cdot(1-\gamma_\ell)$ is at most $\frac{1}{\eta}$.*

Proof of Claim. Let $\{A_1,\ldots,A_m\}$ be the maximal CA domains between $\mathbf{U}$ and $\mathcal{C}$ with size at least $n\cdot(1-\gamma_\ell)$. We have that:

1. For every $i\in[m]$, $|A_i| \geq n\cdot(\rho_{\mathcal{C}}+\eta)^{1/\ell}$.
2. For every $i\neq j\in[m]$, $|A_i\cap A_j| \leq n-\Delta_{\mathcal{C}}$.

The first item follows by definition. Assume the second item does not hold, that is, for some $i\neq j\in[m]$, $|A_i\cap A_j| > n-\Delta_{\mathcal{C}}$. Let $\{\mathbf{c}_{i,k}\}_{k\in[\ell]} \subseteq \mathcal{C}$ be such that for every $k\in[\ell]$, $\mathbf{u}_k|_{A_i} = \mathbf{c}_{i,k}|_{A_i}$, and define $\{\mathbf{c}_{j,k}\}_{k\in[\ell]} \subseteq \mathcal{C}$ similarly. By density of the intersection, it follows that $\mathbf{c}_{i,k} = \mathbf{c}_{j,k}$ for every $k\in[\ell]$. Therefore $A_i = A_j$, a contradiction.

Because $\left((\rho_{\mathcal{C}}+\eta)^{1/\ell}\right)^2 > \rho_{\mathcal{C}}$, we can use the Corrádi bound (Lemma 3.23), truncating the A_i's if necessary, and obtain that:

$$m \leq \frac{1}{(\rho_{\mathcal{C}}+\eta)^{2/\ell}-\rho_{\mathcal{C}}} \leq \frac{1}{\eta} \ .$$

□

Claim 6.8. *For every maximal CA domain A, the size of B_A is at most $n \cdot \gamma_\ell \cdot (\ell - 1)$.*

Proof of Claim. Without loss of generality, we may assume that for every $\mathbf{x} \in B_A$, the set T in the definition of B_A is a maximal agreement domain between $G(\mathbf{x}) \cdot \mathbf{U}$ and $\mathcal{C}$. It follows from Lemma 6.6 that the number of possible sets T is bounded above by $(n - |A|) \cdot (\ell - 1) \leq n \cdot \gamma_\ell \cdot (\ell - 1)$. □

Bounding the size of B^2_{γ,γ_ℓ}. We finish the proof with the following claim.

Claim 6.9. *The size of B^2_{γ,γ_ℓ} is bounded above as:*

$$\left|B^2_{\gamma,\gamma_\ell}\right| \leq \max\left\{\frac{2 \cdot (\ell-1)}{\eta \cdot ((\rho_{\mathcal{C}} + \eta)^{1/(\ell+1)} - (\rho_{\mathcal{C}} + \eta)^{1/\ell})}, \frac{\ell \cdot (\ell+1)}{\eta}\right\} \ .$$

Proof. Suppose for a contradiction that this does not hold, that is:

$$\left|B^2_{\gamma,\gamma_\ell}\right| > \max\left\{\frac{2 \cdot (\ell-1)}{\eta \cdot ((\rho_{\mathcal{C}} + \eta)^{1/(\ell+1)} - (\rho_{\mathcal{C}} + \eta)^{1/\ell})}, \frac{\ell \cdot (\ell+1)}{\eta}\right\} \ .$$

For $\mathbf{x} \in B^2_{\gamma,\gamma_\ell}$, let $T_{\mathbf{x}} \subseteq [n]$ be such that $|T_{\mathbf{x}}| \geq n \cdot (1-\gamma)$ and $(G(\mathbf{x}) \cdot \mathbf{U})|_{T_{\mathbf{x}}} \in \mathcal{C}|_{T_{\mathbf{x}}}$ (w.l.o.g. we may assume these are maximal agreement domains). For $\mathbf{x}_1, \ldots, \mathbf{x}_{\ell+1}$ picked uniformly at random from B^2_{γ,γ_ℓ}, by applying Jensen's inequality:

$$\begin{aligned}
\mathop{\mathbb{E}}_{\mathbf{x}_1,\ldots,\mathbf{x}_{\ell+1}}\left[\left|T_{\mathbf{x}_1} \cap \cdots \cap T_{\mathbf{x}_{\ell+1}}\right|\right] &= n \cdot \mathop{\mathbb{E}}_{j,\mathbf{x}_1,\ldots,\mathbf{x}_{\ell+1}}\left[\mathbf{1}\left\{j \in T_{\mathbf{x}_1} \cap \cdots \cap T_{\mathbf{x}_{\ell+1}}\right\}\right] \\
&= n \cdot \mathop{\mathbb{E}}_{j}\left[\mathop{\mathbb{E}}_{\mathbf{x}}\left[\mathbf{1}\{j \in T_{\mathbf{x}}\}\right]^{\ell+1}\right] \\
&\geq n \cdot \mathop{\mathbb{E}}_{j,\mathbf{x}}\left[\mathbf{1}\{j \in T_{\mathbf{x}}\}\right]^{\ell+1} \\
&\geq n \cdot (1-\gamma)^{\ell+1} \\
&\geq n \cdot (\rho_{\mathcal{C}} + \eta) \ .
\end{aligned}$$

We obtain that $\Pr_{\mathbf{x}_1,\ldots,\mathbf{x}_{\ell+1}}\left[\left|T_{\mathbf{x}_1} \cap \cdots \cap T_{\mathbf{x}_{\ell+1}}\right| > n \cdot \rho_{\mathcal{C}}\right] \geq \eta$. We also have that:

$$\Pr_{\mathbf{x}_1,\ldots,\mathbf{x}_{\ell+1}}\left[\mathbf{x}_1, \ldots, \mathbf{x}_{\ell+1} \text{ are not all distinct}\right] \leq \frac{\binom{\ell+1}{2}}{\left|B^2_{\gamma,\gamma_\ell}\right|} < \frac{\eta}{2} \ .$$

Therefore, we have that the probability over $\mathbf{x}_1, \ldots, \mathbf{x}_{\ell+1}$ that $\left|T_{\mathbf{x}_1} \cap \cdots \cap T_{\mathbf{x}_{\ell+1}}\right| > n \cdot \rho_{\mathcal{C}}$ and that $\mathbf{x}_1, \ldots, \mathbf{x}_{\ell+1}$ are all distinct is at least $\eta/2$. It follows that there exist distinct $\mathbf{x}_1, \ldots, \mathbf{x}_\ell \in B^2_{\gamma,\gamma_\ell}$ such that:

$$\Pr_{\mathbf{x}}\left[|T_{\mathbf{x}_1} \cap \cdots \cap T_{\mathbf{x}_\ell} \cap T_{\mathbf{x}}| > n \cdot \rho_{\mathcal{C}}\right] \geq \frac{\eta}{2} \ . \tag{8}$$

In particular, $|T_{\mathbf{x}_1} \cap \cdots \cap T_{\mathbf{x}_\ell}| > n \cdot \rho_{\mathcal{C}}$ and by Lemma 6.4, $T_{\mathbf{x}_1} \cap \cdots \cap T_{\mathbf{x}_\ell}$ is a maximal CA domain between $\mathbf{U}$ and $\mathcal{C}$. But from the proof of Lemma 6.4, for every $T_{\mathbf{x}}$ such that $|T_{\mathbf{x}_1} \cap \cdots \cap T_{\mathbf{x}_\ell} \cap T_{\mathbf{x}}| > n \cdot \rho_{\mathcal{C}}$, we have that $T_{\mathbf{x}_1} \cap \cdots \cap T_{\mathbf{x}_\ell} \subseteq T_{\mathbf{x}}$. Equation 8 shows that the number of $\mathbf{x}$ such that this can happen is at least:

$$\frac{\ell - 1}{(\rho_{\mathcal{C}} + \eta)^{1/(\ell+1)} - (\rho_{\mathcal{C}} + \eta)^{1/\ell}} \ .$$

Since $\mathbf{x}_1, \ldots, \mathbf{x}_\ell \in B^2_{\gamma,\gamma_\ell}$, we have that $|T_{\mathbf{x}_1} \cap \cdots \cap T_{\mathbf{x}_\ell}| < n \cdot (\rho_{\mathcal{C}} + \eta)^{1/\ell}$. For every $\mathbf{x} \in B^2_{\gamma,\gamma_\ell}$, $|T_{\mathbf{x}}| \geq n \cdot (\rho_{\mathcal{C}} + \eta)^{1/(\ell+1)}$. Modifying the proof of Lemma 6.6, we find that:

$$\begin{aligned}\left|\{\mathbf{x} \in B^2_{\gamma,\gamma_\ell} : T_{\mathbf{x}_1} \cap \cdots \cap T_{\mathbf{x}_\ell} \subseteq T_{\mathbf{x}}\}\right| &\leq \frac{(n - |T_{\mathbf{x}_1} \cap \cdots \cap T_{\mathbf{x}_\ell}|) \cdot (\ell - 1)}{n \cdot (\rho_{\mathcal{C}} + \eta)^{1/(\ell+1)} - n \cdot (\rho_{\mathcal{C}} + \eta)^{1/\ell}} \\ &< \frac{\ell - 1}{(\rho_{\mathcal{C}} + \eta)^{1/(\ell+1)} - (\rho_{\mathcal{C}} + \eta)^{1/\ell}} \ ,\end{aligned}$$

a contradiction. □

□

7 The affine space generator

We prove that MCA for the affine line generator implies MCA for the affine space generator, at the cost of a small multiplicative overhead, but independently of the number of combined words and with no requirements on the field size.

Lemma 7.1 (MCA for lines implies MCA for affine spaces)**.** *Let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code. Assume that the affine line generator $\mathbb{F} \to \mathbb{F}^2$, $x \mapsto (1, x)$ has MCA error for $\mathcal{C}$ with error ϵ_{MCA}. Then, for every $s \geq 2$, the affine space generator $\mathbb{F}^s \to \mathbb{F}^{s+1}$, $\mathbf{x} \mapsto (1, \mathbf{x})$ has MCA for $\mathcal{C}$ with error $\epsilon_{\mathsf{MCA}} \cdot \left(1 - \frac{1}{|\mathbb{F}|}\right)^{-1}$.*

Proof. For $\gamma \in [0, 1]$, define the set of bad seeds as:

$$\mathsf{B} \coloneqq \left\{ \mathbf{x} \in \mathbb{F}^s : \begin{array}{l} \exists\, T_{\mathbf{x}} \subseteq [n] \\ \quad |T_{\mathbf{x}}| \geq n \cdot (1 - \gamma) \\ \quad \wedge\, ((1, \mathbf{x}) \cdot \mathbf{U})|_{T_{\mathbf{x}}} \in \mathcal{C}|_{T_{\mathbf{x}}} \\ \quad \wedge\, \exists\, j \in \{0, \ldots, s\}, (\mathbf{u}_j)|_{T_{\mathbf{x}}} \notin \mathcal{C}|_{T_{\mathbf{x}}} \end{array} \right\} .$$

Assume for a contradiction that $|\mathsf{B}| > |\mathbb{F}|^s \cdot \left(1 - \frac{1}{|\mathbb{F}|}\right)^{-1} \cdot \epsilon_{\mathsf{MCA}}(\gamma)$. For $\mathbf{x} \coloneqq (x_1, \ldots, x_s) \in \mathsf{B}$, we fix $T_{\mathbf{x}}$ to be a witness subset appearing in the definition of B. Define the quotient $Q_{\mathbf{x}} \coloneqq \Sigma^{T_{\mathbf{x}}}/\mathcal{C}|_{T_{\mathbf{x}}}$. For $\mathbf{u} \in \Sigma^n$, denote by $[\mathbf{u}]$ the coset of $\mathbf{u}|_{T_{\mathbf{x}}}$ in $Q_{\mathbf{x}}$, and consider the linear map:

$$\mathcal{L} : (\lambda_1, \ldots, \lambda_s) \in \mathbb{F}^s \longmapsto \sum_{j \in [s]} \lambda_j [\mathbf{u}_j] \in Q_{\mathbf{x}} \ .$$

Observe that:

$$\left[\begin{array}{l} (\mathbf{u}_0 + \sum_{i \in [s]} x_i \mathbf{u}_i)|_{T_{\mathbf{x}}} \in \mathcal{C}|_{T_{\mathbf{x}}} \\ \wedge\, \exists\, j \in \{0, \ldots, s\}, \mathbf{u}_j|_{T_{\mathbf{x}}} \notin \mathcal{C}|_{T_{\mathbf{x}}} \end{array} \right] \implies [\exists\, j \in \{1, \ldots, s\}, \mathbf{u}_j|_{T_{\mathbf{x}}} \notin \mathcal{C}|_{T_{\mathbf{x}}}] \ ,$$

if not, we can deduce that $\mathbf{u}_0|_{T_{\mathbf{x}}} = \left(\mathbf{u}_0 + \sum_{i \in [s]} x_i \mathbf{u}_i - \sum_{i \in [s]} x_i \mathbf{u}_i\right)\Big|_{T_{\mathbf{x}}} \in \mathcal{C}|_{T_{\mathbf{x}}}$, a contradiction. Since there exists $j \in [s]$ such that $\mathbf{u}_j|_{T_{\mathbf{x}}} \notin \mathcal{C}|_{T_{\mathbf{x}}}$, which implies $[\mathbf{u}_j] \neq 0$, the image of $\mathcal{L}$ is at least one-dimensional. Therefore, we find that $\Pr_{(\lambda_1, \ldots, \lambda_s)}[\sum_{j \in [s]} \lambda_j [\mathbf{u}_j] = 0] \leq 1/|\mathbb{F}|$. Averaging over all $\mathbf{x}$, we obtain that:

$$\mathop{\mathbb{E}}_{\lambda_1, \ldots, \lambda_s} \left[\Pr_{\mathbf{x} \in \mathbb{F}^s} \left[\sum_{j \in [s]} \lambda_j [\mathbf{u}_j] = 0 \ \wedge\ \mathbf{x} \in \mathsf{B} \right] \right] \leq \frac{1}{|\mathbb{F}|} \cdot \frac{|\mathsf{B}|}{|\mathbb{F}|^s} \ .$$

Therefore there exists $\lambda^\star \coloneqq (\lambda_1^\star, \ldots, \lambda_s^\star) \in \mathbb{F}^s$ such that:

$$\Pr_{\mathbf{x} \in \mathbb{F}^s} \left[\sum_{j \in [s]} \lambda_j^\star [\mathbf{u}_j] = 0 \ \wedge\ \mathbf{x} \in \mathsf{B} \right] \leq \frac{1}{|\mathbb{F}|} \cdot \frac{|\mathsf{B}|}{|\mathbb{F}|^s} \ .$$

Set $\tilde{\mathbf{u}} \coloneqq \sum_{j \in [s]} \lambda_j^\star \mathbf{u}_j$, we have just found that if $\mathsf{B}' \coloneqq \{\mathbf{x} \in \mathsf{B} : \tilde{\mathbf{u}}|_{T_{\mathbf{x}}} \notin \mathcal{C}|_{T_{\mathbf{x}}}\}$, then $|\mathsf{B}'| \geq \left(1 - \frac{1}{|\mathbb{F}|}\right) \cdot |\mathsf{B}|$. For $\mathbf{v} \in \mathbb{F}^s$, define the following line in seed space:

$$\mathbf{x}_{\mathbf{v}}(t) \coloneqq \mathbf{v} + t \cdot \lambda^\star \ , \text{ for } t \in \mathbb{F} \ .$$

Note that:

$$\mathop{\mathbb{E}}_{\mathbf{v}\in\mathbb{F}^s}\left[\Pr_{t\in\mathbb{F}}[\mathbf{x}_{\mathbf{v}}(t)\in\mathsf{B}']\right]=\Pr_{\mathbf{x}\in\mathbb{F}^s}[\mathbf{x}\in\mathsf{B}']\ .$$

This implies that there exists $\mathbf{v}^\star\in\mathbb{F}^s$ such that:

$$\Pr_{t\in\mathbb{F}}[\mathbf{x}_{\mathbf{v}^\star}(t)\in\mathsf{B}']\geq\Pr_{\mathbf{x}\in\mathbb{F}^s}[\mathbf{x}\in\mathsf{B}']>\epsilon_{\mathsf{MCA}}(\gamma)\ .$$

Define another word in Σ^n as:

$$\mathbf{u}'\coloneqq\mathbf{u}_0+\sum_{j\in[s]}\mathbf{v}^\star_j\mathbf{u}_j\ .$$

Note that for any $t\in\mathbb{F}$, if we set $\mathbf{x}_{\mathbf{v}^\star}(t)=(x_1(t),\ldots,x_s(t))$, then:

$$\mathbf{u}_0+\sum_{j\in[s]}x_j(t)\mathbf{u}_j=\mathbf{u}'+t\cdot\tilde{\mathbf{u}}\ .$$

Fixing any $t\in\mathbb{F}$ such that $\mathbf{x}_{\mathbf{v}^\star}(t)\in\mathsf{B}'$, we have by definition of B' that there exists $T_t\subseteq[n]$ with $|T_t|\geq n\cdot(1-\gamma)$ such that:

$$(\mathbf{u}'+t\cdot\tilde{\mathbf{u}})|_{T_t}\in\mathcal{C}|_{T_t}\quad\text{and}\quad\tilde{\mathbf{u}}|_{T_t}\notin\mathcal{C}|_{T_t}\ .$$

We find that:

$$\Pr_{t\in\mathbb{F}}\left[\begin{array}{l}\exists\, T\subseteq[n]\text{ s.t.}\\ \quad|T|\geq n\cdot(1-\gamma)\\ \quad\wedge\,(\mathbf{u}'+t\cdot\tilde{\mathbf{u}})|_T\in\mathcal{C}|_T\\ \quad\wedge\,\tilde{\mathbf{u}}|_T\notin\mathcal{C}|_T\end{array}\right]\geq\Pr_{t\in\mathbb{F}}[\mathbf{x}_{\mathbf{v}^\star}(t)\in\mathsf{B}']>\epsilon_{\mathsf{MCA}}(\gamma)\ .$$

This contradicts the fact that the affine line generator $\mathbb{F}\to\mathbb{F}^2$, $x\mapsto(1,x)$ has MCA for $\mathcal{C}$ with error ϵ_{MCA}. $\square$

8 Applications of Theorem 6.1

We present several applications of Theorem 6.1. First, we show that our result implies that all polynomial generators have MCA for any linear code. We then specialize to the case of multilinear generators, which contains the affine space generator and the generator in [BMMS25b]. Finally, we apply our results to the generators in [AGHP92; MZ25].

Definition 8.1 (MCA error for univariate powers)**.** *Let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code with relative distance $\delta_{\mathcal{C}}$, and set $\rho_{\mathcal{C}} \coloneqq 1 - \delta_{\mathcal{C}}$. For every $d, m \in \mathbb{N}$ and $\eta \in (0,1)$, define the error function $\xi_{\mathcal{C},d,m} \colon [0,1] \to [0,1]$ as*

$$\xi_{\mathcal{C},d,m}(\gamma) \coloneqq \begin{cases} \max\{n \cdot \gamma, 1\} \cdot \frac{d}{m} & \text{if } \gamma < \delta_{\mathcal{C}}/(d+2) \\ \frac{n \cdot \gamma_d}{\eta} \cdot \frac{d}{m} + \max\left\{ \frac{2 \cdot d}{\eta \cdot ((\rho_{\mathcal{C}}+\eta)^{1/(d+2)} - (\rho_{\mathcal{C}}+\eta)^{1/(d+1)}) \cdot m}, \frac{(d+1)\cdot(d+2)}{\eta \cdot m} \right\} & \text{if } \gamma \leq 1 - (\rho_{\mathcal{C}}+\eta)^{1/(d+2)} \\ 1 & \text{otherwise} \end{cases} .$$

Here, $\gamma_d \coloneqq 1 - (\rho_{\mathcal{C}} + \eta)^{\frac{1}{d+1}}$.

8.1 Polynomial generators

From Theorem 6.1, we are able to obtain MCA for all polynomial generators.

Theorem 8.2 (MCA for polynomial generators)**.** *Fix $\ell \geq 2$. Let $G \colon S_1 \times \cdots \times S_s \subseteq \mathbb{F}^s \to \mathbb{F}^\ell$ be a polynomial generator where $G(\mathbf{x}) \coloneqq (P_j(\mathbf{x}))_{j \in [\ell]}$. Let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code. For every $i \in [s]$, let $d_i \coloneqq \max_{j \in [\ell]}\{\deg_{\mathtt{X}_i}(P_j)\}$. Assume that $|S_i| \geq d_i + 1$ for every $i \in [s]$. Then G has MCA for $\mathcal{C}$ with error*

$$\epsilon_{\mathsf{MCA}}(\gamma) \coloneqq \sum_{i \in [s]} \xi_{\mathcal{C},d_i,|S_i|}(\gamma) \ ,$$

where $\xi_{\mathcal{C},d_i,|S_i|}$ is defined in Definition 8.1.

Proof. Consider the generator $G_d \colon S \subseteq \mathbb{F} \to \mathbb{F}^{d+1}$ where $G(x) = (1, x, \ldots, x^d)$ for $d \geq 1$, and $|S| \geq d+1$. Note that its corresponding code $\mathcal{C}_{G_d}$ (see Definition 3.12) is the Reed–Solomon code of length $|S|$, degree d, and dimension $d+1$, which is an MDS code (see, e.g., [GRS23]). By Theorem 6.1, G_d has mutual correlated agreement for any $\mathbb{F}$-linear code $\mathcal{C}$ with error $\xi_{\mathcal{C},d,|S|}(\gamma)$ as defined in Definition 8.1.

Let $k \coloneqq \prod_{i \in [s]}(d_i + 1)$. Since the entries in G are linear combinations of monomials of individual degree in $\mathtt{X}_i$ at most d_i, there exists a matrix $\mathbf{A} \in \mathbb{F}^{\ell \times k}$ such that $G^\intercal = \mathbf{A} \cdot \widetilde{G}^\intercal$, where

$$\widetilde{G}(x_1, \ldots, x_s) \coloneqq \bigotimes_{i \in [s]} (1, x_i, \ldots, x_i^{d_i}) \ .$$

Further, since G is a multivariate generator, $(P_j(\mathbf{x}))_{j \in [\ell]}$ are linearly independent over $\mathbb{F}$ (see Definition 3.19), which implies that $k \geq \ell$ and $\mathbf{A}$ has rank ℓ (any nontrivial linear dependence relation between the rows of $\mathbf{A}$ gives a nontrivial linear dependence between the entries of G). Therefore, $G = \widetilde{G} \cdot \mathbf{A}^\intercal$ and $\mathbf{A}^\intercal$ has a left inverse. Applying Lemmas 4.1 and 4.4 concludes our proof. □

8.2 Multilinear generators

Applying Theorem 6.1 to multilinear generators leads to a particularly simple MCA error, because the individual degrees in each variable are equal to 1. Note that the affine space generator pertains to this class. Additionally, this allows us to obtain a better MCA error for the multilinear equality generator in [BMMS25b].

Corollary 8.3. *Let $G\colon S_1 \times \cdots \times S_s \subseteq \mathbb{F}^s \to \mathbb{F}^\ell$ be a multilinear polynomial generator, i.e., $G(\mathbf{x}) \coloneqq (P_j(\mathbf{x}))_{j\in[\ell]}$ where $P_1, \ldots, P_\ell \in \mathbb{F}[\mathtt{X}_1, \ldots, \mathtt{X}_s]$ are multilinear polynomials. Let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code. Then G has MCA for $\mathcal{C}$ with error $\epsilon_{\mathsf{MCA}}(\gamma) \coloneqq \sum_{i\in[s]} \xi_{\mathcal{C},1,|S_i|}(\gamma)$.*

Proof. This follows from applying Theorem 8.2, taking into account that the individual degrees in each variable is exactly 1. □

Corollary 8.4 (MCA for affine spaces)**.** *Let $G\colon \mathbb{F}^s \to \mathbb{F}^{s+1}$ be defined as $G(x_1, \ldots, x_s) \coloneqq (1, x_1, \ldots, x_s)$. Let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code. Then G has mutual correlated agreement for $\mathcal{C}$ with error $\epsilon_{\mathsf{MCA}} = \epsilon_{\mathsf{MCA}}(\gamma)$ such that:*

$$\epsilon_{\mathsf{MCA}}(\gamma) \coloneqq \begin{cases} \max\{n\cdot\gamma, 1\} \cdot \frac{1}{|\mathbb{F}|} & \text{if } \gamma < \delta_{\mathcal{C}}/3 \\ \left(1 - \frac{1}{|\mathbb{F}|}\right)^{-1} \cdot \left(\frac{n\cdot\gamma_1}{\eta} \cdot \frac{1}{|\mathbb{F}|} + \max\left\{\frac{2}{\eta\cdot((\rho_{\mathcal{C}}+\eta)^{1/3} - (\rho_{\mathcal{C}}+\eta)^{1/2})\cdot|\mathbb{F}|}, \frac{6}{\eta\cdot|\mathbb{F}|}\right\}\right) & \text{if } \gamma \le 1 - (\rho_{\mathcal{C}}+\eta)^{1/3} \\ 1 & \text{otherwise} \end{cases} .$$

Here $\gamma_1 \coloneqq 1 - (\rho_{\mathcal{C}} + \eta)^{1/2}$.

Proof. In the unique decoding setting, it is shown in [ACFY25, Lemma 4.10] that correlated agreement can be generically lifted to MCA. We prove that the correlated agreement error of the affine space generator up to $\gamma \le \delta_{\mathcal{C}}/3$ is $\max\{n\cdot\gamma, 1\} \cdot \frac{1}{|\mathbb{F}|}$ in Lemmas C.3 and C.5.

For the list-decoding setting, we use Lemma 7.1, noting that the affine line generator has MCA error $\xi_{\mathcal{C},1,|\mathbb{F}|}(\gamma)$. □

Remark 8.5. Note that this corollary immediately gives the same MCA error for the linear space generator $G\colon \mathbb{F}^s \to \mathbb{F}^s$ defined as $G(\mathbf{x}) \coloneqq \mathbf{x}$, by applying Corollary 4.2.

Corollary 8.6. *Let $\widehat{\mathsf{eq}}$ be the multilinear equality polynomial in $2s$ variables:*

$$\widehat{\mathsf{eq}}(\mathtt{X}_1, \ldots, \mathtt{X}_s, \mathtt{Y}_1, \ldots, \mathtt{Y}_s) \coloneqq \prod_{i\in[s]} (\mathtt{X}_i \cdot \mathtt{Y}_i + (1 - \mathtt{X}_i)(1 - \mathtt{Y}_i)) \ .$$

Let $G\colon \mathbb{F}^s \to \mathbb{F}^{2^s}$ be the multilinear equality generator defined (up to reordering) as $G(\mathbf{x}) \coloneqq (\widehat{\mathsf{eq}}(\mathbf{v}, \mathbf{x}))_{\mathbf{v}\in\{0,1\}^s}$. Let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code. Then G has MCA for $\mathcal{C}$ with error $\epsilon_{\mathsf{MCA}}(\gamma) \coloneqq s \cdot \xi_{\mathcal{C},1,|\mathbb{F}|}(\gamma)$.

8.3 Univariate powers on Reed–Muller codes

We apply our results to the univariate powers generator for the Reed–Muller codes as in [MZ25].

Corollary 8.7. *Fix a finite field extension $\mathbb{B} \subseteq \mathbb{F}$ and $d \in \mathbb{N}$. Let $\mathcal{C} \coloneqq \mathsf{RM}[\mathbb{F}, \mathbb{B}^2, k]$. Set $\rho_{\mathcal{C}} \coloneqq 1 - \delta_{\mathcal{C}} = k/|\mathbb{B}|$. Consider the univariate powers generator $G\colon \mathbb{F} \to \mathbb{F}^{d+1}$ defined as $G(x) \coloneqq (1, x, \ldots, x^d)$. Then, for every $\eta \in (0, 1)$, G has MCA for $\mathcal{C}$ with error*

$$\epsilon_{\mathsf{MCA}}(\gamma) \coloneqq \begin{cases} \max\{|\mathbb{B}|^2\cdot\gamma, 1\} \cdot \frac{d}{|\mathbb{F}|} & \text{if } \gamma < \delta_{\mathcal{C}}/(d+2) \\ \frac{|\mathbb{B}|^2\cdot\gamma_d}{\eta\cdot|\mathbb{F}|} + \max\left\{\frac{2\cdot d}{\eta\cdot((\rho_{\mathcal{C}}+\eta)^{1/(d+2)} - (\rho_{\mathcal{C}}+\eta)^{1/(d+1)})\cdot|\mathbb{F}|}, \frac{(d+1)(d+2)}{\eta\cdot|\mathbb{F}|}\right\} & \text{if } \gamma \le 1 - (\rho_{\mathcal{C}}+\eta)^{1/(d+2)} \\ 1 & \text{otherwise} \end{cases} .$$

Here, $\gamma_d \coloneqq 1 - (\rho_{\mathcal{C}} + \eta)^{\frac{1}{d+1}}$.

Proof. Note that the length of $\mathcal{C}$ is $|\mathbb{B}|^2$, and since $k \leq |\mathbb{B}| - 1$, its relative distance is $\frac{|\mathbb{B}|(|\mathbb{B}|-k)}{|\mathbb{B}|^2} = 1 - \frac{k}{|\mathbb{B}|}$. The result then follows from Theorem 8.2. □

8.4 A non-polynomial generator with MCA

We study a generator in [AGHP92] that is *not a polynomial generator*. Let q be a prime power, $k, \ell \in \mathbb{N}$, and $\phi \colon \mathbb{F}_{q^k} \to \mathbb{F}_q^k$ an isomorphism of vector spaces over $\mathbb{F}$. Define $G \colon \mathbb{F}_{q^k} \times \mathbb{F}_q^k \to \mathbb{F}_q^\ell$ such that

$$G(x, \mathbf{v}) \coloneqq (\phi(x^i) \cdot \mathbf{v}^\mathsf{T})_{i \in [\ell]} \ . \tag{9}$$

As proven in [AGHP92], G is a zero-evading generator with error $\epsilon_{\mathsf{ZE}} = \frac{\ell-1}{q^k} + \frac{1}{q}$.

Corollary 8.8. *The generator in Equation 9 has MCA for any $\mathbb{F}$-linear code $\mathcal{C} \subseteq \Sigma^n$ with error*

$$\epsilon_{\mathsf{MCA}} + \xi_{\mathcal{C}, d, |\mathbb{F}_{q^k}|}(\gamma) \ ,$$

where ϵ_{MCA} is the error defined as follows:

$$\epsilon_{\mathsf{MCA}}(\gamma) \coloneqq \begin{cases} \max\{n \cdot \gamma, 1\} \cdot \frac{1}{|\mathbb{F}|} & \text{if } \gamma < \delta_{\mathcal{C}}/3 \\ \left(1 - \frac{1}{|\mathbb{F}|}\right)^{-1} \cdot \left(\frac{n \cdot \gamma_1}{\eta} \cdot \frac{1}{|\mathbb{F}|} + \max\left\{\frac{2}{\eta \cdot ((\rho_{\mathcal{C}}+\eta)^{1/3} - (\rho_{\mathcal{C}}+\eta)^{1/2}) \cdot |\mathbb{F}|}, \frac{6}{\eta \cdot |\mathbb{F}|}\right\}\right) & \text{if } \gamma \leq 1 - (\rho_{\mathcal{C}} + \eta)^{1/3} \\ 1 & \text{otherwise} \end{cases} \ .$$

Proof. This generator is the matrix-vector composition of two generators: the inner one is given by the map $\mathbb{F}_q^k \to \mathbb{F}_q^k$, $\mathbf{v} \mapsto \mathbf{v}$, and the outer one is given by the map $\mathbb{F}_{q^k} \to (\mathbb{F}_{q^k})^\ell$, $x \mapsto (x, x^2, \ldots, x^\ell)$. We conclude by using Corollary 4.2, Lemma 4.5, Theorem 8.2, and Remark 8.5. □

9 Polynomial generators have MCA for RS codes up to list-decoding

We prove that all polynomial generators have MCA for Reed–Solomon codes up to the list-decoding bound.

Definition 9.1. *Fix a field $\mathbb{F}$ and $n, k \in \mathbb{N}$. Consider $\mathcal{C} \coloneqq \mathsf{RS}[\mathbb{F}, D, k]$, $d \in \mathbb{N}$, and $m \geq 3$. We define $\epsilon_{\mathsf{MCA},\mathsf{RS},d} \colon [0,1] \to [0,1]$ as follows:*

$$\epsilon_{\mathsf{MCA},\mathsf{RS},d}(\gamma) \coloneqq \begin{cases} \frac{1}{|\mathbb{F}|} \cdot \frac{(m+1/2)^7}{3\rho^{3/2}} \cdot d \cdot n^2 & \text{if } \gamma \leq 1 - \left(1 + \frac{1}{2m}\right) \cdot \sqrt{\rho} \\ 1 & \text{otherwise} \end{cases} .$$

Theorem 9.2. *Let $\mathcal{C} \coloneqq \mathsf{RS}[\mathbb{F}, D, k]$ be a Reed–Solomon code and define $\rho \coloneqq k/n$. Let $G \colon \mathbb{F}^s \to \mathbb{F}^\ell$ be a polynomial generator where $G(\mathbf{x}) \coloneqq (P_j(\mathbf{x}))_{j \in [\ell]}$. For every $i \in [s]$, let $d_i \coloneqq \max_{j \in [\ell]} \{\deg_{\mathtt{X}_i}(P_j)\}$. Then, for every $m \geq 3$, G has MCA for $\mathcal{C}$ with error $\sum_{i \in [s]} \epsilon_{\mathsf{MCA},\mathsf{RS},d_i}$ (as defined in Definition 9.1).*

We proceed in two steps: first in Lemma 9.3 we establish that the univariate powers generator has MCA for Reed–Solomon codes up to the list-decoding bound; second we rely on our MCA-preserving transformations to extend the result to all polynomial generators. Lemma 9.3 modifies a proof in [Hab25] that establishes that the affine line generator has MCA for Reed–Solomon codes up to the list-decoding bound (in turn by extending the correlated agreement analysis for the affine line generator in [BCIKS20]); our extension for the univariate powers generator relies on results about the zero-evading error of generators (see Lemma 5.3 and Remark 5.4).

Lemma 9.3. *Fix a field $\mathbb{F}$ and $n, k \in \mathbb{N}$. Consider $\mathcal{C} \coloneqq \mathsf{RS}[\mathbb{F}, D, k]$, $d \in \mathbb{N}$, and $m \geq 3$. Then, the generator $G_d \colon \mathbb{F} \to \mathbb{F}^{d+1}$ where $G_d(x) \coloneqq (1, x, \ldots, x^d)$ has MCA for $\mathsf{RS}[\mathbb{F}, D, k]$ with error $\epsilon_{\mathsf{MCA},\mathsf{RS},d}$.*

Proof of Theorem 9.2. Consider the generator $G_d \colon \mathbb{F} \to \mathbb{F}^{d+1}$ where $G(x) = (1, x, \ldots, x^d)$ for $d \geq 1$. By Lemma 9.3, G_d has mutual correlated agreement for any linear code $\mathcal{C}$ with error $\epsilon_{\mathsf{MCA},\mathsf{RS},d}$ as defined in Definition 9.1.

Let $k \coloneqq \prod_{i \in [s]} (d_i + 1)$. Since the entries in G are linear combinations of monomials of individual degree in $\mathtt{X}_i$ at most d_i, there exists a matrix $\mathbf{A} \in \mathbb{F}^{\ell \times k}$ such that $G^\intercal = \mathbf{A} \cdot \widetilde{G}^\intercal$, where

$$\widetilde{G}(x_1, \ldots, x_s) \coloneqq \bigotimes_{i \in [s]} (1, x_i, \ldots, x_i^{d_i}) \ .$$

Further, since G is a multivariate generator, $(P_j(\mathbf{x}))_{j \in [\ell]}$ are linearly independent over $\mathbb{F}$ (see Definition 3.19), which implies that $k \geq \ell$ and $\mathbf{A}$ has rank ℓ (any nontrivial linear dependence relation between the rows of A gives a nontrivial linear dependence between the entries of G). Therefore, $G = \widetilde{G} \cdot \mathbf{A}^\intercal$ since $\mathbf{A}^\intercal$ has a left inverse. Applying Lemmas 4.1 and 4.4 concludes our proof. □

Proof of Lemma 9.3. Let $\theta \coloneqq \left(1 + \frac{1}{2m}\right) \cdot \sqrt{\rho}$. To prove the theorem, it suffices to show that for every $\mathbf{u}_0, \ldots, \mathbf{u}_d \in \mathbb{F}^D$, if we define:

$$B \coloneqq \left\{ z \in \mathbb{F} : \begin{array}{l} \exists\, A \subseteq D \text{ s.t.} \\ \quad |A| \geq n \cdot \theta \\ \quad \wedge \ \left(\sum_{j=0}^{d} z^j \cdot \mathbf{u}_j \right) \Big|_A \in \mathcal{C}|_A \\ \quad \wedge \exists\, j \in \{0, \ldots, d\}, \mathbf{u}_j|_A \notin \mathcal{C}|_A \end{array} \right\} ,$$

then:

$$|B| \leq \frac{(m+1/2)^7}{3\rho^{3/2}} \cdot d \cdot n^2 \ .$$

Let $Q(X,Y,Z)$ be the Guruswami–Sudan polynomial constructed in Step 1 of the proof of [BCIKS20, Theorem 5.1], but adapted to the parametrized curve of degree d as explained in [BCIKS20, Section 6.2]. As remarked there, the degrees D_X, D_Y of Q in X, Y (respectively) are unchanged. Only the degree D_Z of Q in the variable Z is multiplied by d, so that:

$$\begin{aligned} D_X &< (m+1/2)\sqrt{\rho}n \\ D_Y &< \frac{m+1/2}{\sqrt{\rho}} \\ D_{YZ} &< \frac{(m+1/2)^3}{6\sqrt{\rho}}nd \ . \end{aligned}$$

We recall that D_{YZ} is the total degree in the variables Y, Z of the polynomial Q. Consider the decomposition:

$$Q(X,Y,Z) = C(X,Z) \cdot \prod_{\mu} R(X, Y^{p^{f_\mu}}, Z)^{e_\mu}$$

into separable irreducibles, where $p \coloneqq \mathrm{char}(\mathbb{F})$, $f_\mu \geq 0$ and $e_\mu \geq 1$. As in [BCIKS20], set:

$$\mathsf{disc}^*_Y(Q)(X,Z) \coloneqq \prod_{\mu} \mathsf{disc}_Y(R_\mu(X,Y,Z)) \ .$$

It follows from our initial remarks and the proof of [BCIKS20, Claim 5.6] that

$$\deg_X(\mathsf{disc}^*_Y(Q)) < (m+1/2)^2 n \ .$$

We can therefore assume that $|\mathbb{F}| > (m+1/2)^2 n$ (the result of the theorem follows trivially otherwise), and conclude similarly that there must exist $x_0 \in \mathbb{F}$ such that $\mathsf{disc}^*_Y(Q)(x_0, Z)$ is not identically zero, in particular for every μ we have that $\mathsf{disc}_Y(R_\mu(x_0, Y, Z)) \not\equiv 0$. We can further consider the decomposition $R_\mu(x_0, Y, Z) = C_\mu(Z) \cdot \prod_\nu H_{\mu,\nu}(Y,Z)$ into irreducibles, and define:

$$B_{\mu,\nu} \coloneqq \left\{ z \in \mathbb{F} : \begin{array}{l} \exists\, A \subseteq D \text{ s.t.} \\ \quad |A| \geq n \cdot \theta \\ \quad \wedge \exists\, p \in \mathcal{C},\ \left(\sum_{j=0}^{d} z^j \cdot \mathbf{u}_j\right)\Big|_A = p|_A \\ \quad \wedge\, R_\mu(X, p(X), z) \equiv 0 \\ \quad \wedge\, H_{\mu,\nu}(p(x_0), z) = 0 \\ \quad \wedge \exists\, j \in \{0,\ldots,d\}, \mathbf{u}_j|_A \notin \mathcal{C}|_A \end{array} \right\} \ .$$

The number of possible indices (μ, ν) is bounded above by D_Y, this can be seen by writing the factorization:

$$Q(x_0, Y, Z) = C(x_0, Z) \cdot \prod_{\mu} C_\mu(Z) \cdot \prod_{\mu,\nu} H_{\mu,\nu}(Y^{p^{f_\mu}}, Z)^{e_\mu} \ ,$$

and noting that the degree of $H_{\mu,\nu}$ in Y is positive. Hence, in order to prove the theorem it suffices to show that for every possible (μ, ν):

$$|B_{\mu,\nu}| \leq \frac{(m+1/2)^6}{3\rho} \cdot d \cdot n^2 \ .$$

Assume for a contradiction that this does not hold, so that $|B_{\mu,\nu}| > \frac{(m+1/2)^6}{3\rho} \cdot d \cdot n^2$ for some (μ, ν). For every $z \in B_{\mu,\nu}$, set $p_z \in \mathcal{C}$ satisfying the conditions in the definition of $B_{\mu,\nu}$. Then R_μ is a useful factor in the sense of [BCIKS20], since the irreducible factor $H_{\mu,\nu}$ is such that the set:

$$S_{x_0,R_\mu,H_{\mu,\nu}} \coloneqq \{z \in \mathbb{F} : R_\mu(X, p_z(X), z) \equiv 0 \ \wedge \ H_{\mu,\nu}(p(x_0), z) = 0\}$$

has size:

$$|S_{x_0,R_\mu,H_{\mu,\nu}}| \geq |B_{\mu,\nu}| > \frac{(m+1/2)^6}{3\rho} \cdot d \cdot n^2 > 2D_Y^2 D_X D_{YZ} \ .$$

Following the proof of [BCIKS20, Theorem 5.1] (and its extension in [BCIKS20, Section 6.2]), we conclude similarly that there exist $\mathbf{c}_0, \ldots, \mathbf{c}_d \in \mathcal{C}$ such that for every $z \in S_{x_0,R_\mu,H_{\mu,\nu}}$, $p_z(X) \equiv \sum_{j=0}^{d} z^j \cdot \mathbf{c}_j$, and further:

$$R_\mu(X, Y, Z) \equiv Y - \sum_{j=0}^{d} Z^j \cdot \mathbf{c}_j(X) \ .$$

In particular, the definition of $B_{\mu,\nu}$ can be modified to:

$$B_{\mu,\nu} = \left\{ z \in \mathbb{F} : \begin{array}{l} \exists\, A \subseteq D \text{ s.t.} \\ \quad |A| \geq n \cdot \theta \\ \quad \wedge \ \left(\sum_{j=0}^{d} z^j \cdot \mathbf{u}_j\right)\Big|_A = \left(\sum_{j=0}^{d} z^j \cdot \mathbf{c}_j\right)\Big|_A \\ \quad \wedge \exists\, j \in \{0, \ldots, d\}, \mathbf{u}_j|_A \notin \mathcal{C}|_A \end{array} \right\} \ .$$

This means that any $z \in B_{\mu,\nu}$ must necessarily improve the agreement between $\sum_{j=0}^{d} z^j \cdot \mathbf{u}_j$ and $\sum_{j=0}^{d} z^j \cdot \mathbf{c}_j$ beyond the correlated agreement set $T \coloneqq \{i \in D : \bigwedge_j \mathbf{u}_j[i] = \mathbf{c}_j[i]\}$. But by Remark 5.4, we see that because $z \mapsto (1, z, \ldots, z^d)$ has zero-evading error $d/|\mathbb{F}|$, the set of $z \in B_{\mu,\nu}$ for which this happens has size bounded above by $d \cdot |D \setminus T| \leq d \cdot n$. Finally, note that:

$$\frac{(m+1/2)^6}{3\rho} \cdot d \cdot n^2 > d \cdot n \ ,$$

a contradiction. □

10 Operations on codes

We show how, for a given arbitrary generator, the MCA error is affected by common operations on codes: *interleaving* in Section 10.1, *direct sum* in Section 10.2, and *concatenation* in Section 10.3.

10.1 Interleaving

Lemma 10.1. *Let $G\colon S \to \mathbb{F}^\ell$ be a generator that has MCA for an $\mathbb{F}$-linear code $\mathcal{C} \subseteq \Sigma^n$ with error ϵ_{MCA}. For every $k \in \mathbb{N}$, G has MCA for the k-interleaving $\mathcal{C}^k$ with error $k \cdot \epsilon_{\mathsf{MCA}}$.*

Proof. For $j \in [\ell]$, let $\mathbf{U}_j \coloneqq \mathsf{Mat}(\mathbf{u}_{j,1}, \ldots, \mathbf{u}_{j,k}) \in \Sigma^{k \times n}$. Note that for $\mathbf{x} \in S$:

$$\sum_{j \in [\ell]} G(\mathbf{x})[j] \cdot \mathbf{U}_j = \mathsf{Mat}(\sum_{j \in [\ell]} G(\mathbf{x})[j] \cdot \mathbf{u}_{j,1}, \ldots, \sum_{j \in [\ell]} G(\mathbf{x})[j] \cdot \mathbf{u}_{j,k}) \ .$$

For every $\gamma \in [0,1]$,

$$\begin{aligned}
&\Pr_{\mathbf{x} \in S}\left[\begin{array}{l} \exists\, T \subseteq [n], \\ \quad |T| \geq n \cdot (1-\gamma) \\ \quad \wedge \ \left(\sum_{j \in [\ell]} G(\mathbf{x})[j] \cdot \mathbf{U}_j\right)|_T \in (\mathcal{C}^k)|_T \\ \quad \wedge\, \exists\, j \in [\ell],\ \mathbf{U}_j|_T \notin (\mathcal{C}^k)|_T \end{array}\right] \\
&= \Pr_{\mathbf{x} \in S}\left[\begin{array}{l} \exists\, T \subseteq [n], \\ \quad |T| \geq n \cdot (1-\gamma) \\ \quad \wedge\, \forall\, j' \in [k],\ \left(\sum_{j \in [\ell]} G(\mathbf{x})[j] \cdot \mathbf{u}_{j,j'}\right)|_T \in \mathcal{C}|_T \\ \quad \wedge\, \exists\, (j, j_0) \in [\ell] \times [k],\ \mathbf{u}_{j,j_0}|_T \notin \mathcal{C}|_T \end{array}\right] \\
&\leq \sum_{j_0 \in [k]} \Pr_{\mathbf{x} \in S}\left[\begin{array}{l} \exists\, T \subseteq [n], \\ \quad |T| \geq n \cdot (1-\gamma) \\ \quad \wedge \ \left(\sum_{j \in [\ell]} G(\mathbf{x})[j] \cdot \mathbf{u}_{j,j_0}\right)|_T \in \mathcal{C}|_T \\ \quad \wedge\, \exists\, j \in [\ell],\ \mathbf{u}_{j,j_0}|_T \notin \mathcal{C}|_T \end{array}\right] \\
&\leq k \cdot \epsilon_{\mathsf{MCA}}(\gamma) \ .
\end{aligned}$$

□

10.2 Direct sum

Definition 10.2. *Let $\mathcal{C} \subseteq \Sigma^n$, $\mathcal{C}' \subseteq \Sigma^{n'}$ be $\mathbb{F}$-linear codes. The* **direct sum code** *$\mathcal{C} \oplus \mathcal{C}' \subseteq \Sigma^{n+n'}$ is*

$$\mathcal{C} \oplus \mathcal{C}' \coloneqq \{(\mathbf{c}_1, \mathbf{c}_2) : \mathbf{c}_1 \in \mathcal{C}, \mathbf{c}_2 \in \mathcal{C}'\} \ .$$

Lemma 10.3. *Let $G\colon S \to \mathbb{F}^\ell$ be a generator that has MCA for an $\mathbb{F}$-linear code $\mathcal{C} \subseteq \Sigma^n$ (resp. $\mathcal{C}' \subseteq \Sigma^{n'}$) with error ϵ_{MCA} (resp. ϵ'_{MCA}). Then G has MCA for $\mathcal{C} \oplus \mathcal{C}'$ with error $\epsilon_{\mathsf{MCA}} + \epsilon'_{\mathsf{MCA}}$.*

Proof. Let $T \subseteq [n+n']$ be such that $|T| \geq (n+n') \cdot (1-\gamma)$ for some $\gamma \in [0,1]$. Letting $T_1 \coloneqq T \cap [n]$ and $T_2 \coloneqq T \cap \{n+1, \ldots, n+n'\}$, it follows that either $|T_1| \geq n \cdot (1-\gamma)$ or $|T_2| \geq n' \cdot (1-\gamma)$. For any word $\mathbf{u} \in \mathbb{F}^{n+n'}$, define $\mathbf{u}^{(1)} \coloneqq \mathbf{u}|_{[n]}$ and $\mathbf{u}^{(2)} \coloneqq \mathbf{u}|_{\{n+1,\ldots,n+n'\}}$.

We compute, for any words $\mathbf{u}_1, \ldots, \mathbf{u}_\ell \in \mathbb{F}^{n+n'}$, and any $\gamma \in [0,1]$:

$$\begin{aligned}
&\Pr_{\mathbf{x}\in S}\left[\begin{array}{l} \exists\, T \subseteq [n+n'], \\ \quad |T| \geq (n+n')\cdot(1-\gamma) \\ \quad \wedge\ \left(\sum_{j\in[\ell]} G(\mathbf{x})[j]\cdot \mathbf{u}_j\right)|_T \in (\mathcal{C}\oplus\mathcal{C}')|_T \\ \quad \wedge \exists j \in [\ell],\ \mathbf{u}_j|_T \notin (\mathcal{C}\oplus\mathcal{C}')|_T \end{array}\right] \\
&\leq \Pr_{\mathbf{x}\in S}\left[\begin{array}{l} \exists\, T_1 \subseteq [n], \\ \quad |T_1| \geq n\cdot(1-\gamma) \\ \quad \wedge\ \left(\sum_{j\in[\ell]} G(\mathbf{x})[j]\cdot \mathbf{u}_j^{(1)}\right)|_{T_1} \in \mathcal{C}|_{T_1} \\ \quad \wedge \exists j \in [\ell],\ \mathbf{u}_j^{(1)}|_{T_1} \notin \mathcal{C}|_{T_1} \end{array}\right] \\
&+ \Pr_{\mathbf{x}\in S}\left[\begin{array}{l} \exists\, T_2 \subseteq [n'], \\ \quad |T_2| \geq n'\cdot(1-\gamma) \\ \quad \wedge\ \left(\sum_{j\in[\ell]} G(\mathbf{x})[j]\cdot \mathbf{u}_j^{(2)}\right)|_{T_2} \in \mathcal{C}'|_{T_2} \\ \quad \wedge \exists j \in [\ell],\ \mathbf{u}_j^{(2)}|_{T_2} \notin \mathcal{C}'|_{T_2} \end{array}\right] \\
&\leq \epsilon_{\mathsf{MCA}}(\gamma) + \epsilon'_{\mathsf{MCA}}(\gamma)\ .
\end{aligned}$$

□

10.3 Concatenation

Definition 10.4. *Let $\mathcal{C}_{\mathsf{out}}$ be a code over alphabet Σ_{out}, with blocklength n_{out}, and injective encoding function $\mathsf{Enc}_{\mathsf{out}}\colon D_{\mathsf{out}} \to \Sigma_{\mathsf{out}}^{n_{\mathsf{out}}}$. Let $\mathcal{C}_{\mathsf{in}}$ be a code over alphabet Σ_{in}, with blocklength n_{in}, and (injective) encoding function $\mathsf{Enc}_{\mathsf{in}}\colon D_{\mathsf{in}} \to \Sigma_{\mathsf{in}}^{n_{\mathsf{in}}}$. Suppose that $|\Sigma_{\mathsf{out}}| = |D_{\mathsf{in}}|$, and let $\phi\colon \Sigma_{\mathsf{out}} \to D_{\mathsf{in}}$ be a bijection. The* **concatenation of $\mathcal{C}_{\mathsf{out}}$ with $\mathcal{C}_{\mathsf{in}}$** *is the code $\mathcal{C}_{\mathsf{concat}}$ whose encoding function is defined as:*

$$D_{\mathsf{out}} \to \Sigma_{\mathsf{in}}^{n_{\mathsf{in}}n_{\mathsf{out}}}, \quad m \mapsto (\mathsf{Enc}_{\mathsf{in}}(\phi(\mathsf{Enc}_{\mathsf{out}}(m)[i])))_{i\in[n_{\mathsf{out}}]}\ .$$

The notion of MCA deals with linear codes, so we focus on the case where $\mathcal{C}_{\mathsf{out}}$ is $\mathbb{K}$-linear and $\mathcal{C}_{\mathsf{in}}$ is $\mathbb{F}$-linear, for some finite fields $\mathbb{F}, \mathbb{K}$, and the bijection ϕ is an isomorphism of vector spaces. Say that $\Sigma_{\mathsf{out}} = \mathbb{F}_{q^r}^{k'}$ and $\Sigma_{\mathsf{in}} = \mathbb{F}_{p^s}^{k}$, for some primes p, q and integers r, s, k, k'. The equality $|\Sigma_{\mathsf{out}}| = |D_{\mathsf{in}}|$ implies that $p = q$, and $rk' = skk_{\mathsf{in}}$. In the special case $r = s = 1$ and $k = k' = 1$, we have $\mathbb{F} = \mathbb{F}_q$ and $\mathbb{K} = \mathbb{F}_{q^{k_{\mathsf{in}}}}$.

Lemma 10.5. *Let $G\colon S \to \mathbb{F}^\ell$ be a generator that has MCA for $\mathcal{C}_{\mathsf{in}}$ with error ϵ_{MCA}. Then G has MCA for $\mathcal{C}_{\mathsf{concat}}$ with error $n_{\mathsf{out}}\cdot\epsilon_{\mathsf{MCA}}$.*

Proof. Let $\mathbf{u}_1, \ldots, \mathbf{u}_\ell \in \Sigma_{\mathsf{in}}^{n_{\mathsf{in}}n_{\mathsf{out}}}$ be any words, and $\gamma \in [0,1]$. For any $i \in [n_{\mathsf{out}}]$, let $\mathbf{u}_{j,i} \in \Sigma_{\mathsf{in}}^{n_{\mathsf{in}}}$ be the projection of $\mathbf{u}_j$ onto the coordinates $\{i\cdot n_{\mathsf{in}}+1, \ldots, (i+1)\cdot n_{\mathsf{in}}\}$. Let $T \subseteq [n_{\mathsf{in}}n_{\mathsf{out}}]$ be such that $|T| \geq n_{\mathsf{in}}n_{\mathsf{out}}\cdot(1-\gamma)$ and for some $\mathbf{x} \in S$:

$$\left(\sum_{j\in[\ell]} G(\mathbf{x})[j]\cdot\mathbf{u}_j\right)\Bigg|_T \in (\mathcal{C}_{\mathsf{concat}})|_T\ .$$

Consider the projections:

$$T_i \coloneqq T \cap \{i\cdot n_{\mathsf{in}}+1, \ldots, (i+1)\cdot n_{\mathsf{in}}\}\ .$$

Then $T = \sqcup_i T_i$, so that if there exists $j \in [\ell]$ such that $\mathbf{u}_j|_T \notin (\mathcal{C}_{\mathsf{concat}})|_T$, there exists $i \in [n_{\mathsf{out}}]$ such that $\mathbf{u}_{j,i}|_{T_i} \notin \mathcal{C}_{\mathsf{in}}|_{T_i}$. Note that T_i can naturally be considered as a subset of $[n_{\mathsf{in}}]$, by considering the shift by $i \cdot n_{\mathsf{in}}$. Further, for all $i \in [n_{\mathsf{out}}]$:

$$\sum_{j \in [\ell]} G(\mathbf{x})[j] \cdot \mathbf{u}_{j,i}|_{T_i} \in \mathcal{C}_{\mathsf{in}}|_{T_i} \ .$$

Moreover, since $|T| \geq n_{\mathsf{in}} n_{\mathsf{out}} \cdot (1-\gamma)$, there exists $i \in [n_{\mathsf{out}}]$ such that $|T_i| \geq n_{\mathsf{in}} \cdot (1-\gamma)$. We compute:

$$\begin{aligned}
&\Pr_{\mathbf{x} \in S} \left[\begin{array}{l} \exists\, T \subseteq [n_{\mathsf{in}} n_{\mathsf{out}}], \\ \quad |T| \geq n_{\mathsf{in}} n_{\mathsf{out}} \cdot (1-\gamma) \\ \quad \wedge \left(\sum_{j \in [\ell]} G(\mathbf{x})[j] \cdot \mathbf{u}_j \right)|_T \in (\mathcal{C}_{\mathsf{concat}})|_T \\ \quad \wedge\, \exists\, j \in [\ell],\, \mathbf{u}_j|_T \notin (\mathcal{C}_{\mathsf{concat}})|_T \end{array} \right] \\
&\leq \Pr_{\mathbf{x} \in S} \left[\begin{array}{l} \exists i \in [n_{\mathsf{out}}],\, \exists\, T_i \subseteq [n_{\mathsf{in}}], \\ \quad |T_i| \geq n_{\mathsf{in}} \cdot (1-\gamma) \\ \quad \wedge \left(\sum_{j \in [\ell]} G(\mathbf{x})[j] \cdot \mathbf{u}_{j,i} \right)|_{T_i} \in \mathcal{C}_{\mathsf{in}}|_{T_i} \\ \quad \wedge\, \exists\, j \in [\ell],\, \mathbf{u}_{j,i}|_{T_i} \notin \mathcal{C}_{\mathsf{in}}|_{T_i} \end{array} \right] \\
&\leq \sum_{i \in [n_{\mathsf{out}}]} \Pr_{\mathbf{x} \in S} \left[\begin{array}{l} \exists\, T_i \subseteq [n_{\mathsf{in}}], \\ \quad |T_i| \geq n_{\mathsf{in}} \cdot (1-\gamma) \\ \quad \wedge \left(\sum_{j \in [\ell]} G(\mathbf{x})[j] \cdot \mathbf{u}_{j,i} \right)|_{T_i} \in \mathcal{C}_{\mathsf{in}}|_{T_i} \\ \quad \wedge\, \exists\, j \in [\ell],\, \mathbf{u}_{j,i}|_{T_i} \notin \mathcal{C}_{\mathsf{in}}|_{T_i} \end{array} \right] \\
&\leq n_{\mathsf{out}} \cdot \epsilon_{\mathsf{MCA}}(\gamma) \ .
\end{aligned}$$

□

A MCA preserves list-decoding

Definition A.1. *For every code $\mathcal{C} \subseteq \Sigma^n$, word $\mathbf{u} \in \Sigma^n$, and $\gamma \in [0,1]$, we define*

$$\Lambda(\mathcal{C}, \mathbf{u}, \gamma) \coloneqq \{\mathbf{c} \in \mathcal{C} : \delta(\mathbf{u}, \mathbf{c}) \leq \gamma\} \quad .$$

Lemma A.2. *Let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code with relative distance $\delta_{\mathcal{C}}$. Let $G\colon S \to \mathbb{F}^\ell$ have MCA for $\mathcal{C}$ with error ϵ_{MCA}. Then for every $\mathbf{U} \coloneqq \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell) \in \mathbb{F}^{\ell \times n}$ and $\gamma < \delta_{\mathcal{C}}$,*

$$\Pr_{\mathbf{x} \in S}\left[\Lambda(\mathcal{C}, G(\mathbf{x}) \cdot \mathbf{U}, \gamma) \neq \left\{G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{c}_1, \ldots, \mathbf{c}_\ell) : \mathsf{Mat}(\mathbf{c}_1, \ldots, \mathbf{c}_\ell) \in \Lambda\left(\mathcal{C}^\ell, \mathbf{U}, \gamma\right)\right\}\right] \leq \epsilon_{\mathsf{MCA}}(\gamma) \quad .$$

Proof. We follow the proof in [ACFY25]. For $\mathbf{x} \in S$, define

$$\begin{aligned} A_{\mathbf{x}} &\coloneqq \Lambda(\mathcal{C}, G(\mathbf{x}) \cdot \mathbf{U}, \gamma) \quad , \\ B_{\mathbf{x}} &\coloneqq \left\{G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{c}_1, \ldots, \mathbf{c}_\ell) : \mathsf{Mat}(\mathbf{c}_1, \ldots, \mathbf{c}_\ell) \in \Lambda\left(\mathcal{C}^\ell, \mathbf{U}, \gamma\right)\right\} \quad . \end{aligned}$$

We prove the inclusions $A_{\mathbf{x}} \subseteq B_{\mathbf{x}}$ and $B_{\mathbf{x}} \subseteq A_{\mathbf{x}}$ separately.

$B_{\mathbf{x}} \subseteq A_{\mathbf{x}}$ always. If $\mathsf{Mat}(\mathbf{c}_1, \ldots, \mathbf{c}_\ell) \in \Lambda\left(\mathcal{C}^\ell, \mathbf{U}, e\right)$, there exists $T \subseteq [n]$ with $|T| \geq n \cdot (1-\gamma)$ such that $\mathbf{u}_j|_T = \mathbf{c}_j|_T$ for all $j \in [\ell]$. Then, $(G(\mathbf{x}) \cdot \mathbf{U})|_T = (G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{c}_1, \ldots, \mathbf{c}_\ell))|_T$, that is, $G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{c}_1, \ldots, \mathbf{c}_\ell) \in \Lambda(\mathcal{C}, G(\mathbf{x}) \cdot \mathbf{U}, \gamma)$.

$A_{\mathbf{x}} \subseteq B_{\mathbf{x}}$ with high probability. Fix $\mathbf{x} \in S$ such that for every $T \subseteq [n]$ with $|T| \geq n \cdot (1-\gamma)$, one of the following holds:

1. $G(\mathbf{x}) \cdot \mathbf{U}|_T \notin \mathcal{C}|_T$.
2. For every $j \in [\ell]$, $\mathbf{u}_j|_T \in \mathcal{C}|_T$.

Since G has MCA with error ϵ_{MCA}, there are at least $(1 - \epsilon_{\mathsf{MCA}}(\gamma)) \cdot |S|$ such $\mathbf{x}$. Consider $\mathbf{c} \in A_{\mathbf{x}}$, by definition point 1 above cannot hold: indeed there exists $T \subseteq [n]$ such that $|T| \geq n \cdot (1-\gamma) > n \cdot (1 - \delta_{\mathcal{C}})$ and $G(\mathbf{x}) \cdot \mathbf{U}|_T = \mathbf{c}|_T$. Therefore, it must be that point 2 holds: there exists $\mathbf{c}_1, \ldots, \mathbf{c}_\ell \in \mathcal{C}$ such that $\mathbf{u}_j|_T = \mathbf{c}_j|_T$, for every $j \in [\ell]$. For every $i \in T$,

$$\mathbf{c}[i] = (G(\mathbf{x}) \cdot \mathbf{U})[i] = (G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{c}_1, \ldots, \mathbf{c}_\ell))[i] \quad .$$

By definition of the code distance, $\mathbf{c} \equiv G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{c}_1, \ldots, \mathbf{c}_\ell)$, so $\mathbf{c} \in B_{\mathbf{x}}$. This argument holds simultaneously for every $\mathbf{c} \in A_{\mathbf{x}}$, as long as $\mathbf{x} \in S$ is such that either 1 or 2 above holds. As we remarked, there are at least $(1 - \epsilon_{\mathsf{MCA}}(\gamma)) \cdot |S|$ such $\mathbf{x}$. □

B Lower bounds for distance-preservation

Note that for $d \geq 1$, applying Lemma C.5 to the MDS generator of dimension $d+1$ given by the univariate powers $G\colon x \in \mathbb{F} \mapsto (1, x, \ldots, x^d) \in \mathbb{F}^{d+1}$ and any linear code $\mathcal{C} \in \mathbb{F}^n$ with distance $\Delta_{\mathcal{C}}$ gives us the following. For any $e < \Delta_{\mathcal{C}}/(d+2)$ and $\mathbf{U} \in \mathbb{F}^{(d+1)\times n}$ with $\Delta_{\sum^{d+1}}(\mathbf{U}, \mathcal{C}^{d+1}) \geq e$,

$$\Pr_{x\in\mathbb{F}}[\Delta(G(\mathbf{x}) \cdot \mathbf{U}, \mathcal{C}) < e] \leq e \cdot \frac{d}{|\mathbb{F}|} \ .$$

We construct an example that shows that this bound is tight. We prove it for the weakest form of distance preservation, which gives the strongest statement.

Lemma B.1. *Let $d \geq 1$, and let $\mathbb{F}$ be a finite field such that d divides $|\mathbb{F}| - 1$. For a linear code $\mathcal{C} \subseteq \mathbb{F}^n$ with distance $\Delta_{\mathcal{C}}$, let $0 < e \leq (\Delta_{\mathcal{C}} - 1)/2$, and assume $|\mathbb{F}| \geq e \cdot d + 1$. Then, there exists $\mathbf{u}_0, \ldots, \mathbf{u}_d \in \mathbb{F}^n$ such that $\max_j\{\Delta(\mathbf{u}_j, \mathcal{C})\} \geq e$ and*

$$\Pr_{x\in\mathbb{F}}[\Delta(\textstyle\sum_{j=0}^{d} x^j \cdot \mathbf{u}_j, \mathcal{C}) < e] \geq e \cdot \frac{d}{|\mathbb{F}|} \ .$$

Proof. Let $\omega \in \mathbb{F}^\times$ be an element of order d, and let $\mathcal{G} \coloneqq \langle\omega\rangle$ be the multiplicative subgroup generated by ω. By assumption, $e \leq (|\mathbb{F}| - 1)/d$, and so by Lagrange's theorem, we can consider e distinct cosets of $\mathcal{G}$ in $\mathbb{F}^\times$:

$$\mathcal{G}_1 \coloneqq a_1 \cdot \mathcal{G}, \ \ldots, \ \mathcal{G}_e \coloneqq a_e \cdot \mathcal{G} \ .$$

Define:

$$\begin{aligned}
\mathbf{u}_0 &\coloneqq (a_1^d, \ldots, a_e^d, \mathbf{0}^{n-e}) \\
\mathbf{u}_j &\coloneqq \mathbf{0}^n, \ \ 1 \leq j \leq d-1 \\
\mathbf{u}_d &\coloneqq (-1, -1, \ldots, -1, \mathbf{0}^{n-e}) \ .
\end{aligned}$$

We have $\Delta(\mathbf{u}_0, \mathbf{0}^n) = e \leq (\Delta_{\mathcal{C}} - 1)/2$, and so $\Delta(\mathbf{u}_0, \mathcal{C}) = \Delta(\mathbf{u}_0, \mathbf{0}^n) = e$. It follows that $\max_j\{\Delta(\mathbf{u}_j, \mathcal{C})\} \geq e$. For every $i \in [e]$, any element $x \in \mathcal{G}_i$ satisfies $x^d = a_i^d$. Therefore, for any $x \in \bigcup_i \mathcal{G}_i$, we have:

$$\begin{aligned}
\Delta\left(\sum_{j=0}^{d} x^j \cdot \mathbf{u}_j, \mathcal{C}\right) &= \Delta\left(\mathbf{u}_0 + x^d \cdot \mathbf{u}_d, \mathcal{C}\right) \\
&= \Delta\left(\mathbf{u}_0 + x^d \cdot \mathbf{u}_d, \mathbf{0}^n\right) \\
&< e \ .
\end{aligned}$$

Finally, notice that $|\bigcup_i \mathcal{G}_i| = e \cdot d$. □

C Correlated agreement

In this section we show that one can:

1. Prove a lossless reduction from the fact that random linear combinations have CA to the fact that lines have CA.
2. Prove that MDS generators are CA generators.

The first item is folklore, and was only recently written down formally in [GG25]. The second item is new, and leads to a better error parameter than Theorem 6.1, for the weaker guarantee of CA in the list-decoding setting.

C.1 Reducing random linear combinations to lines

We adapt an idea used in [BCIKS20] to propagate the distance-preservation error from lines to random linear combinations without loss. This allows us to obtain distance-preservation error bounds for random linear combinations that are linear in the blocklength n, and do not depend on the number of words that appear in the linear combination.

Lemma C.1. *Let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code. Let $\mathbf{u}_0, \ldots, \mathbf{u}_{\ell-1} \in \Sigma^n$ and $U \coloneqq \mathbf{u}_0 + \mathsf{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_{\ell-1}\}$. Assume that the generator $x \in \mathbb{F} \mapsto (1, x)$ has CA for $\mathcal{C}$ with error ϵ_{CA}. Let $U' \coloneqq \mathsf{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_{\ell-1}\}$. Then, for $e \leq n$,*

$$\Pr_{\mathbf{u} \in U}[\Delta(\mathbf{u}, \mathcal{C}) < e] > \epsilon_{\mathsf{CA}}(e, 1) \Rightarrow \max_{\mathbf{u}' \in U'} \Delta\left(\mathbf{u}', \mathcal{C}\right) < e.$$

Proof. Let $\mathbf{u}' \in U'$. If $\mathbf{u}'$ is the zero vector, then $\Delta(\mathbf{u}', \mathcal{C}) = 0 < e$. Otherwise, write $U = \widetilde{U} \oplus \mathsf{span}\{\mathbf{u}'\}$, where $\widetilde{U} \subset U$ is a direct complement of $\mathsf{span}\{\mathbf{u}'\}$ in U' shifted by $\mathbf{u}_0$. Then

$$\mathop{\mathbb{E}}_{\widetilde{\mathbf{u}} \in \widetilde{U}} \left[\Pr_{x \in \mathbb{F}} \left[\Delta(\widetilde{\mathbf{u}} + x\mathbf{u}', \mathcal{C}) < e \right] \right] = \Pr_{u \in U}[\Delta(u, \mathcal{C}) < e] > \epsilon_{\mathsf{CA}}(e, 1) \ .$$

In particular, there exists $\widetilde{u} \in \widetilde{U}$ such that

$$\Pr_{x \in \mathbb{F}} \left[\Delta(\widetilde{\mathbf{u}} + x\mathbf{u}', \mathcal{C}) < e \right] > \epsilon_{\mathsf{CA}}(e, 1).$$

By assumption, we obtain that $\Delta(\mathbf{u}', \mathcal{C}) < e$. □

Lemma C.2. *Let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code. Let $\mathbf{u}_0, \ldots, \mathbf{u}_{\ell-1} \in \Sigma^n$ and $U \coloneqq \mathbf{u}_0 + \mathsf{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_{\ell-1}\}$. Let ϵ_{CA} be defined as in Lemma C.1. We have, for $e \leq n$,*

$$\Pr_{\mathbf{u} \in U}[\Delta(\mathbf{u}, \mathcal{C}) < e] > \epsilon_{\mathsf{CA}}(e, 1) \implies \max_{\mathbf{u} \in U} \Delta\left(\mathbf{u}, \mathcal{C}\right) < e \ .$$

Proof. Let $V = \mathsf{span}(U)$. If $\mathbf{0} \in U$, we have $V = U$. Otherwise, V can be described as the disjoint union

$$V = U' \cup \{x \cdot U : x \in \mathbb{F} \setminus \{0\}\} \ ,$$

where $U' = \mathsf{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_{\ell-1}\}$. Since U' is a linear space, Lemma C.1 ensures that for every $\mathbf{u}' \in U'$, $\Delta(\mathbf{u}', \mathcal{C}) < e$. Moreover, for every $x \in \mathbb{F} \setminus \{0\}$ and $\mathbf{u} \in U$, $\Delta(x \cdot \mathbf{u}, \mathcal{C}) = \Delta(\mathbf{u}, \mathcal{C})$. Therefore, we have

$$\Pr_{\mathbf{v} \in V}[\Delta(\mathbf{v}, \mathcal{C}) < e] \geq \Pr_{\mathbf{u} \in U}[\Delta(\mathbf{u}, \mathcal{C}) < e] > \epsilon_{\mathsf{CA}}(e, 1) \ .$$

We apply Lemma C.1 to V. We get that for every $\mathbf{v} \in V$, $\Delta(\mathbf{v}, \mathcal{C}) < e$, and the result follows from the fact that $U \subseteq V$. □

Finally, we arrive at the statement that for every $\mathbb{F}$-linear code, correlated agreement over lines implies correlated agreement over affine spaces. In other words, obtaining correlated agreement for a random linear combination of two words implies correlated agreement for a random linear combination of any number of words *with the same error* (and the same proximity parameter).

Lemma C.3. *Let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code. Let $\mathbf{U} = \mathsf{Mat}(\mathbf{u}_0, \ldots, \mathbf{u}_{\ell-1}) \in \Sigma^{\ell \times n}$ and $U \coloneqq \mathbf{u}_0 + \mathsf{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_{\ell-1}\}$. Let ϵ_{CA} be defined as in Lemma C.1, and let $e \leq n$. Suppose $\mathcal{C}$ satisfies the following property: for every $\mathbf{u} \in \Sigma^n$, $|\{\mathbf{c} \in \mathcal{C} : \Delta(\mathbf{u}, \mathbf{c}) < e\}| < |\Sigma|$. Assume further that*

$$\Pr_{\mathbf{u} \in U}[\Delta(\mathbf{u}, \mathcal{C}) < e] > \epsilon_{\mathsf{CA}}(e, 1) \ .$$

Then, there exists $T \subseteq [n]$ of size $|T| > n - e$ such that for every $\mathbf{u} \in U, \mathbf{u}|_T \in \mathcal{C}|_T$. Consequently $\mathbf{U}|_T \in \mathcal{C}^\ell|_T$, and $\Delta_{\Sigma^\ell}(\mathbf{U}, \mathcal{C}^\ell) < e$.

Proof. Let $e_{\max} \coloneqq \max_{\mathbf{u} \in U} \Delta(\mathbf{u}, \mathcal{C})$, and let $\mathbf{w} \in U$ such that $\Delta(\mathbf{w}, \mathcal{C}) = e_{\max}$. Define $\mathcal{L}$ the set of codewords $\mathcal{L} \coloneqq \{\mathbf{c} \in \mathcal{C} : \Delta(\mathbf{w}, \mathbf{c}) \leq e_{\max}\}$. In fact, by definition of $\mathbf{w}$, we have

$$\mathcal{L} = \{\mathbf{c} \in \mathcal{C} : \Delta(\mathbf{w}, \mathbf{c}) = e_{\max}\} \ .$$

For each $\mathbf{c} \in \mathcal{L}$, we define

$$T_{\mathbf{c}} = \{j \in [n] : \mathbf{w}(j) = \mathbf{c}(j)\} \ ,$$

the agreement set between $\mathbf{w}$ and $\mathbf{c}$, and

$$U_{\mathbf{c}} = \{\mathbf{u} \in U : \mathbf{u}|_{T_{\mathbf{c}}} \in \mathcal{C}|_{T_{\mathbf{c}}}\} \ ,$$

the set of all vectors of U that agree with some codeword on $T_{\mathbf{c}}$. We will show that there exists $\mathbf{c} \in \mathcal{L}$ such that $U_{\mathbf{c}} = U$. Notice that for every $\mathbf{c} \in \mathcal{L}$, $|T_{\mathbf{c}}| = n - e_{\max}$, and $U_{\mathbf{c}}$ is an affine subspace of U. We now show that each element of U belongs to at least one $U_{\mathbf{c}}$. First, observe that $\mathbf{w}$ belongs to every $U_{\mathbf{c}}$. Hence, we consider $\mathbf{u} \in U \setminus \{\mathbf{w}\}$. We look at correlated agreement over the affine line

$$\{\mathbf{w} + x \cdot (\mathbf{u} - \mathbf{w}) : x \in \mathbb{F}\} \subset U$$

that contains both $\mathbf{w}$ and $\mathbf{u}$. We have

$$\Pr_{x \in \mathbb{F}}[\Delta(\mathbf{w} + x \cdot (\mathbf{u} - \mathbf{w}), \mathcal{C}) \leq e_{\max}] = 1 > \epsilon_{\mathsf{CA}}(e, 1) \ .$$

By Lemma C.2, $e_{\max} + 1 \leq e$, which means that we can apply our assumption on the affine line generator. We find $T \subseteq [n]$ with $|T| \geq n - e_{\max}$ and codewords $\mathbf{c}_{\mathbf{w}}, \mathbf{c}_{\mathbf{u}} \in \mathcal{C}$ which respectively agree with $\mathbf{w}$ and $\mathbf{u}$ on T. In particular, $\Delta(\mathbf{w}, \mathbf{c}_{\mathbf{w}}) \leq e_{\max}$. Thus there exists $\mathbf{c} \in \mathcal{L}(\mathbf{w}, e_{\max})$ such that $\mathbf{c}_{\mathbf{w}} = \mathbf{c}$ and $T = T_{\mathbf{c}}$. This implies that $\mathbf{u} \in U_{\mathbf{c}}$, and thus $U \subseteq \bigcup_{\mathbf{c} \in \mathcal{L}} U_{\mathbf{c}}$.

Now, let $\mathbf{c}^* \in \mathcal{L}$ be such that $|U_{\mathbf{c}^*}| = \max_{\mathbf{c} \in \mathcal{L}} |U_{\mathbf{c}}|$. We have $|U_{\mathbf{c}^*}| \geq \frac{|U|}{|\mathcal{L}|} > \frac{|U|}{|\mathbb{F}|}$. Since $U_{\mathbf{c}^*}$ is an $\mathbb{F}$-linear subspace of U, we deduce that $U_{\mathbf{c}^*} = U$. Therefore, we have found $T = T_{\mathbf{c}^*}$ of size $|T| > n - e$ such that for every $\mathbf{u} \in U, \mathbf{u}|_T \in \mathcal{C}|_T$. In particular, the projections of $\mathbf{u}_0, \mathbf{u}_1, \ldots, \mathbf{u}_{\ell-1} \in U$ on $T = T_{\mathbf{c}^*}$ are codewords of $\mathcal{C}|_T$, since they are elements, or differences of elements, in $U_{\mathbf{c}^*} = U$. In other words, $\mathbf{U}|_T \in \mathcal{C}^\ell|_T$. □

Remark C.4. When $\Sigma = \mathbb{F}$, the condition that for every $\mathbf{u} \in \mathbb{F}^n$, $|\{\mathbf{c} \in \mathcal{C} : \Delta(\mathbf{u}, \mathbf{c}) < e\}| < |\mathbb{F}|$ is not very restrictive. If e is within the list-decoding bound, then $|\{\mathbf{c} \in \mathcal{C} : \Delta(\mathbf{u}, \mathbf{c}) < e\}| < n$ (see [GRS23]). So, as long as the field size is at least the blocklength of the code the condition will be satisfied up to the list-decoding bound. This also means that if one were able to extend the distance-preservation property of affine lines up to the list-decoding bound, the previous series of results would still apply, as long as the field size is at least the blocklength of the code.

C.2 CA for MDS generators

Unique decoding analysis.

Lemma C.5. *Let $G\colon S \to \mathbb{F}^\ell$ be an MDS generator such that $\mathcal{C}_G$ has dimension $\ell \geq 2$. Let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code with distance $\Delta_\mathcal{C}$. For every $\mathbf{U} \coloneqq \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell) \in \Sigma^{\ell \times n}$, parameter $\eta \in (0,1)$, integers $e > t > 0$ such that:*

$$e - t < \frac{\Delta_\mathcal{C}}{\ell + 1} \ ,$$

if $\Delta_{\Sigma^\ell}(\mathbf{U}, \mathcal{C}^\ell) \geq e$, then:

$$\Pr_{\mathbf{x} \in S}\left[\Delta\left(G(\mathbf{x}) \cdot \mathbf{U}, \mathcal{C}\right) \leq e - t\right] \leq \frac{e}{t} \cdot \frac{(\ell - 1)}{|S|} \ .$$

Proof. We prove the contrapositive. Define:

$$B \coloneqq \left\{\mathbf{x} \in S : \Delta\left(G(\mathbf{x}) \cdot \mathbf{U}, \mathcal{C}\right) \leq e - t\right\} \ ,$$

and assume that $|B| > \frac{e}{t} \cdot (\ell - 1)$. For any $\mathbf{x} \in B$, define $\mathbf{u}_\mathbf{x} \coloneqq G(\mathbf{x}) \cdot \mathbf{U}$. By assumption $|B| \geq \ell$, therefore, we can choose ℓ distinct elements $\mathbf{x}_1, \ldots, \mathbf{x}_\ell$ in B. Let $\mathbf{c}_1, \ldots, \mathbf{c}_\ell \in \mathcal{C}$ be the unique closest codewords to $\mathbf{u}_{\mathbf{x}_1}, \ldots, \mathbf{u}_{\mathbf{x}_\ell}$, respectively (these codewords exist by definition of B). By assumption on G, the matrix $\mathbf{M} \coloneqq \mathsf{Mat}(G(\mathbf{x}_1), \ldots, G(\mathbf{x}_\ell))$ is invertible. Define $\mathbf{c}^*_1, \ldots, \mathbf{c}^*_\ell \in \mathcal{C}$ as follows:

$$\mathsf{Mat}(\mathbf{c}^*_1, \ldots, \mathbf{c}^*_\ell) \coloneqq \mathbf{M}^{-1} \cdot \mathsf{Mat}(\mathbf{c}_1, \ldots, \mathbf{c}_\ell)$$

Note that by applying the triangle inequality we obtain the following, for any $\mathbf{x} \in B$:

$$\begin{aligned}
\Delta(\mathbf{u}_\mathbf{x}, G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{c}^*_1, \ldots, \mathbf{c}^*_\ell)) &= \Delta(G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell), G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{c}^*_1, \ldots, \mathbf{c}^*_\ell)) \\
&= \Delta(G(\mathbf{x}) \cdot \mathbf{M}^{-1} \cdot \mathsf{Mat}(\mathbf{u}_{\mathbf{x}_1}, \ldots, \mathbf{u}_{\mathbf{x}_\ell}), G(\mathbf{x}) \cdot \mathbf{M}^{-1} \cdot \mathsf{Mat}(\mathbf{c}_1, \ldots, \mathbf{c}_\ell)) \\
&\leq \sum_{i=1}^{\ell} \Delta(\mathbf{u}_{\mathbf{x}_i}, \mathbf{c}_i) \\
&\leq \ell \cdot (e - t) \ .
\end{aligned}$$

For any $\mathbf{x} \in B$, let $\mathbf{c}_\mathbf{x}$ be the unique closest codeword to $\mathbf{u}_\mathbf{x}$. By applying the triangle inequality again:

$$\begin{aligned}
\Delta(\mathbf{c}_\mathbf{x}, G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{c}^*_1, \ldots, \mathbf{c}^*_\ell)) &\leq \Delta(\mathbf{c}_\mathbf{x}, \mathbf{u}_\mathbf{x}) + \Delta(\mathbf{u}_\mathbf{x}, G(\mathbf{x}) \cdot \mathsf{Mat}(\mathbf{c}^*_1, \ldots, \mathbf{c}^*_\ell)) \\
&\leq (\ell + 1) \cdot (e - t) \\
&< \Delta_\mathcal{C} \ .
\end{aligned}$$

This last inequality holds by assumption on γ. By definition of the minimal distance, we find that for any $\mathbf{x} \in B$, $\mathbf{c}_\mathbf{x} = \mathsf{Span}_G(\mathbf{c}^*_1, \ldots, \mathbf{c}^*_\ell)[\mathbf{x}]$. By Lemma 5.3, this implies that $\Delta_{\Sigma^\ell}(\mathbf{U}, \mathcal{C}^\ell) < e$. □

List-decoding analysis.

Lemma C.6. *Let $G\colon S \to \mathbb{F}^\ell$ be an MDS generator such that $\mathcal{C}_G$ has dimension $\ell \geq 2$. Let $\mathcal{C} \subseteq \Sigma^n$ be an $\mathbb{F}$-linear code with distance $\Delta_\mathcal{C}$. For every $\mathbf{U} \coloneqq \mathsf{Mat}(\mathbf{u}_1, \ldots, \mathbf{u}_\ell) \in \Sigma^{\ell \times n}$, parameter $\eta \in (0,1)$, integers $e > t > 0$ such that:*

$$e - t < n - n\left(1 - \frac{\Delta_\mathcal{C}}{n} + \eta\right)^{\frac{1}{\ell+1}} \ ,$$

if $\Delta_{\Sigma^\ell}(\mathbf{U}, \mathcal{C}^\ell) \geq e$*, then:*

$$\Pr_{\mathbf{x}\in S}\left[\Delta\left(G(\mathbf{x})\cdot\mathbf{U}, \mathcal{C}\right) \leq e - t\right] \leq \frac{1}{|S|}\cdot\frac{e\cdot\ell\cdot(\ell+3)}{2\cdot t\cdot\eta} \ .$$

Proof. For every $\mathbf{x} \in S$, denote by $\mathbf{u}_\mathbf{x}$ the word $G(\mathbf{x})^\intercal \cdot \mathbf{U}$. We prove the contrapositive of the statement in the theorem, so let $\alpha \coloneqq \frac{2\cdot(\ell-1)\cdot\eta}{\ell\cdot(\ell+3)}$ and assume that the set $A \coloneqq \{\mathbf{x} \in S : \Delta\left(G(\mathbf{x})^\intercal \cdot \mathbf{U}, \mathcal{C}\right) \leq e - t\}$ has size $|A| > \frac{e\cdot(\ell-1)}{\alpha\cdot t}$. For each $\mathbf{x} \in A$, denote by $\mathbf{c}_\mathbf{x} \in \mathcal{C}$ the closest codeword to $\mathbf{u}_\mathbf{x}$ (breaking ties arbitrarily). Denote the agreement set between $\mathbf{u}_\mathbf{x}$ and $\mathbf{c}_\mathbf{x}$ by $T_\mathbf{x}$.

Our plan is fairly straightforward: we try to find a large enough subset A' of A such that we can apply both Lemma 5.2 and Lemma 5.3.

For $\mathbf{x}_0, \ldots, \mathbf{x}_\ell$ sampled uniformly and independently from A and j sampled uniformly from $[n]$, we have:

$$\begin{aligned}
\mathop{\mathbb{E}}_{\mathbf{x}_0,\ldots,\mathbf{x}_\ell}\left[|T_{\mathbf{x}_0} \cap \cdots \cap T_{\mathbf{x}_\ell}|\right] &= n \cdot \mathop{\mathbb{E}}_{j,\mathbf{x}_0,\ldots,\mathbf{x}_\ell}\left[\mathbf{1}\left\{j \in T_{\mathbf{x}_0} \cap \cdots \cap T_{\mathbf{x}_\ell}\right\}\right] \\
&= n \cdot \mathop{\mathbb{E}}_{j}\left[\mathop{\mathbb{E}}_{\mathbf{x}}\left[\mathbf{1}\left\{j \in T_{\mathbf{x}}\right\}\right]^{\ell+1}\right] \\
&\geq n \cdot \mathop{\mathbb{E}}_{j,\mathbf{x}}\left[\mathbf{1}\left\{j \in T_{\mathbf{x}}\right\}\right]^{\ell+1} \\
&\geq n \cdot \left(1 - \frac{e-t}{n}\right)^{\ell+1} \\
&> n - \Delta_\mathcal{C} + \eta n \ .
\end{aligned}$$

The second equality follows from the independence of the random variables, then we applied Jensen's inequality. The second inequality is by definition of A and $T_\mathbf{x}$, and finally the last inequality is by assumption on $e - t$. We deduce that:

$$\Pr_{\mathbf{x}_0,\ldots,\mathbf{x}_\ell\in A}\left[|T_{\mathbf{x}_0} \cap \cdots \cap T_{\mathbf{x}_\ell}| > n - \Delta_\mathcal{C}\right] \geq \frac{\mathbb{E}\left[|T_{\mathbf{x}_0} \cap \cdots \cap T_{\mathbf{x}_\ell}| - n + \Delta_\mathcal{C}\right]}{n} \geq \eta \ .$$

Moreover, the probability that $\mathbf{x}_0, \ldots, \mathbf{x}_\ell$ are not all distinct is at most $\frac{\ell(\ell+1)}{2|A|}$. We compute:

$$\begin{aligned}
\frac{\ell\cdot(\ell+1)}{2\,|A|} &\leq \frac{\ell\cdot(\ell+1)\cdot\alpha}{2\cdot(\ell-1)} \\
&= \frac{\ell\cdot(\ell+3-2)\cdot\alpha}{2\cdot(\ell-1)} \\
&= \frac{\ell\cdot(\ell+3)\cdot\alpha}{2\cdot(\ell-1)} - \frac{\ell\cdot\alpha}{(\ell-1)} \\
&\leq \eta - \alpha \ .
\end{aligned}$$

The first inequality follows by assumption on the size of A and the fact that $e > t$. The second inequality follows by our definition of $\alpha = \frac{2\cdot(\ell-1)\cdot\eta}{\ell\cdot(\ell+3)}$. Hence, the probability over $\mathbf{x}_0, \ldots, \mathbf{x}_\ell$ sampled uniformly and independently from A that $\mathbf{x}_0, \ldots, \mathbf{x}_\ell$ are not all distinct or $|T_{\mathbf{x}_0} \cap \cdots \cap T_{\mathbf{x}_\ell}| \leq n - \Delta_\mathcal{C}$ is at most:

$$\frac{\ell\cdot(\ell+1)}{2\,|A|} + 1 - \eta \leq 1 - \alpha \ .$$

Thus, with probability at least α over the choice of $\mathbf{x}_0, \ldots, \mathbf{x}_\ell$, we simultaneously have that $\mathbf{x}_0, \ldots, \mathbf{x}_\ell$ are all distinct and $|T_{\mathbf{x}_0} \cap \cdots \cap T_{\mathbf{x}_\ell}| > n - \Delta_{\mathcal{C}}$. Therefore, we can consider distinct $\mathbf{x}_1, \ldots, \mathbf{x}_\ell$ such that:

$$\Pr_{\mathbf{x} \in A}[|T_{\mathbf{x}} \cap T_{\mathbf{x}_1} \cap \cdots \cap T_{\mathbf{x}_\ell}| > n - \Delta_{\mathcal{C}}] \geq \alpha \ .$$

Define A' as the set:

$$A' \coloneqq \{\mathbf{x} \in A : |T_{\mathbf{x}} \cap T_{\mathbf{x}_1} \cap \cdots \cap T_{\mathbf{x}_\ell}| > n - \Delta_{\mathcal{C}}\} \ .$$

By Lemma 5.2, for every $\mathbf{x} \in A'$, the codeword $\mathbf{c}_{\mathbf{x}}$ is equal to $\mathsf{Span}_G(\mathbf{c}_1, \ldots, \mathbf{c}_\ell)[\mathbf{x}]$ (see Definition 5.1), where $\mathbf{c}_1, \ldots, \mathbf{c}_\ell \in \mathcal{C}$ are determined uniquely by $\mathbf{x}_1, \ldots, \mathbf{x}_\ell$ and $\mathbf{c}_{\mathbf{x}_1}, \ldots, \mathbf{c}_{\mathbf{x}_\ell}$. Note that by Lemma 3.13, the zero-evading error of G is $\epsilon_{\mathsf{ZE}} \coloneqq (\ell - 1)/|S|$, and that $|A'| \geq \alpha \cdot |A| > \frac{e}{t} \cdot \epsilon_{\mathsf{ZE}} \cdot |S|$. Further by definition, for every $\mathbf{x} \in A'$:

$$e - t \geq \Delta(\mathbf{u}_{\mathbf{x}}, \mathbf{c}_{\mathbf{x}}) = \Delta(\mathbf{u}_{\mathbf{x}}, \mathsf{Span}_G(\mathbf{c}_1, \ldots, \mathbf{c}_\ell)[\mathbf{x}]) \ .$$

By Lemma 5.3, $\Delta_{\Sigma^\ell}(\mathbf{U}, \mathcal{C}^\ell) < e$. □

D Proof of a generalization of the Corrádi lemma

Definition D.1. *For $k \geq 2$ and $\alpha^k > \rho$, the k-wise* **Corrádi coefficient** *is defined as:*

$$C_k(\alpha, \rho) \coloneqq \frac{\frac{k\cdot(k-1)}{2} \cdot \left(\frac{(k-1)^{k-2}\cdot(k-2)^2}{2} + 1\right) \cdot \alpha - \rho}{\alpha^k - \rho} \ .$$

Lemma D.2. *Let $A_1, \ldots, A_m$ be subsets of some set A. Let $k \geq 2$ and $\alpha, \rho \in [0,1]$. Suppose that:*

- *for every $i \in [m]$, $|A_i| = \alpha \cdot |A|$;*
- *for every $i_1 \neq \cdots \neq i_k$, $|A_{i_1} \cap \cdots \cap A_{i_k}| \leq \rho \cdot |A|$.*

If $\alpha^k > \rho$ then

$$m \leq C_k(\alpha, \rho) \ .$$

Proof. Let $\alpha|A| \coloneqq a$ and $\rho|A| \coloneqq r$. We apply Jensen's inequality as follows:

$$\left(\frac{1}{|A|}\sum_{x\in A}\sum_{i=1}^{m}\mathbf{1}_{A_i}(x)\right)^k \leq \frac{1}{|A|}\sum_{x\in A}\left(\sum_{i=1}^{m}\mathbf{1}_{A_i}(x)\right)^k \ .$$

Here, $\mathbf{1}_{A_i}$ denotes the indicator function of the set A_i. The left hand side is greater or equal to $m^k a^k/|A|^k$. The expression in the right hand side can be expanded as follows:

$$\begin{aligned}
\sum_{x\in A}\left(\sum_{i=1}^{m}\mathbf{1}_{A_i}(x)\right)^k &= \sum_{i_1,\ldots,i_k=1}^{m}\sum_{x\in A}\mathbf{1}_{A_{i_1}}(x)\ldots\mathbf{1}_{A_{i_k}}(x) \\
&= \sum_{i_1,\ldots,i_k=1}^{m}|A_{i_1}\cap\cdots\cap A_{i_k}| \\
&= \sum_{j=1}^{k-1}S(k,j)\sum_{i_1\neq\ldots\neq i_j}|A_{i_1}\cap\cdots\cap A_{i_j}| + \sum_{i_1\neq\ldots\neq i_k}|A_{i_1}\cap\cdots\cap A_{i_k}| \ .
\end{aligned}$$

Here, $S(k,j)$ denotes the Stirling numbers of the second kind. We are using the fact that the number of surjective functions from a set of size k to a set of size j is given by $j!\cdot S(k,j)$. An upper bound for $S(k,j)$ is given by $1/2\cdot\binom{k}{j}j^{k-j}$. It follows that the previous sum is upper bounded by:

$$\begin{aligned}
&\frac{a}{2}\cdot\sum_{j=1}^{k-1}j!\binom{k}{j}\binom{m}{j}j^{k-j} + r\cdot k!\binom{m}{k} \\
&\leq \frac{a}{2}\cdot m^{k-1}\sum_{j=1}^{k-1}\binom{k}{j}j^{k-j} + r\cdot(m^k - m^{k-1}) \\
&\leq \frac{k(k-1)}{2}\cdot\left(\frac{(k-1)^{k-2}(k-2)^2}{2}+1\right)\cdot m^{k-1}\cdot a + (m^k - m^{k-1})\cdot r \ .
\end{aligned}$$

The last inequality follows from the following estimate:

$$\sum_{j=1}^{k-1}\binom{k}{j}j^{k-j} = \sum_{j=1}^{k-2}\binom{k}{j}j^{k-j} + k(k-1)$$

$$
\begin{aligned}
&\leq \sum_{j=1}^{k-2} \frac{k(k-1)}{(k-j)(k-j-1)} \binom{k-2}{j} (k-2)^{k-j} + k(k-1) \\
&\leq \frac{k(k-1)(k-2)^2}{2} \cdot \sum_{j=1}^{k-2} \binom{k-2}{j} (k-2)^{k-2-j} + k(k-1) \\
&= \frac{k(k-1)^{k-1}(k-2)^2}{2} + k(k-1)
\end{aligned}
$$

We have found $m \cdot a^k \leq |A|^{k-1} \cdot \left(\frac{k(k-1)}{2} \cdot \left(\frac{(k-1)^{k-2}(k-2)^2}{2} + 1 \right) \cdot a + (m-1) \cdot r \right)$. Dividing through by $|A|^k$ and rearranging gives the desired result, as long as $\alpha^k > \rho$. □

Note that in the case $k = 2$ we recover the expression of the usual lemma (see Lemma 3.23).

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